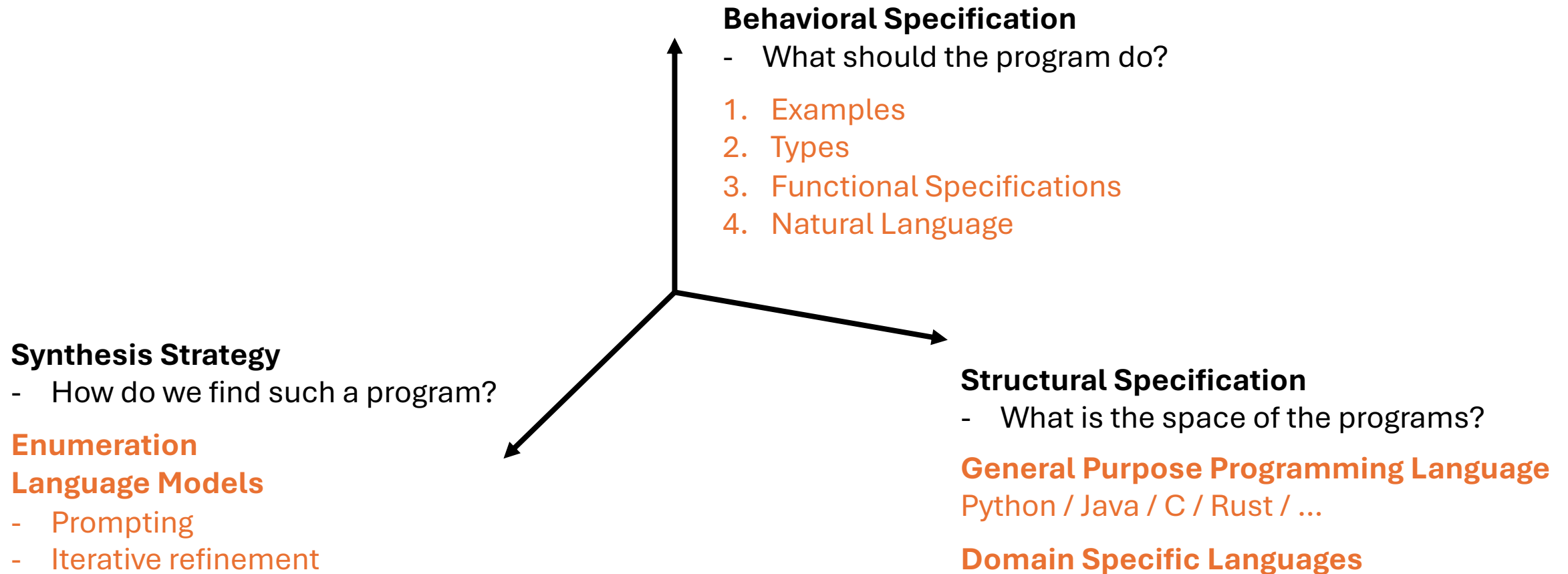


Machine Programming

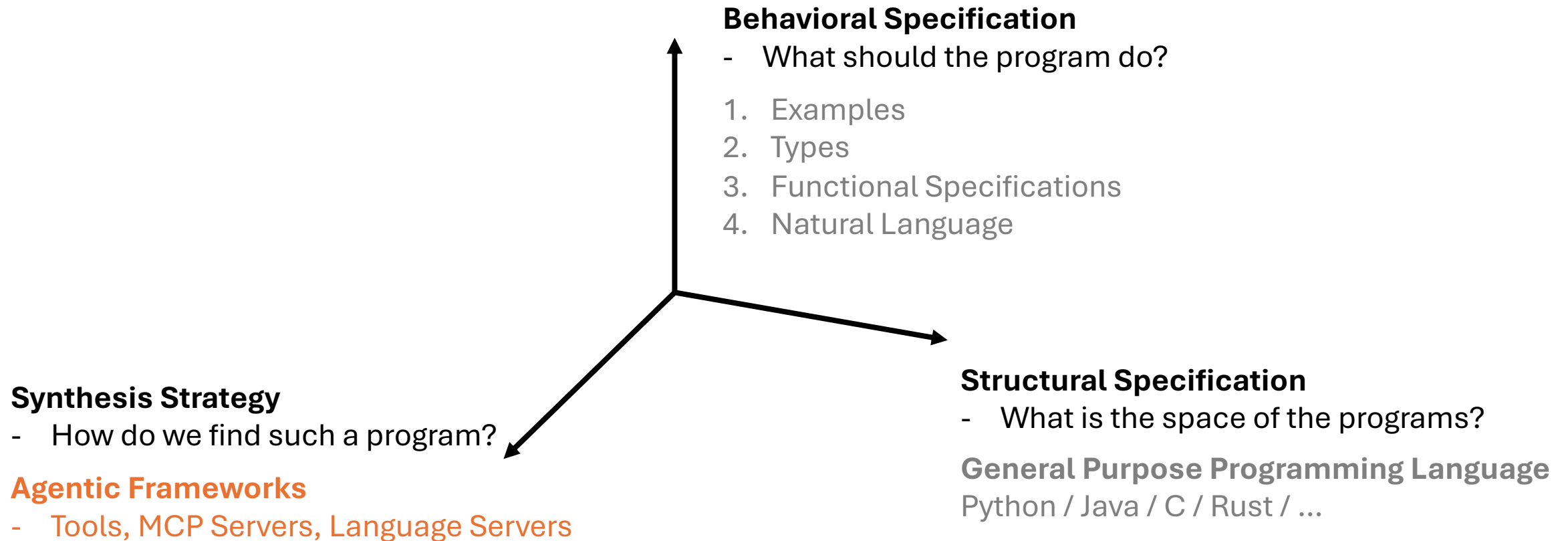
Lecture 8 – Agentic Frameworks: Model Context Protocol

Ziyang Li

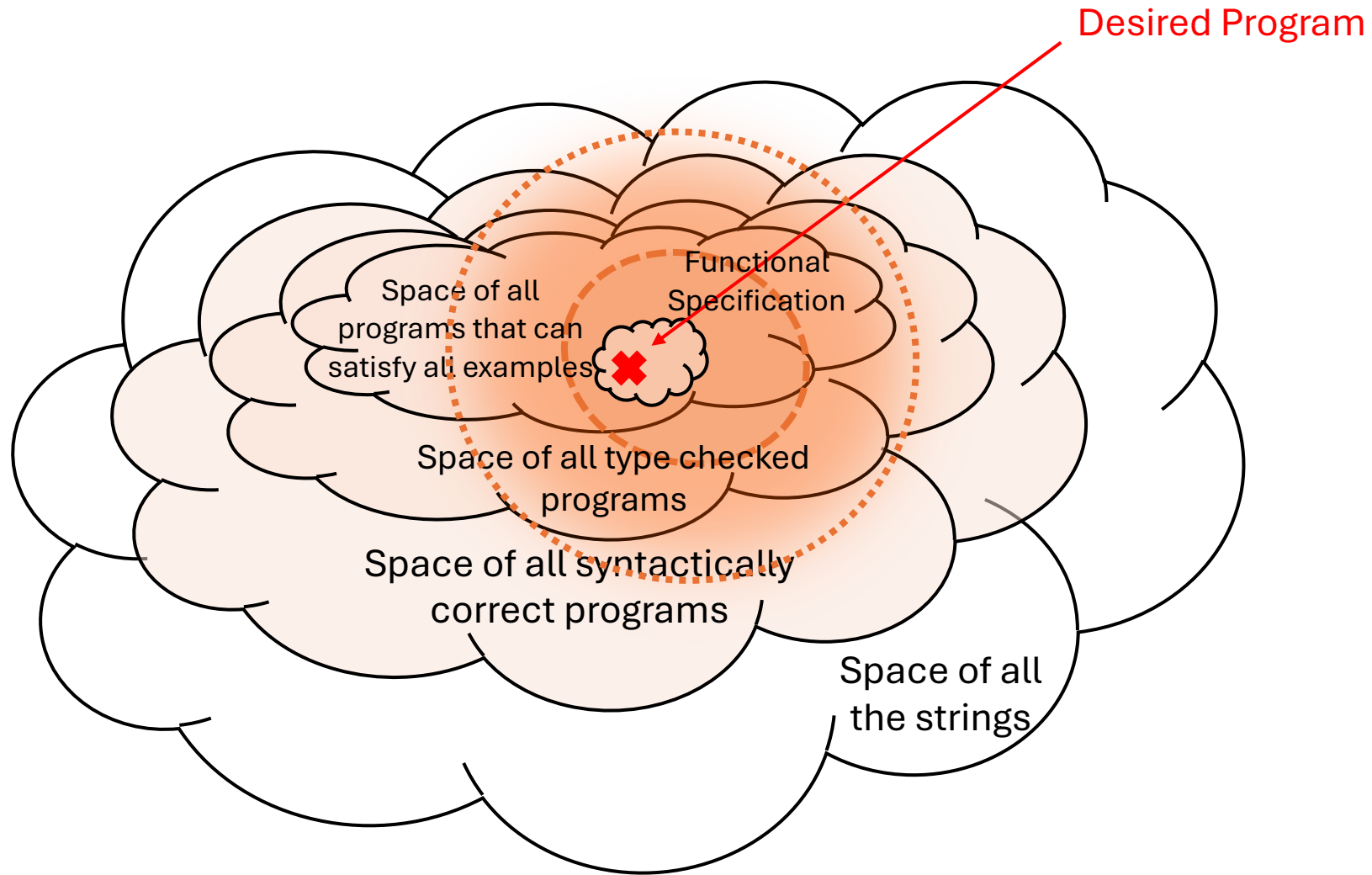
The Course So Far



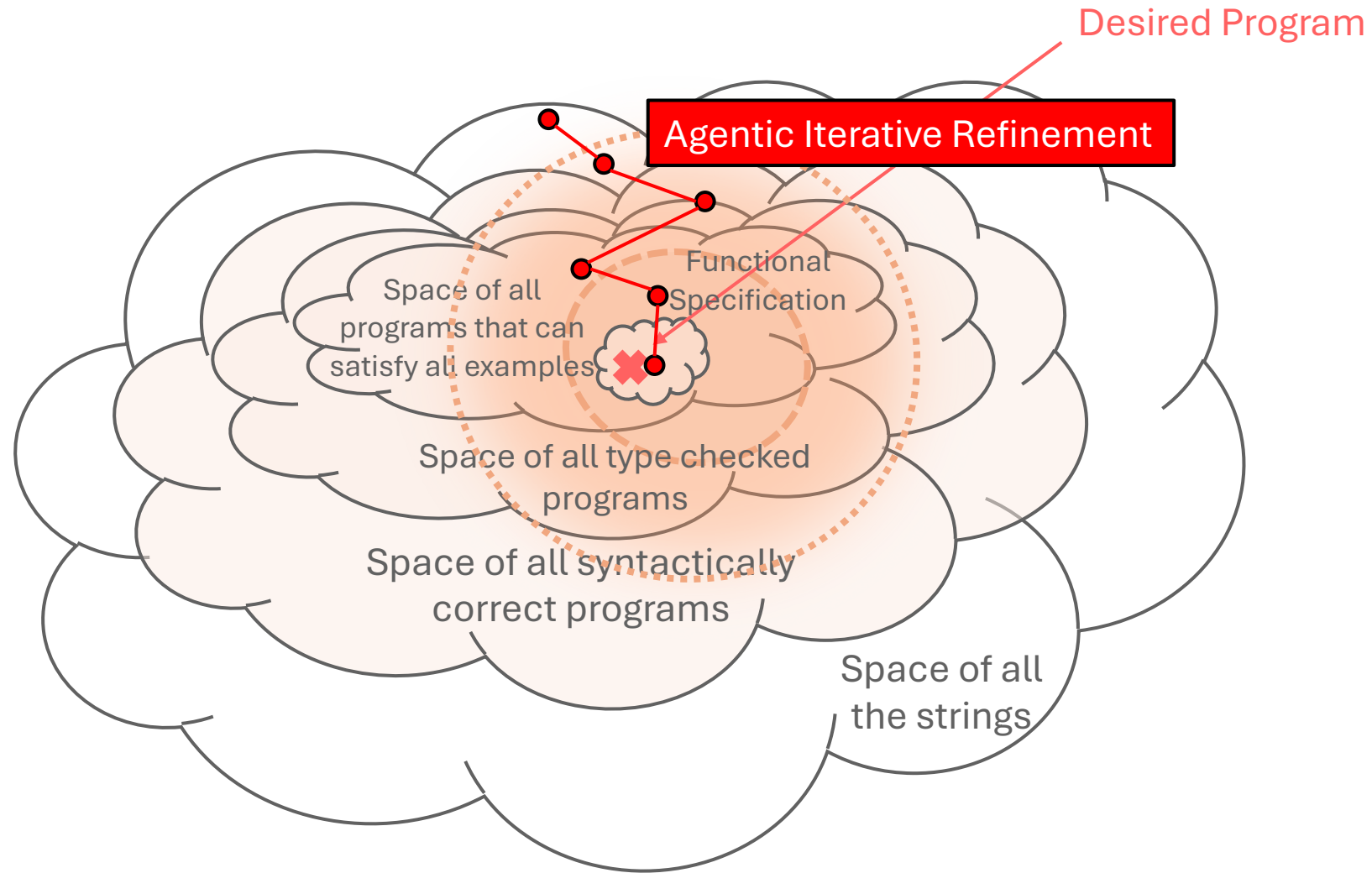
Today



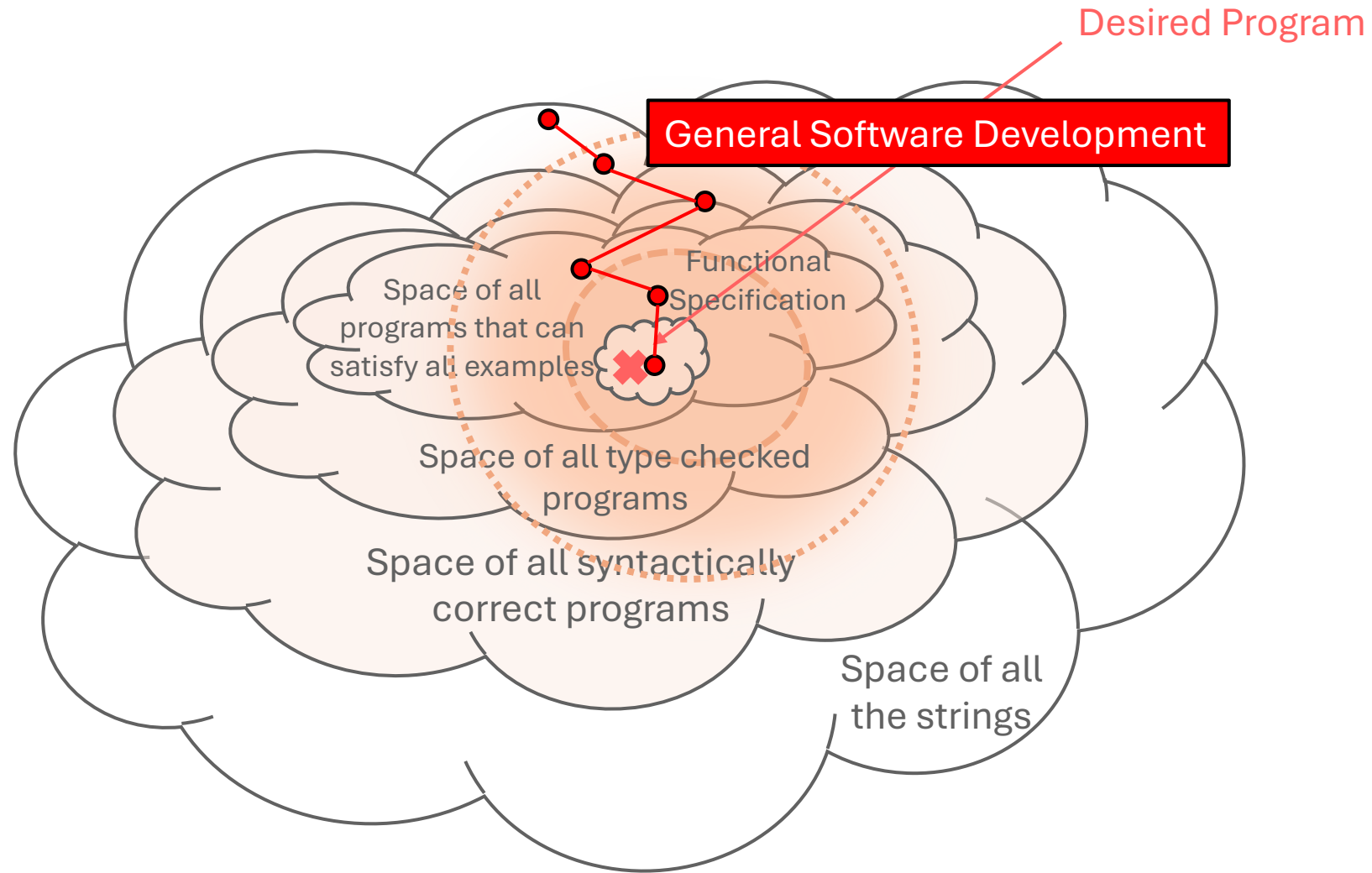
High Level Picture



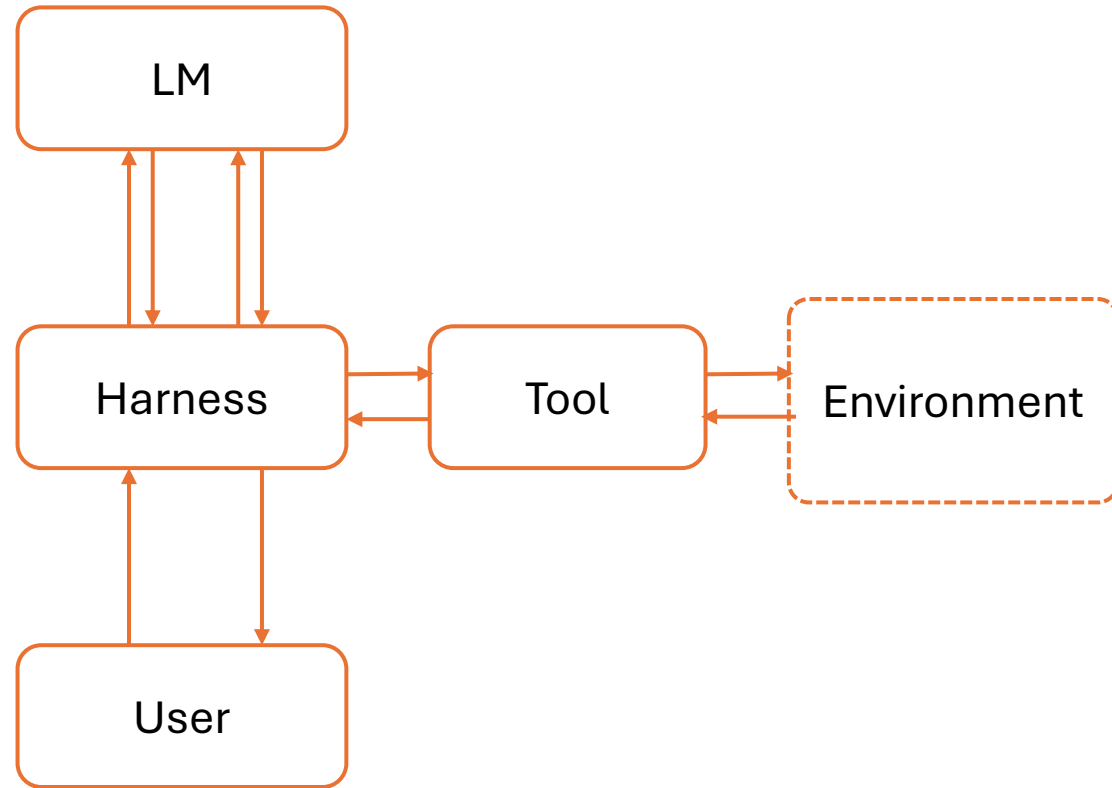
High Level Picture



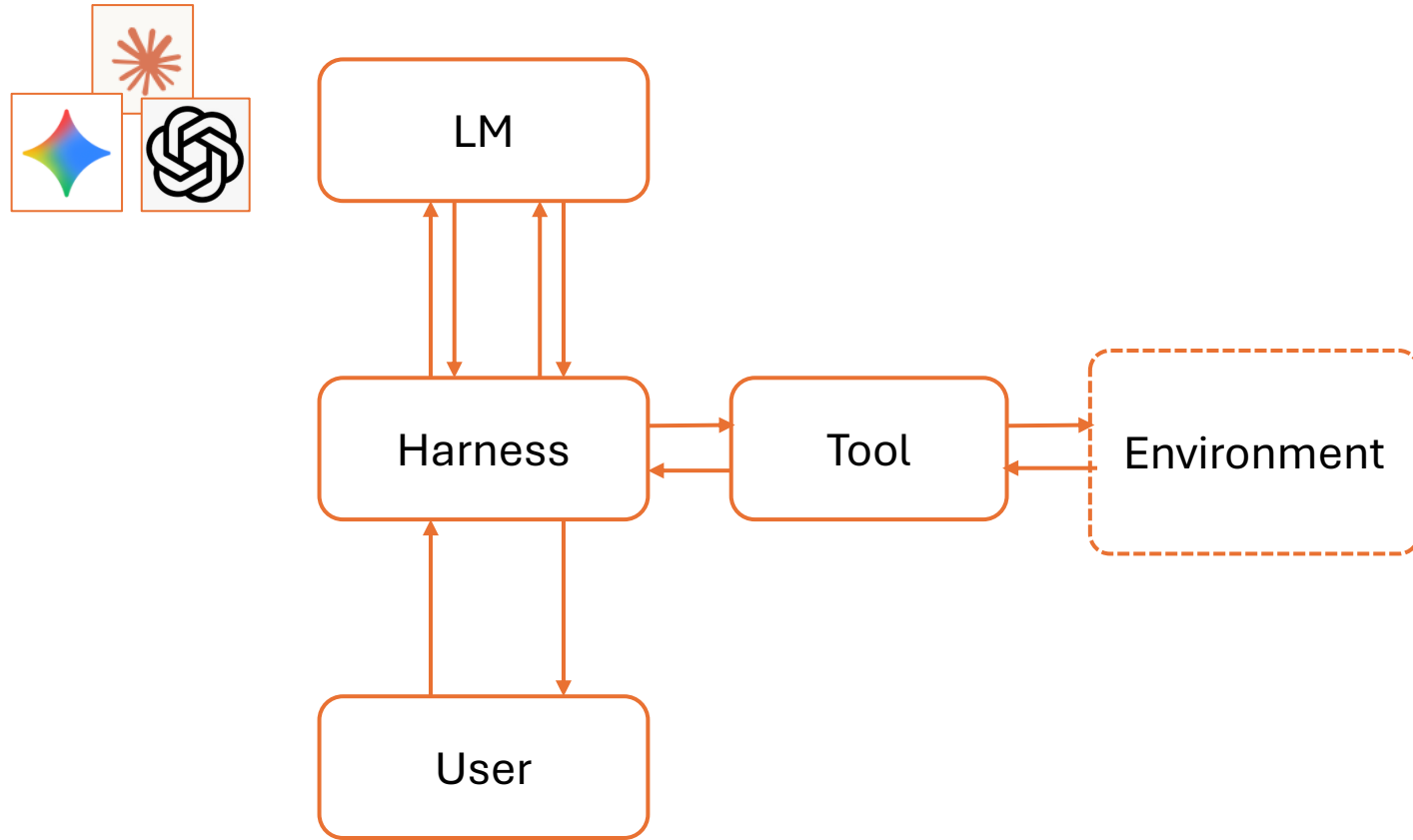
High Level Picture



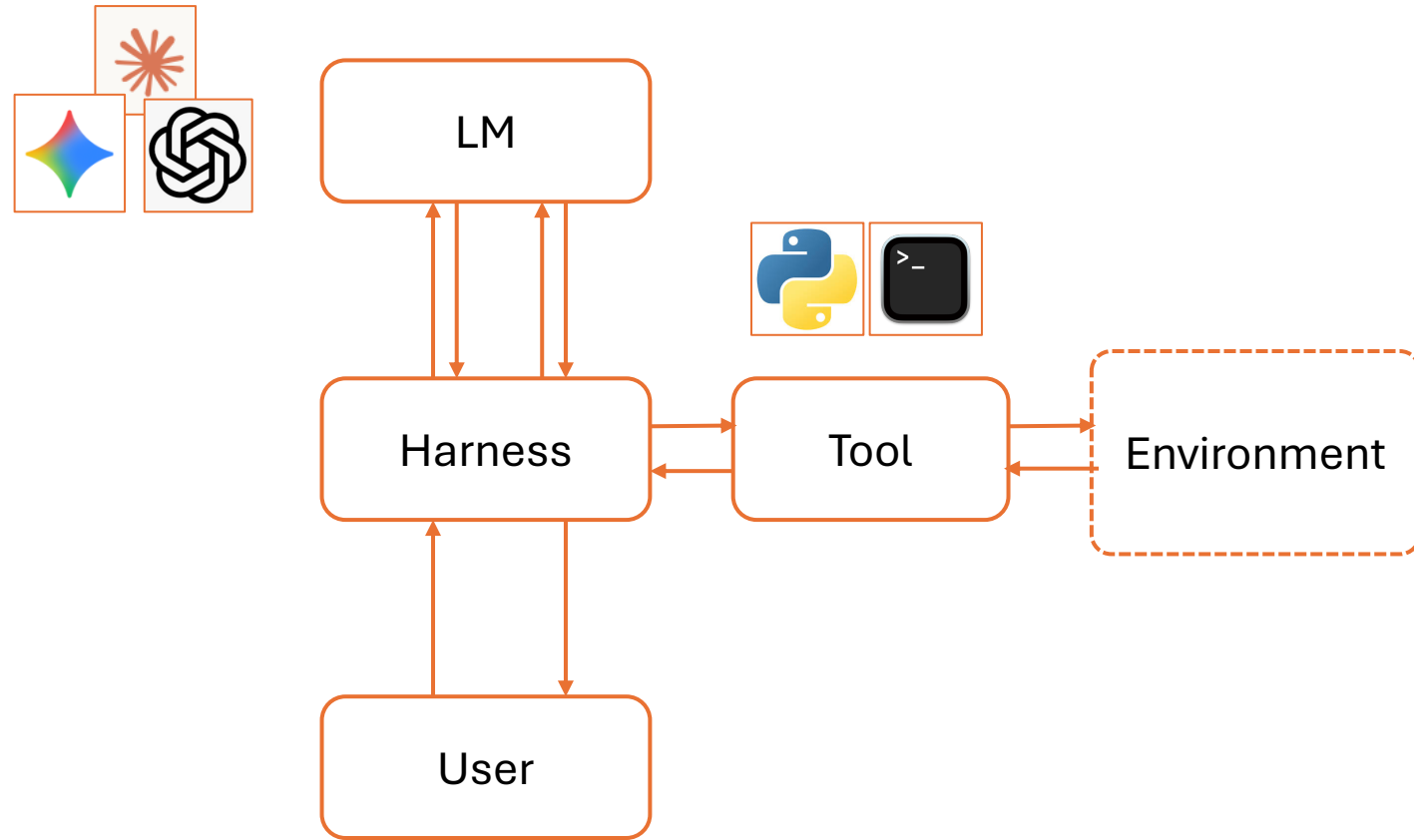
Implication for Program Synthesis



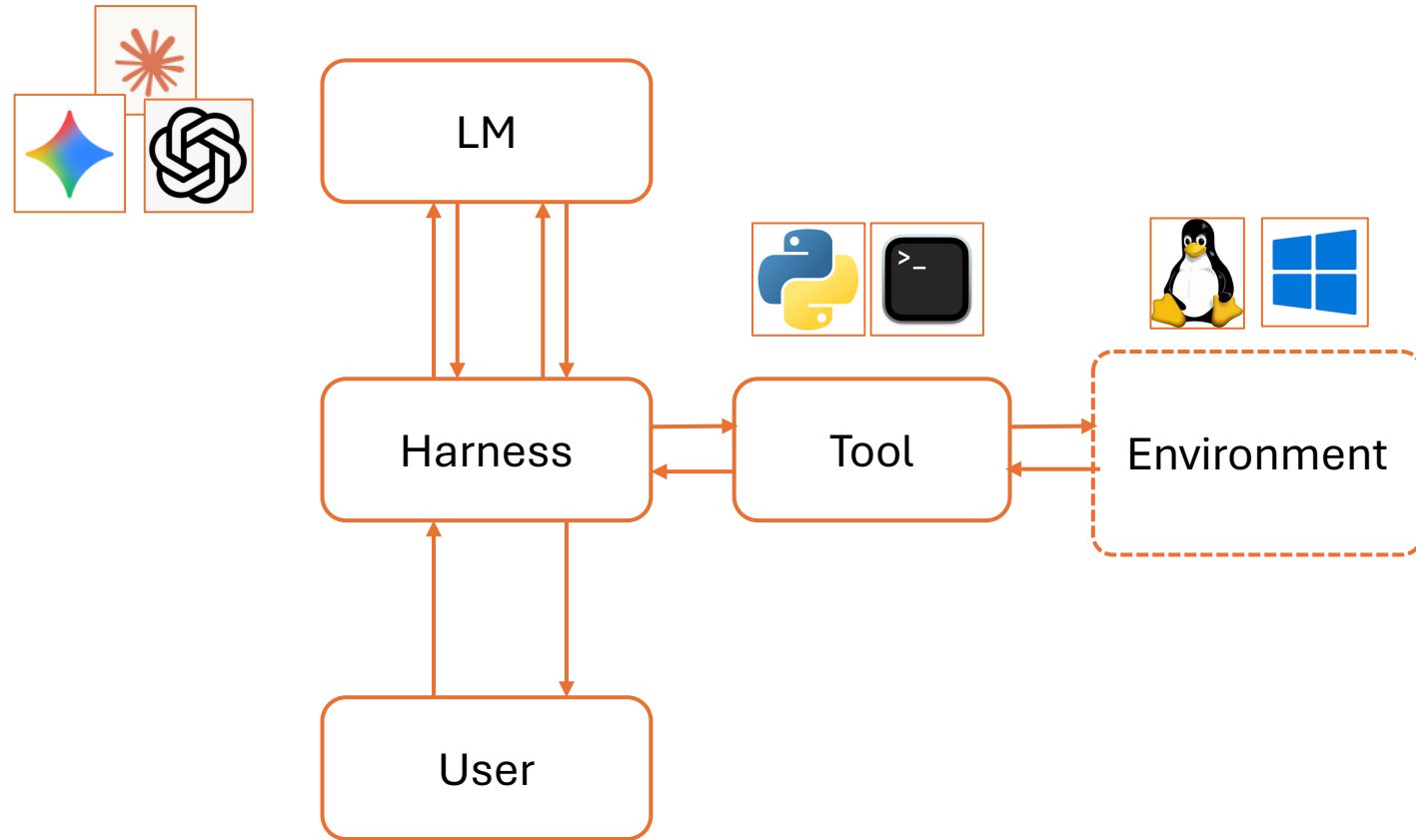
Implication for Program Synthesis



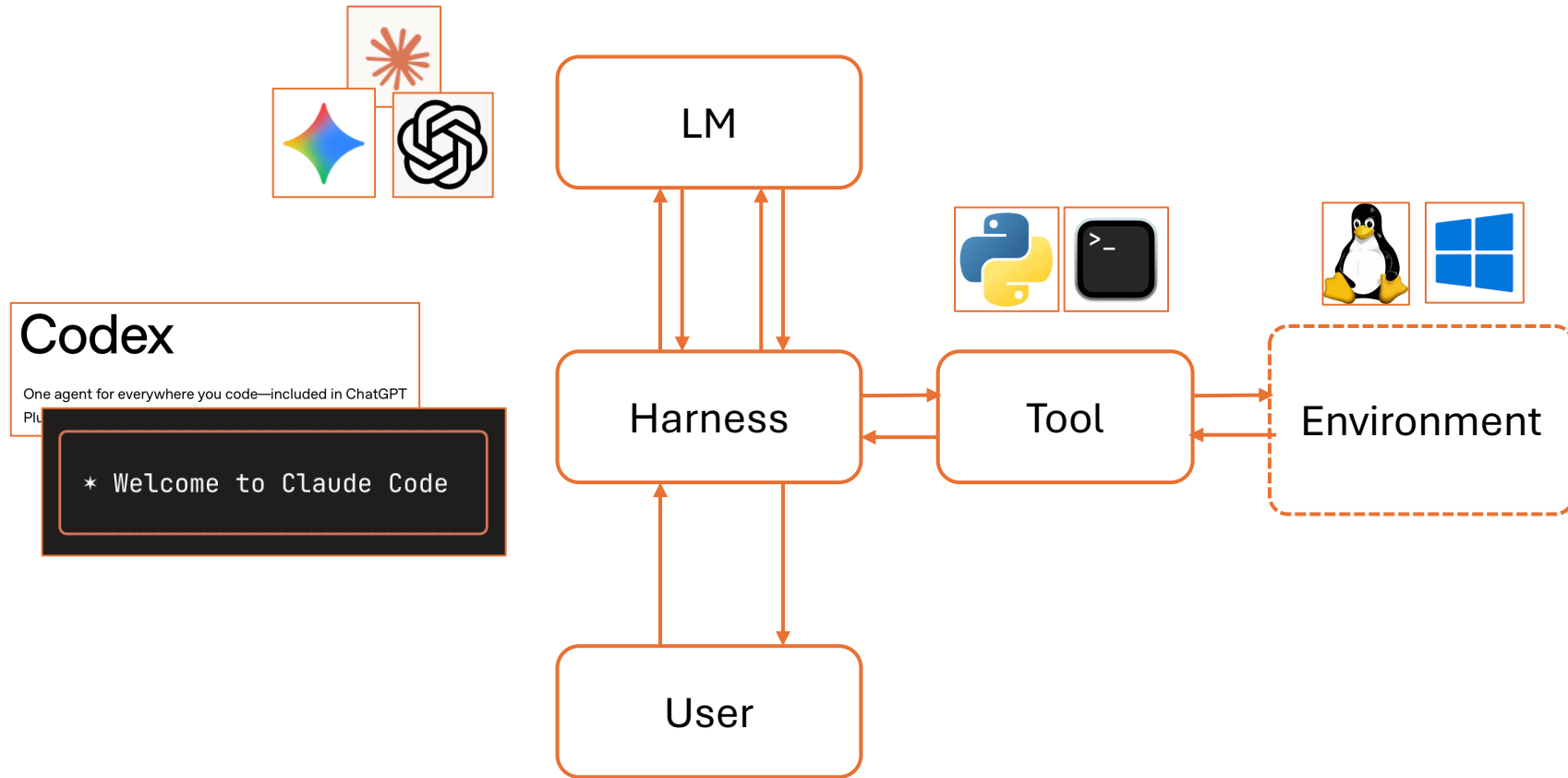
Implication for Program Synthesis



Implication for Program Synthesis

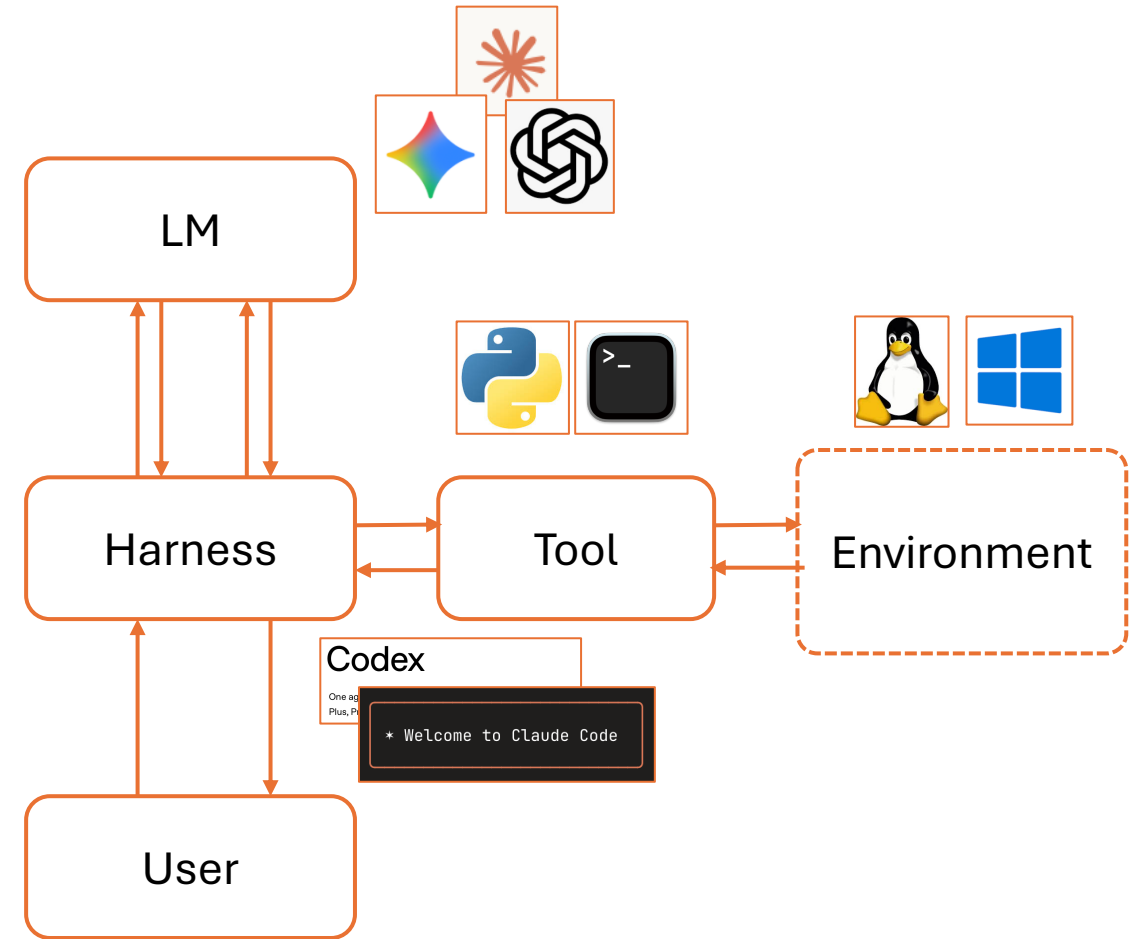


Implication for Program Synthesis



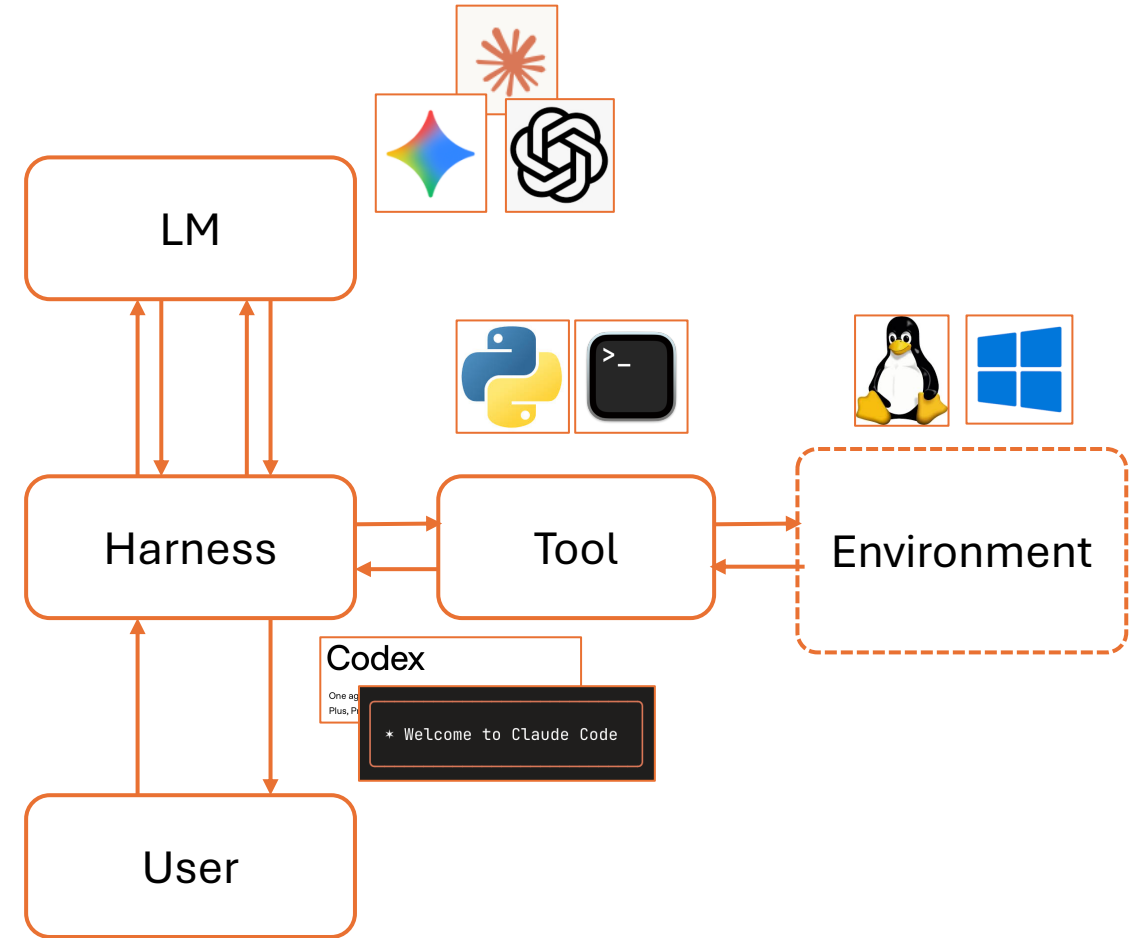
Implication for Program Synthesis

- **Less input required from user**
 - Agent pre-defines a set of general prompts for typical synthesis use cases;
 - Feedback can come from tools and environments.
- **Less responsibility from user**
 - Agent exposes the set of tools to language models;
 - Agent takes the responsibility to invoke the tools and process the results.



Potential Achievements for Program Synthesis

- **Automated testing**
 - Writing and run new test cases
- **Repository level reasoning**
 - Through file systems, read multiple files in the repository
- **Adaptive iterative refinement**
 - Dynamically choose which tool to run (compiler, linter, test, etc.)
 - Creative ways to debug: delta debugging, tracing, etc.



Design and Implementation of Agentic Frameworks

Prompting with a Conversation

System

You are a senior software engineer from a top-tier company...

You

Write this function:
def add(a, b):

Language Model

def add(a, b): return a + b

...

You

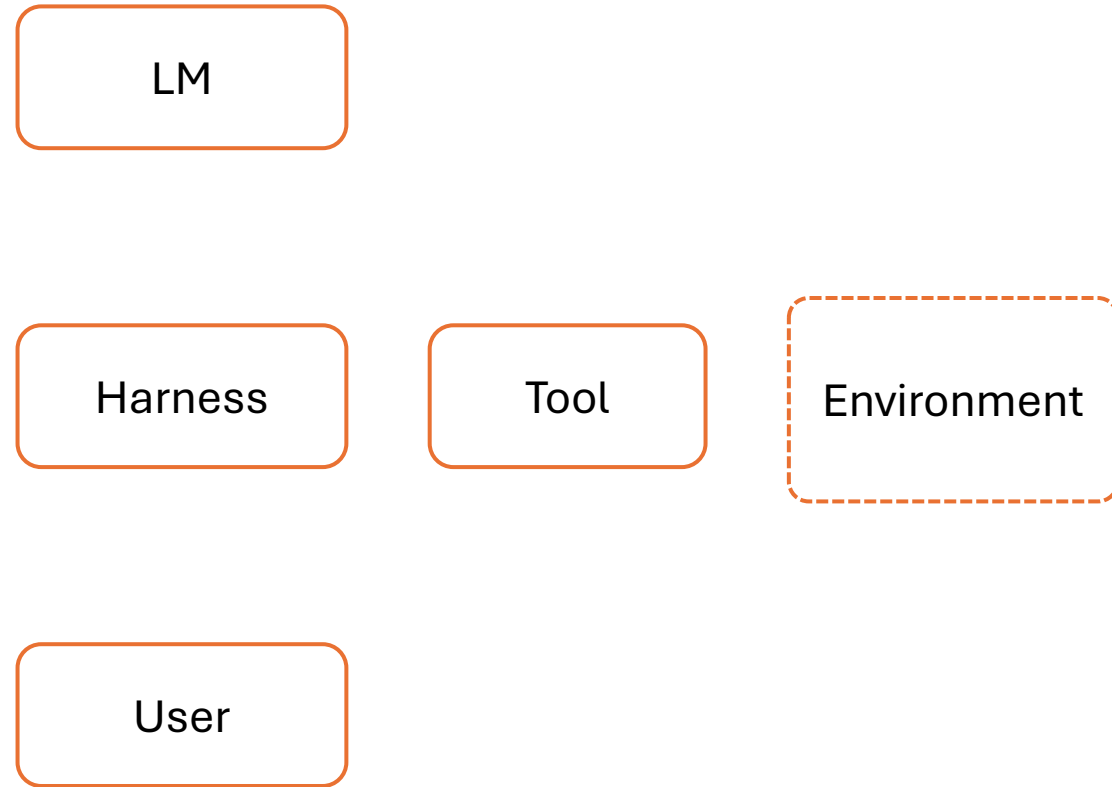
Write this function:
def is_palindrome(s: str)



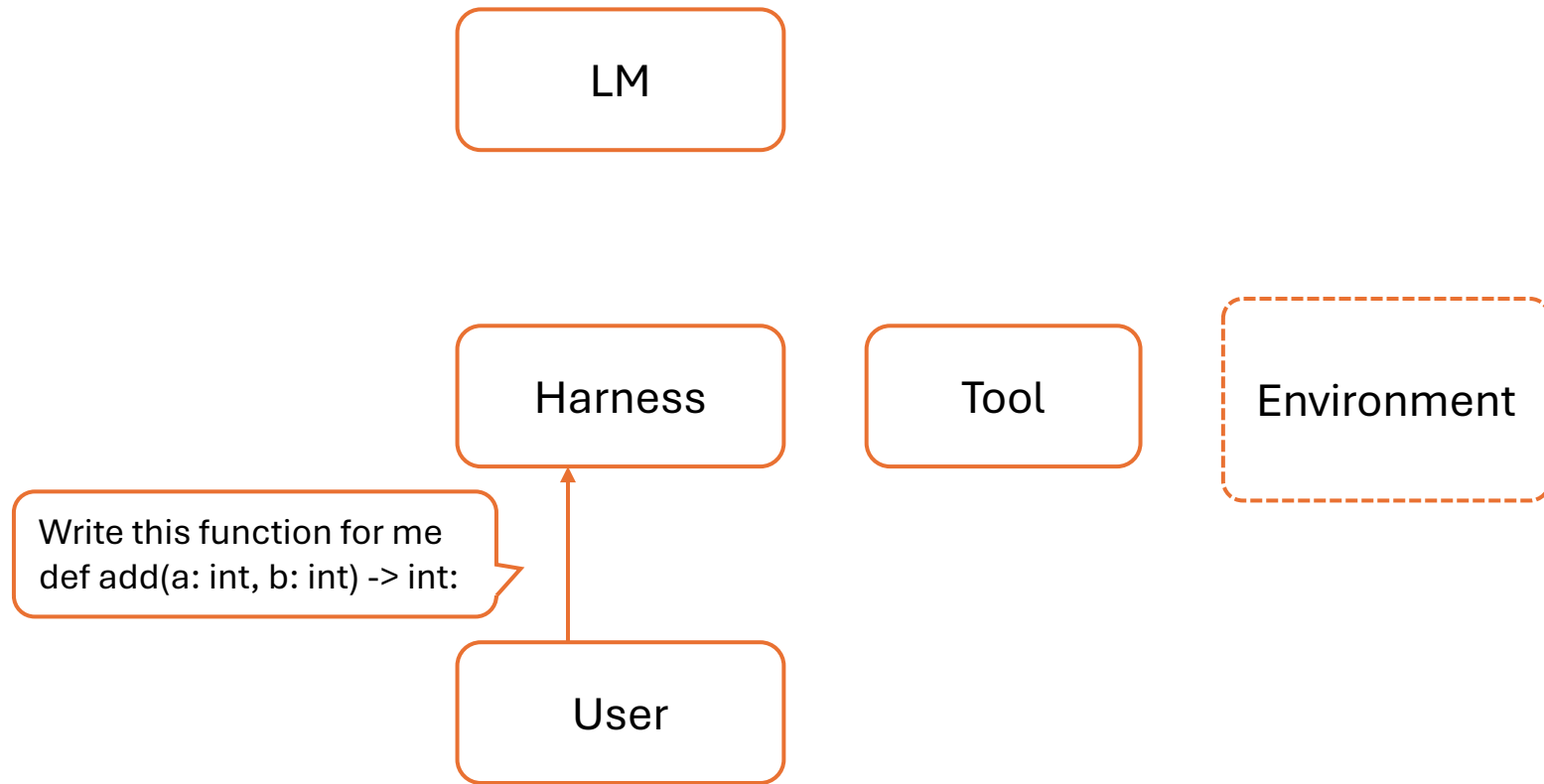
```
{
  "messages": [
    { "role": "system",
      "parts": ["You are a senior software engineer from..."] },
    { "role": "user",
      "parts": ["Write this function:\ndef add(a, b):"] },
    { "role": "assistant",
      "parts": ["def add(a, b): return a + b"] },
    { "role": "user",
      "parts": ["Write this function:\ndef reverse(s: str)..."] },
    { "role": "assistant",
      "parts": ["def reverse(s: str) -> str:\n output = \"\"\n for c in s:\n output = c + output\n return output"] },
    { "role": "user",
      "parts": ["Write this function:\ndef is_palindrome(s: str)"] }
  ]
}
```

```
client.chat.completions.create(
  model="gpt-4o-mini",
  messages=messages
)
```

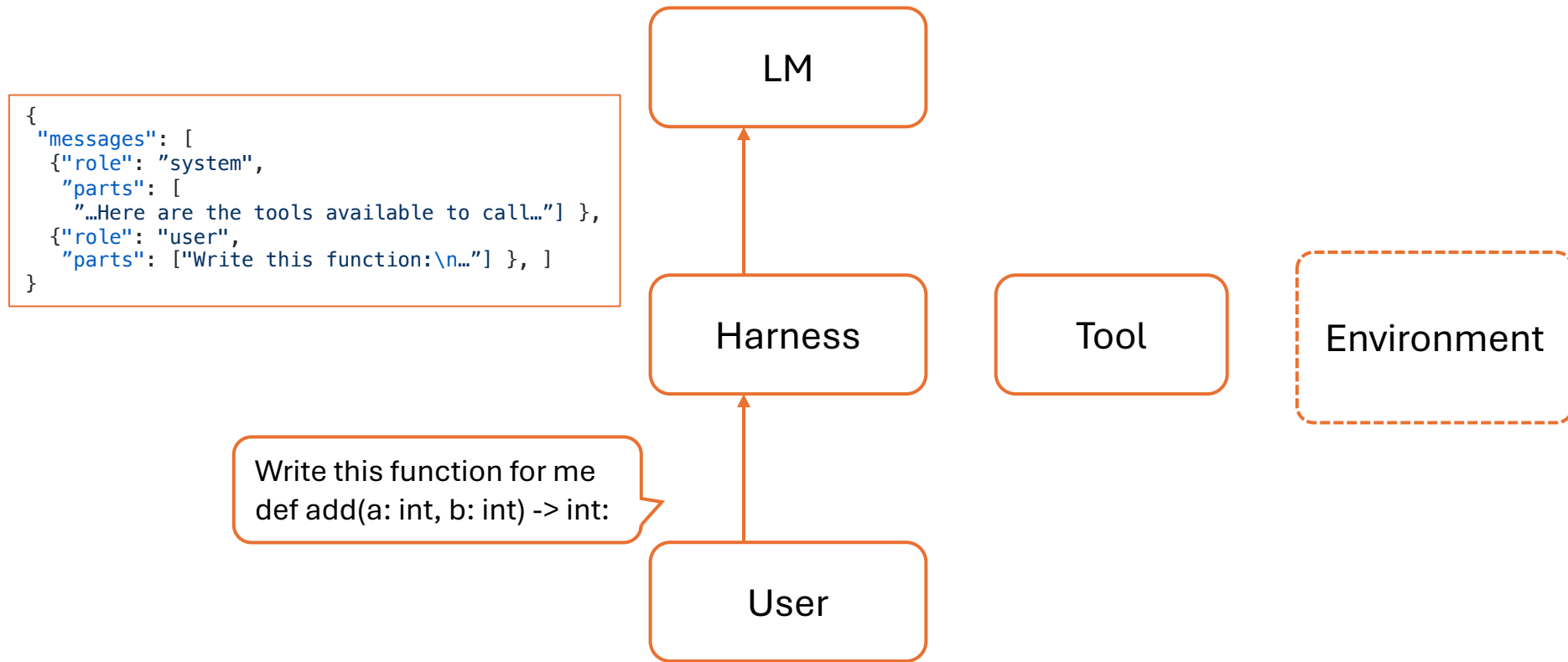
Design and Implementation of Agentic Frameworks



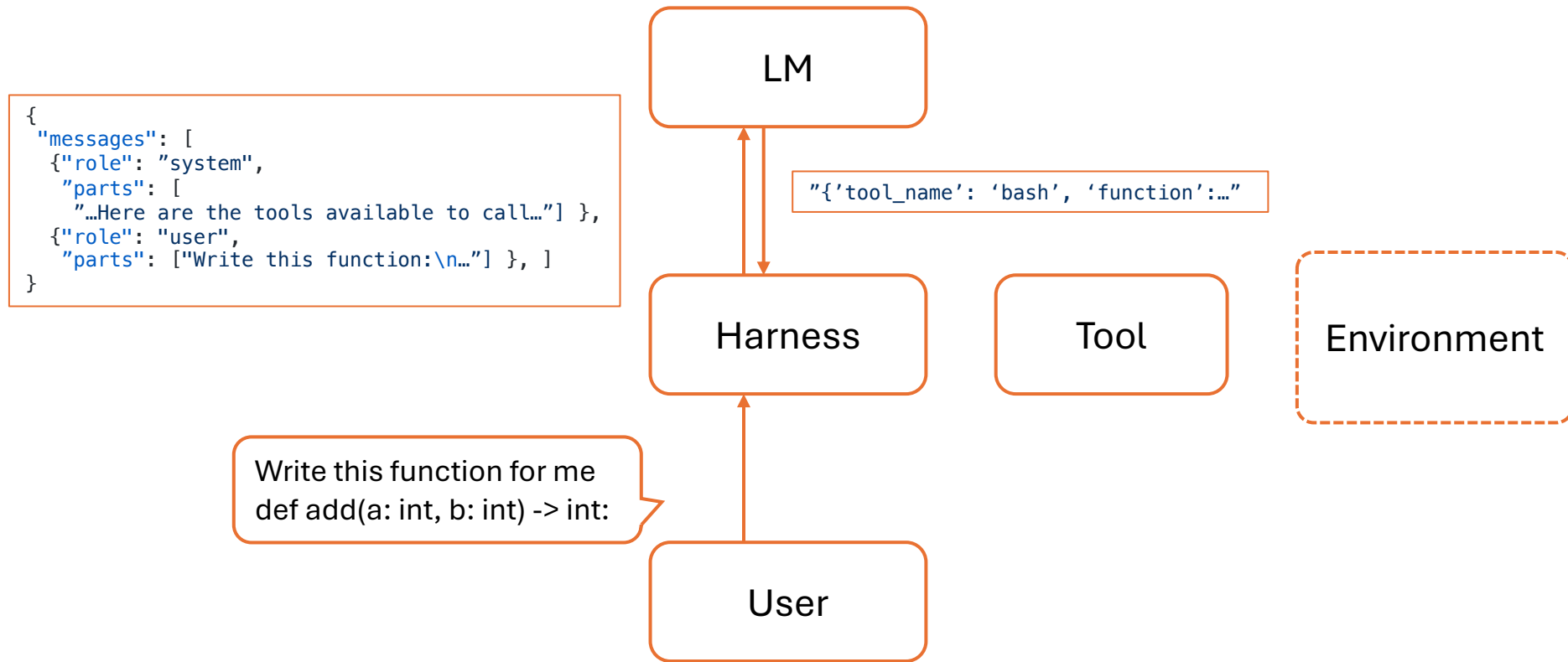
Design and Implementation of Agentic Frameworks



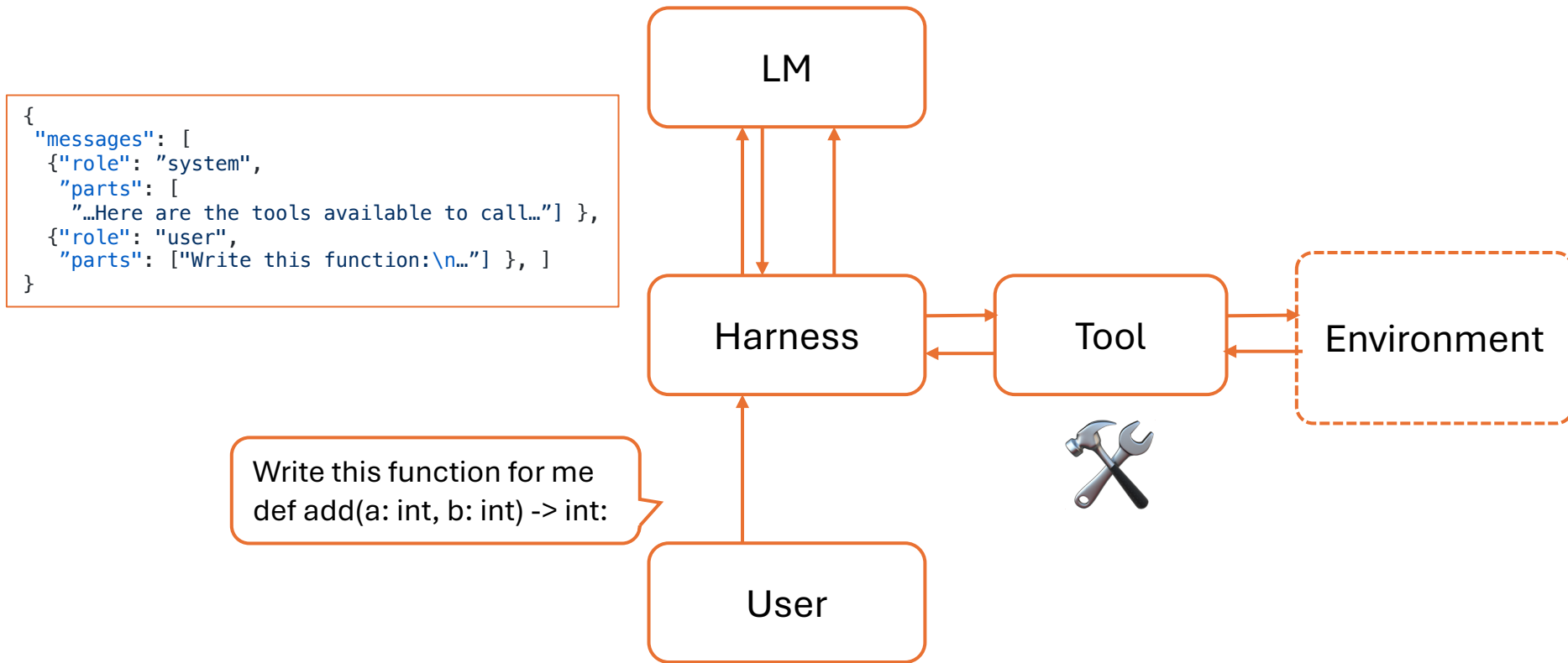
Design and Implementation of Agentic Frameworks



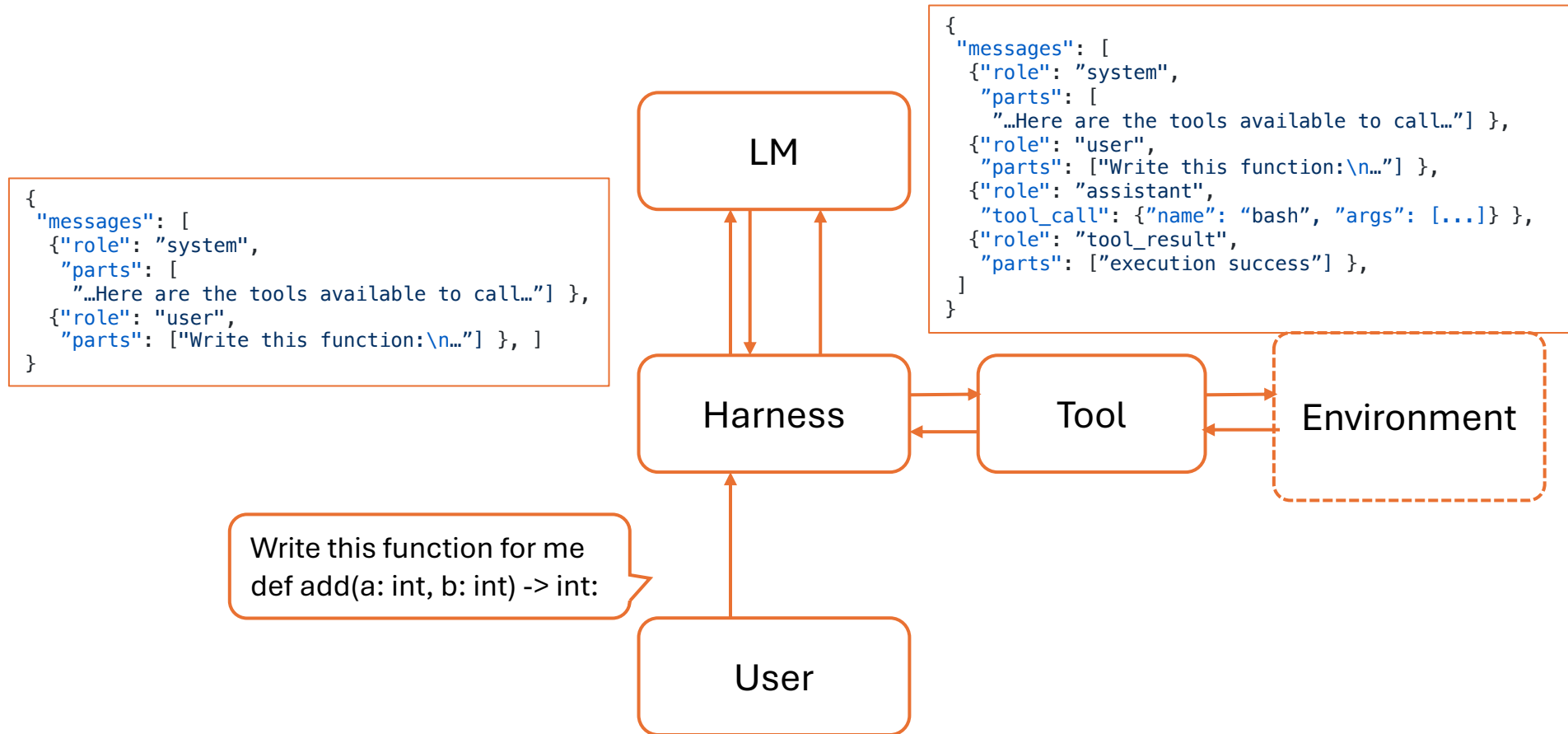
Design and Implementation of Agentic Frameworks



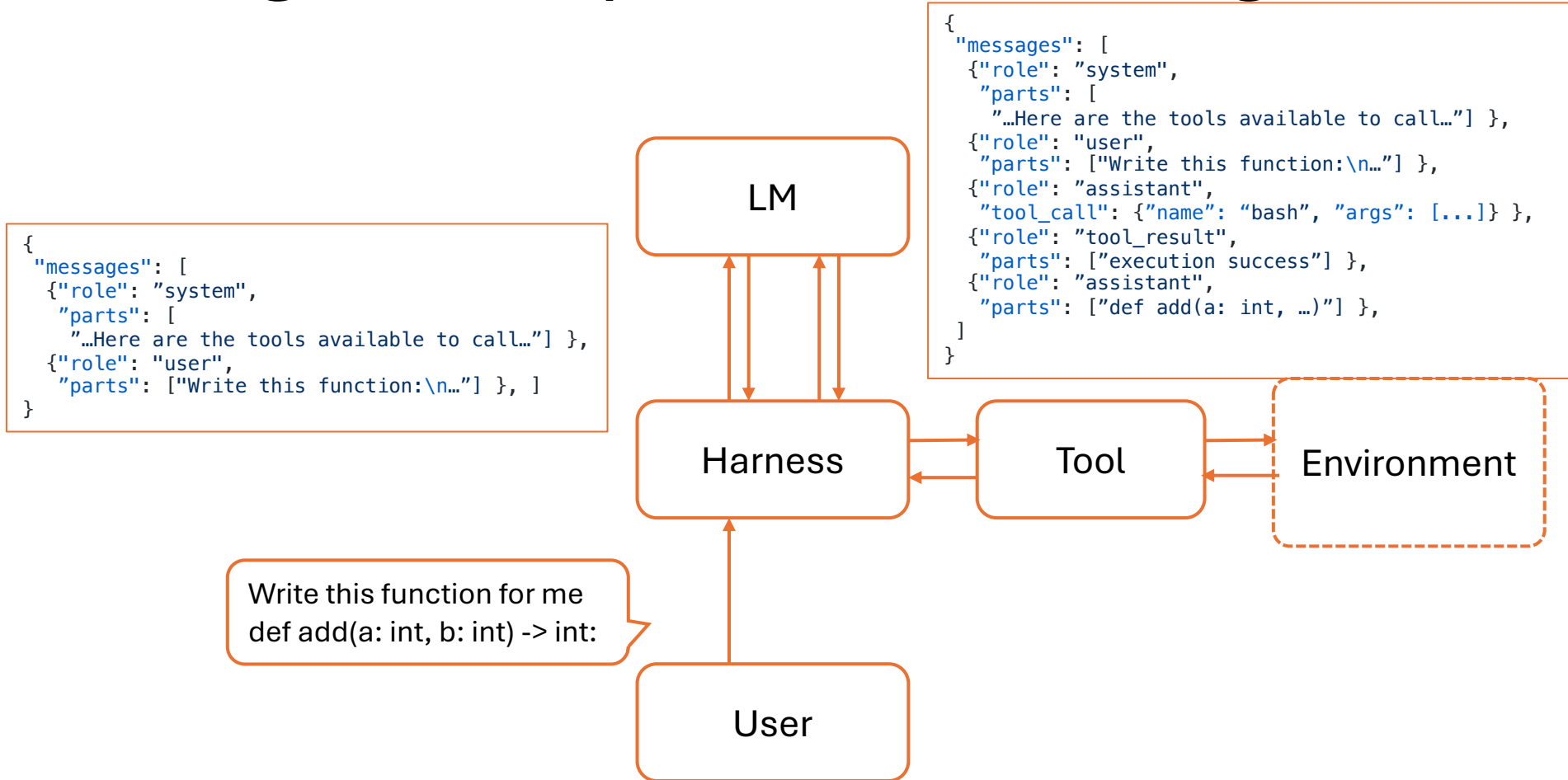
Design and Implementation of Agentic Frameworks



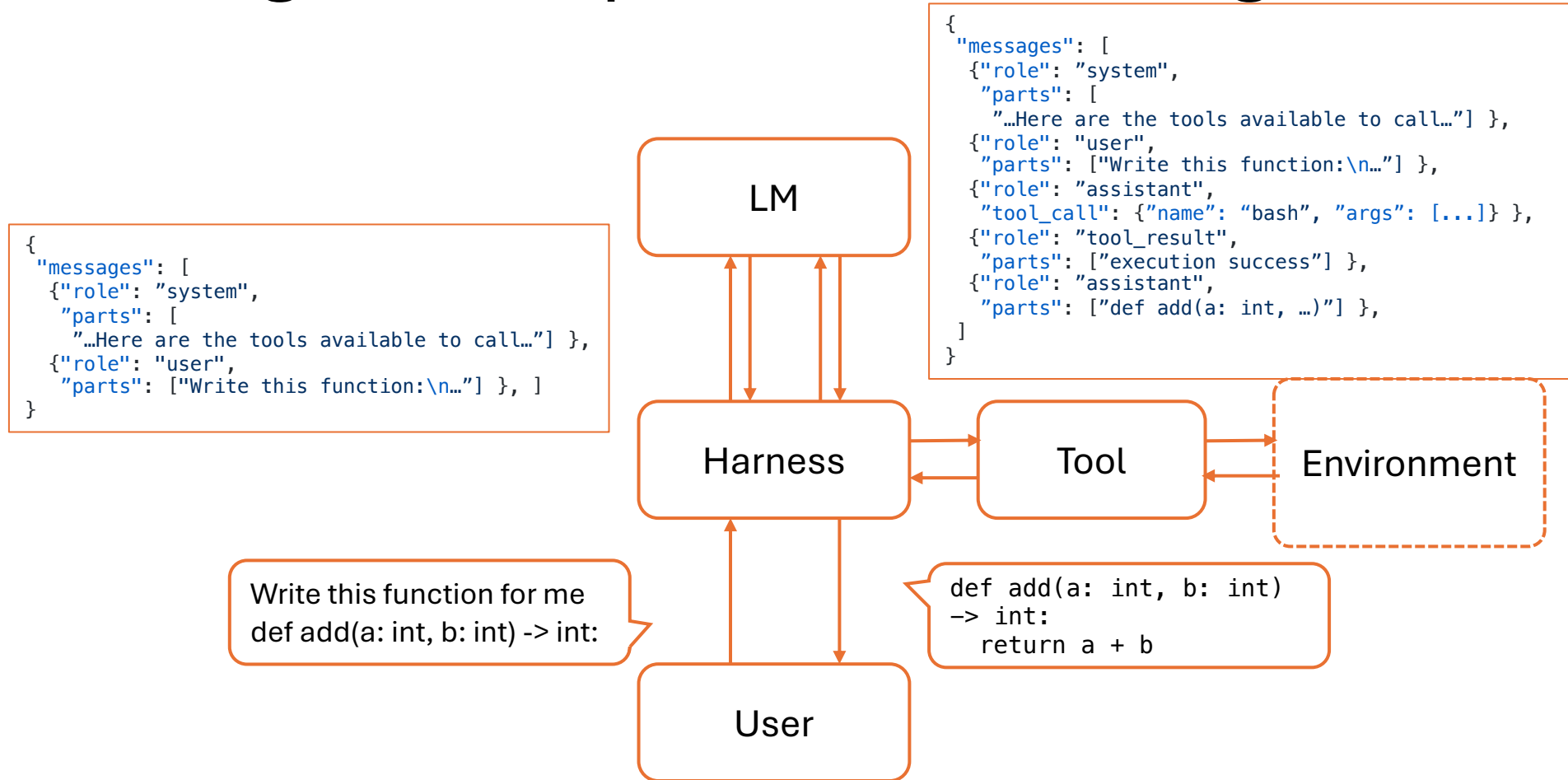
Design and Implementation of Agentic Frameworks



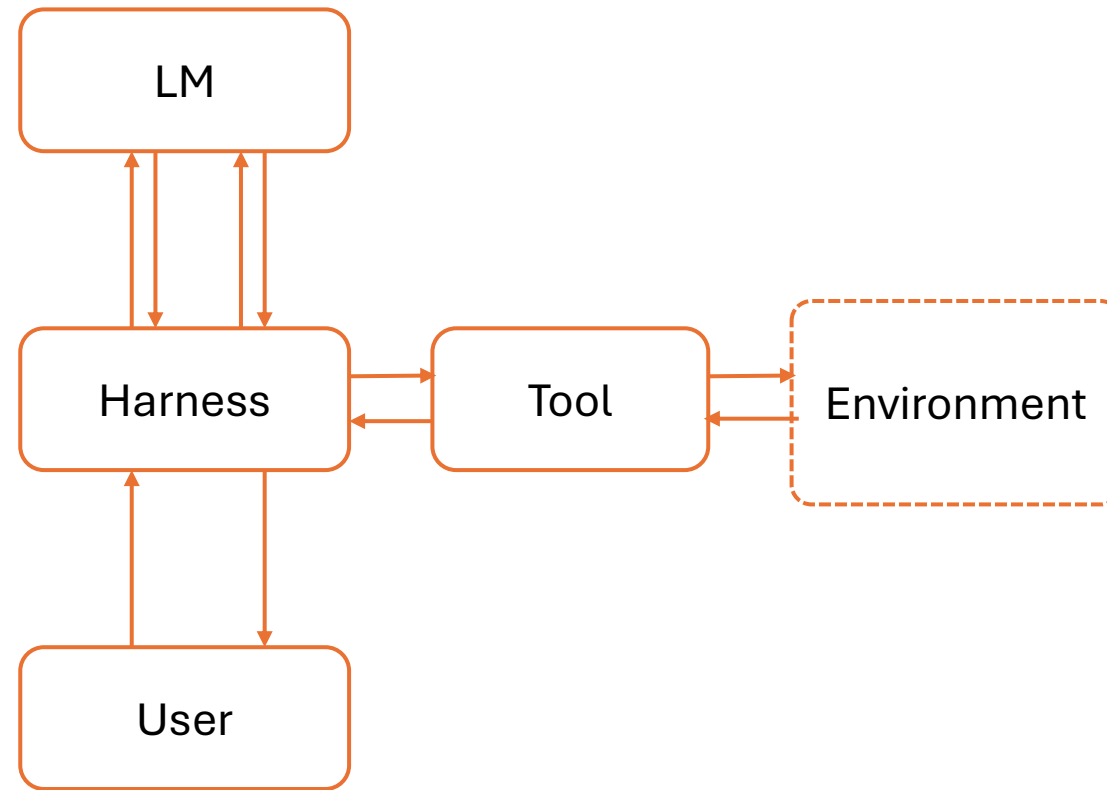
Design and Implementation of Agentic Frameworks



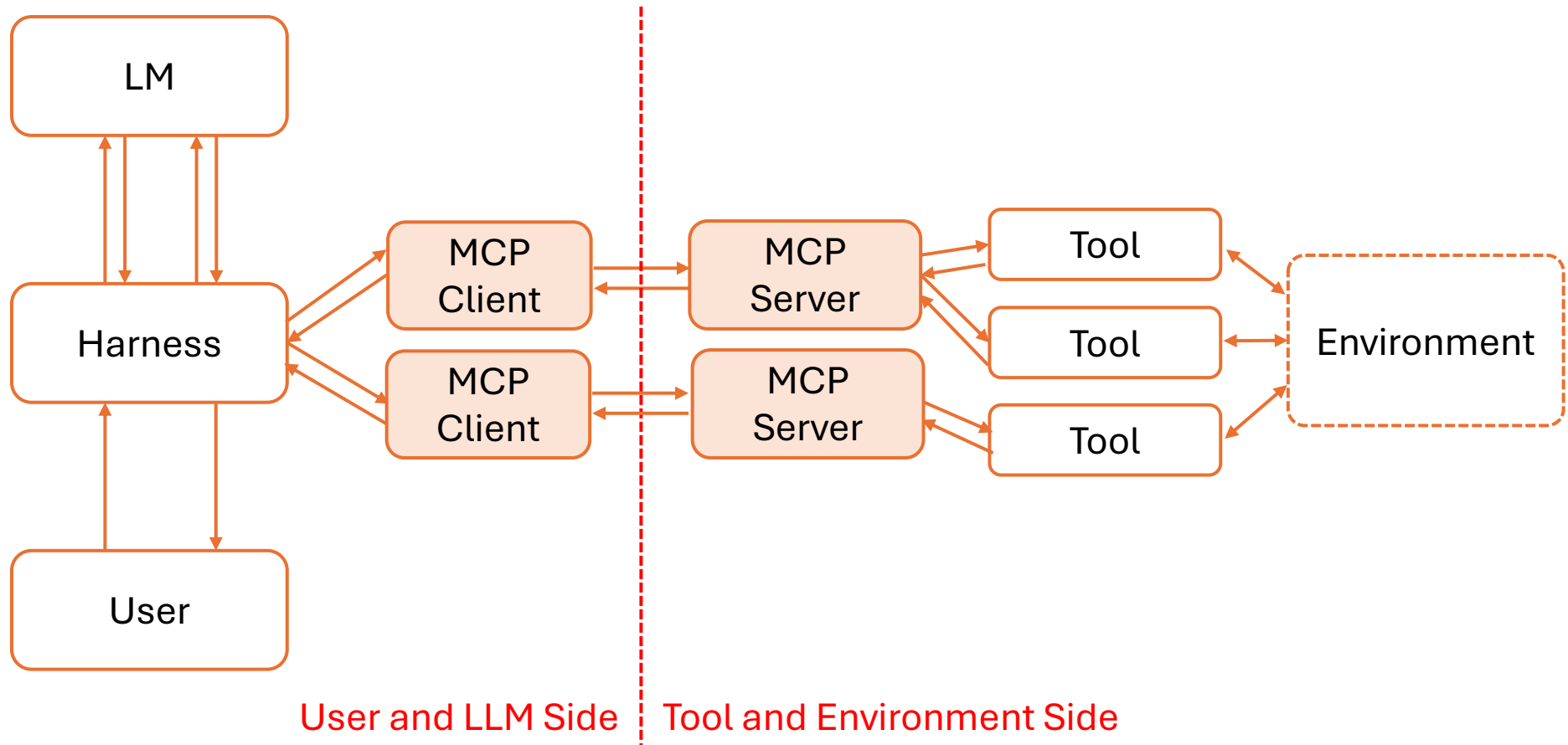
Design and Implementation of Agentic Frameworks



Agentic Framework



Model Context Protocol (MCP)



Model Context Protocol



Model Context Protocol

A protocol for seamless integration between LLM applications and external data sources

[Documentation](#) | [Specification](#) | [Discussions](#)

The Model Context Protocol (MCP) is an open protocol that enables seamless integration between LLM applications and external data sources and tools. Whether you're building an AI-powered IDE, enhancing a chat interface, or creating custom AI workflows, MCP provides a standardized way to connect LLMs with the context they need.

Model Context Protocol

The Mod
sources
provides

main ▾

[modelcontextprotocol](#) / [docs](#) / [specification](#) / 

T

Add file ▾

...

 **localden** Fix "Serves" typo in resources capability text ([#3380](#))  

3f6e5e1 · 2 days ago 

Name	Last commit message	Last commit date
 ..		
 2024-11-05	docs: fix broken links to client/sampling and spec landing	4 months ago
 2025-03-26	docs: fix broken links to client/sampling and spec landing	4 months ago
 2025-06-18	docs: fix invalid JSON in two example code blocks (#3381)	2 days ago
 2025-11-25	Fix typo in 2025-11-25 authorization spec	2 months ago
 2026-07-28	Fix "Serves" typo in resources capability text (#3380)	2 days ago
 draft	Fix "Serves" typo in resources capability text (#3380)	2 days ago

Model Context Protocol



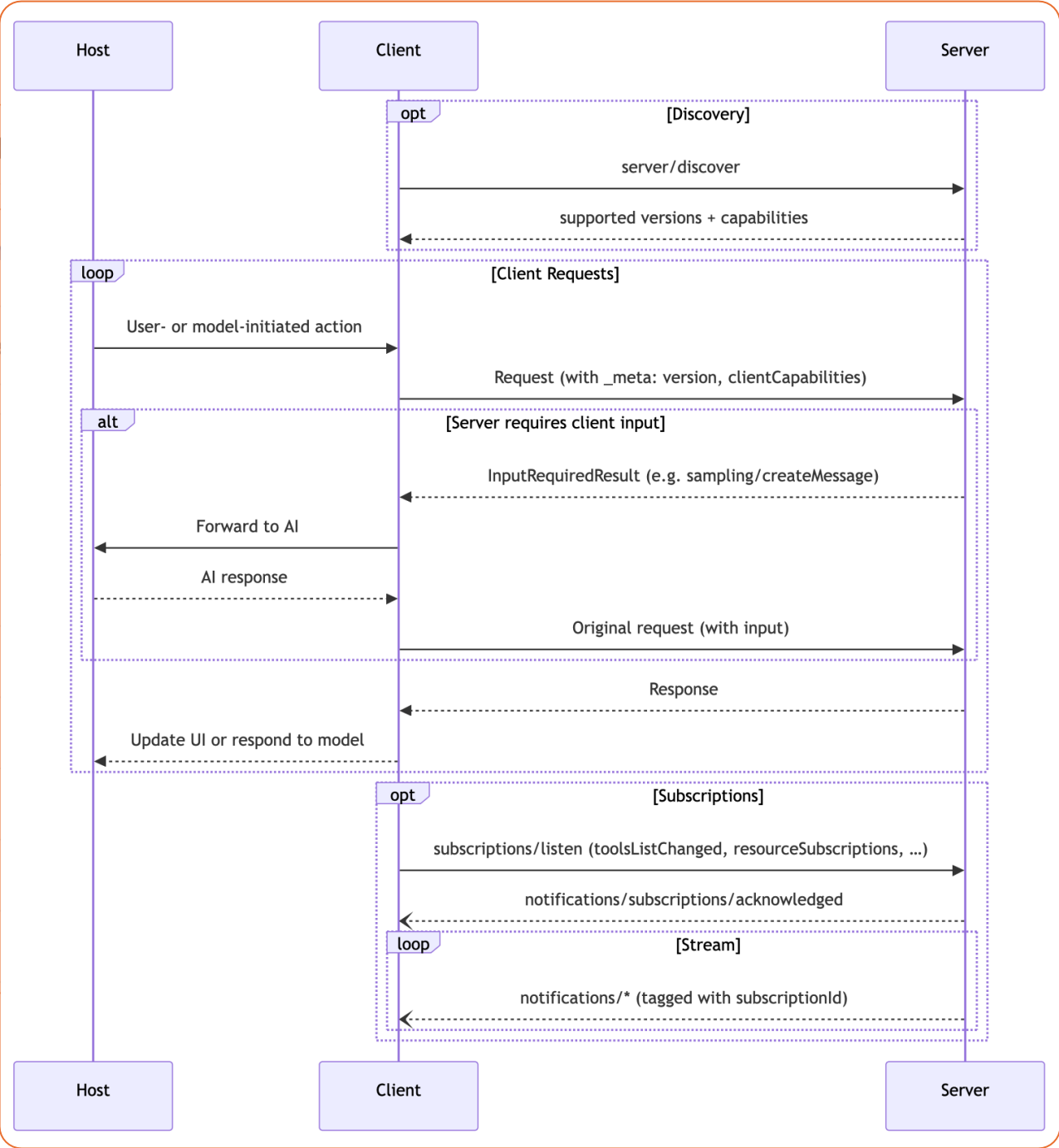
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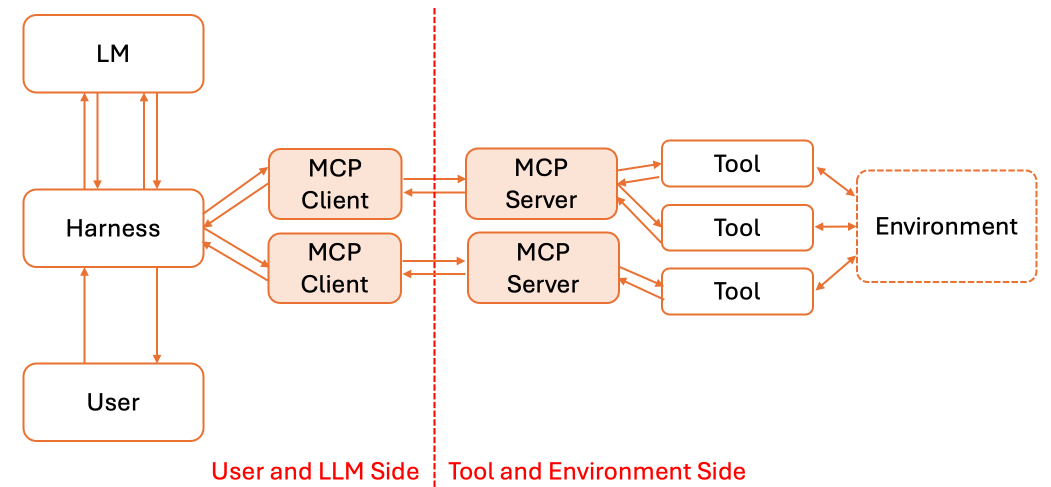
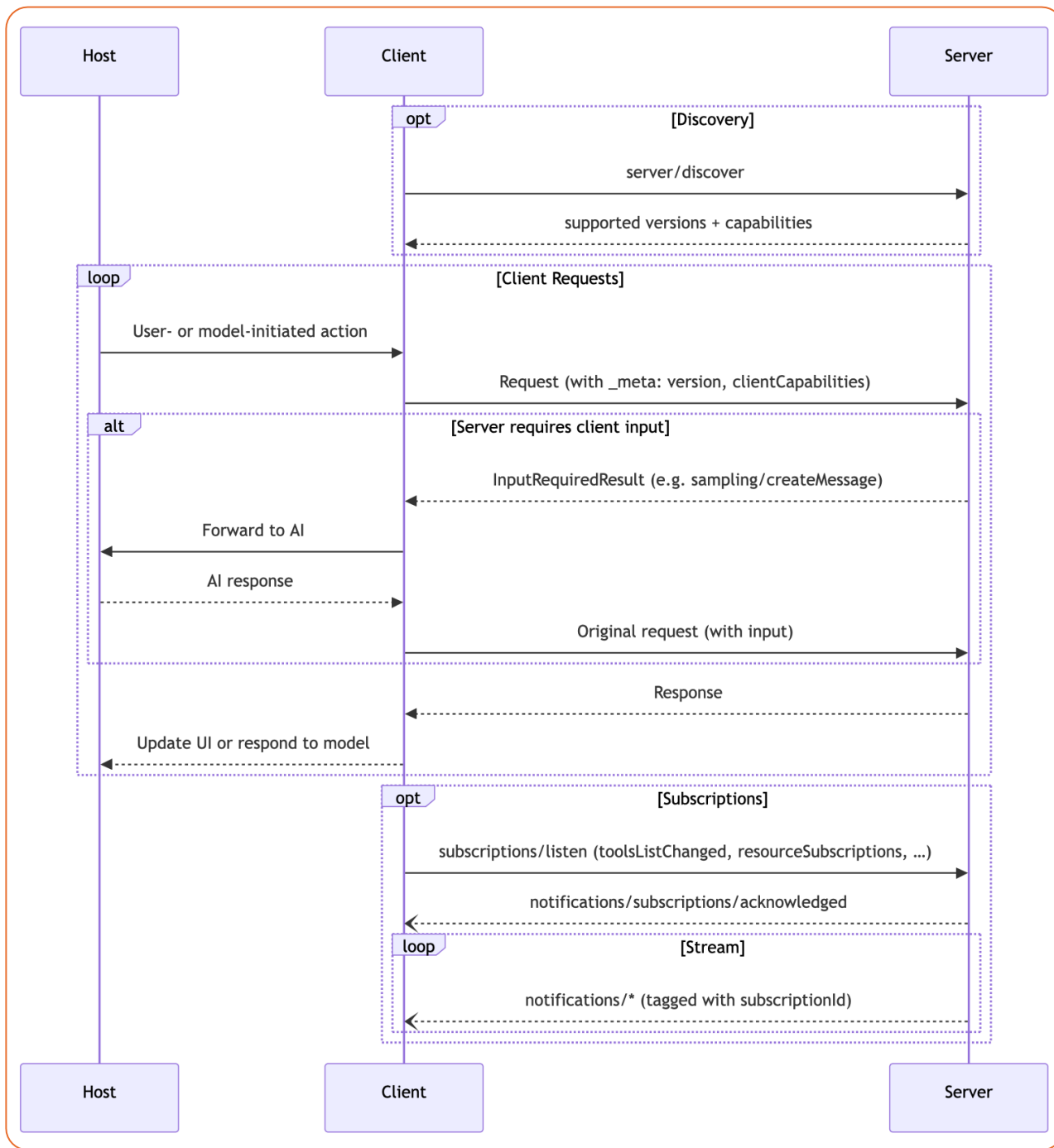
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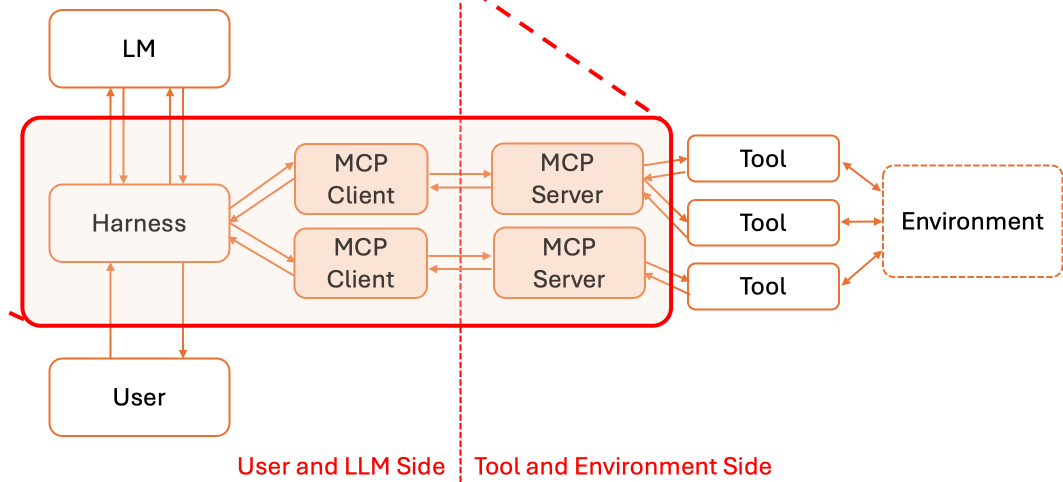
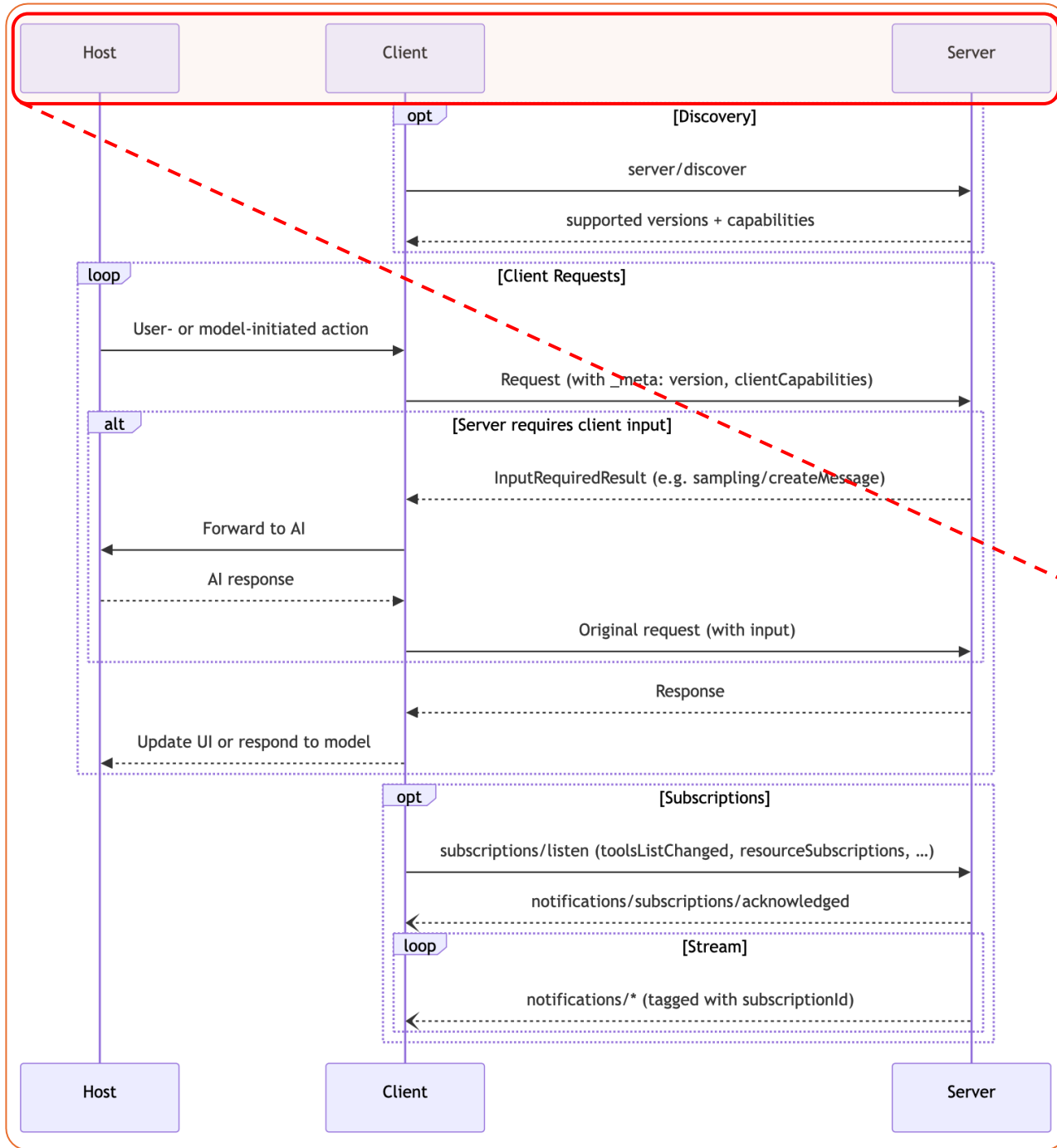
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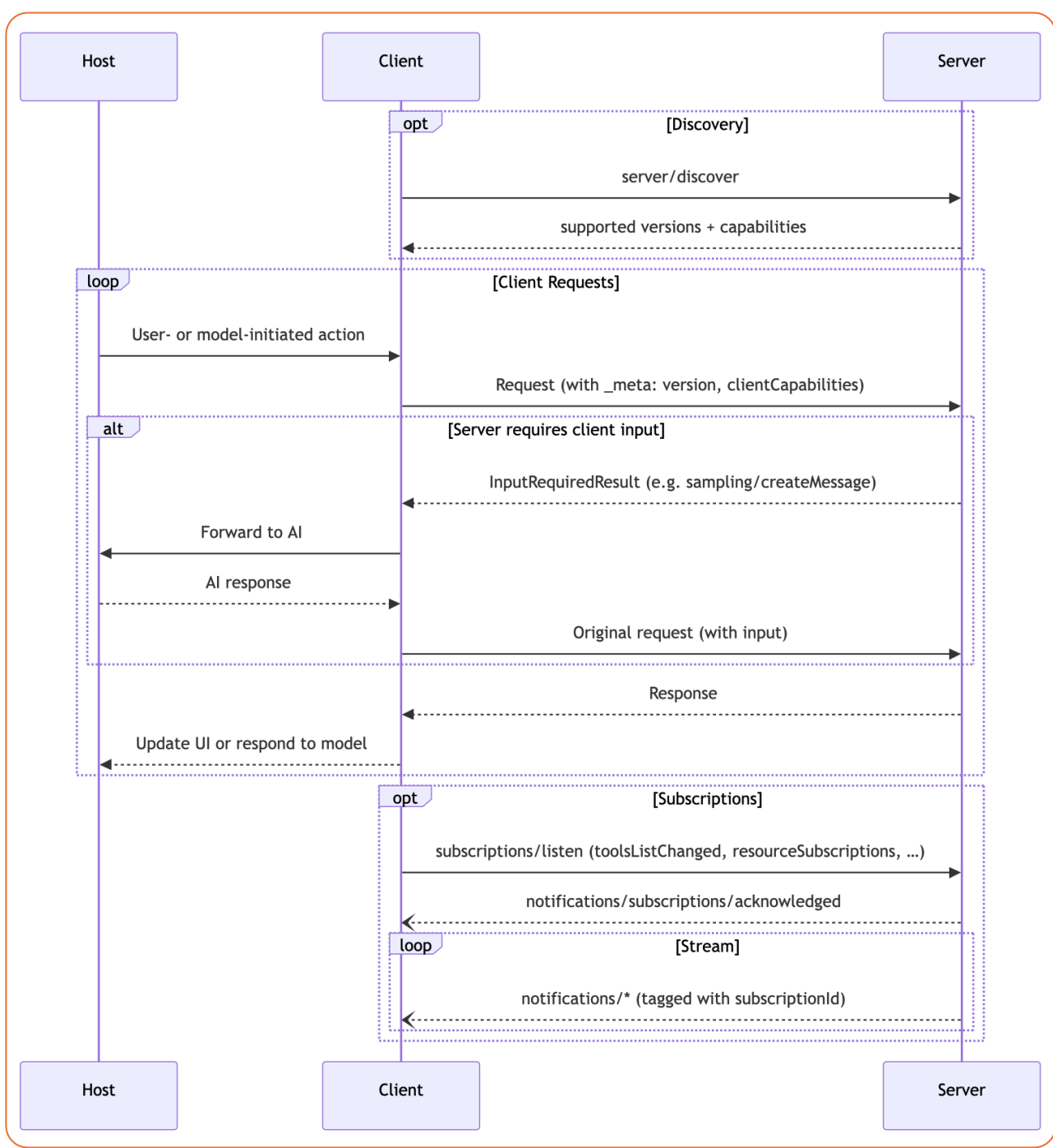
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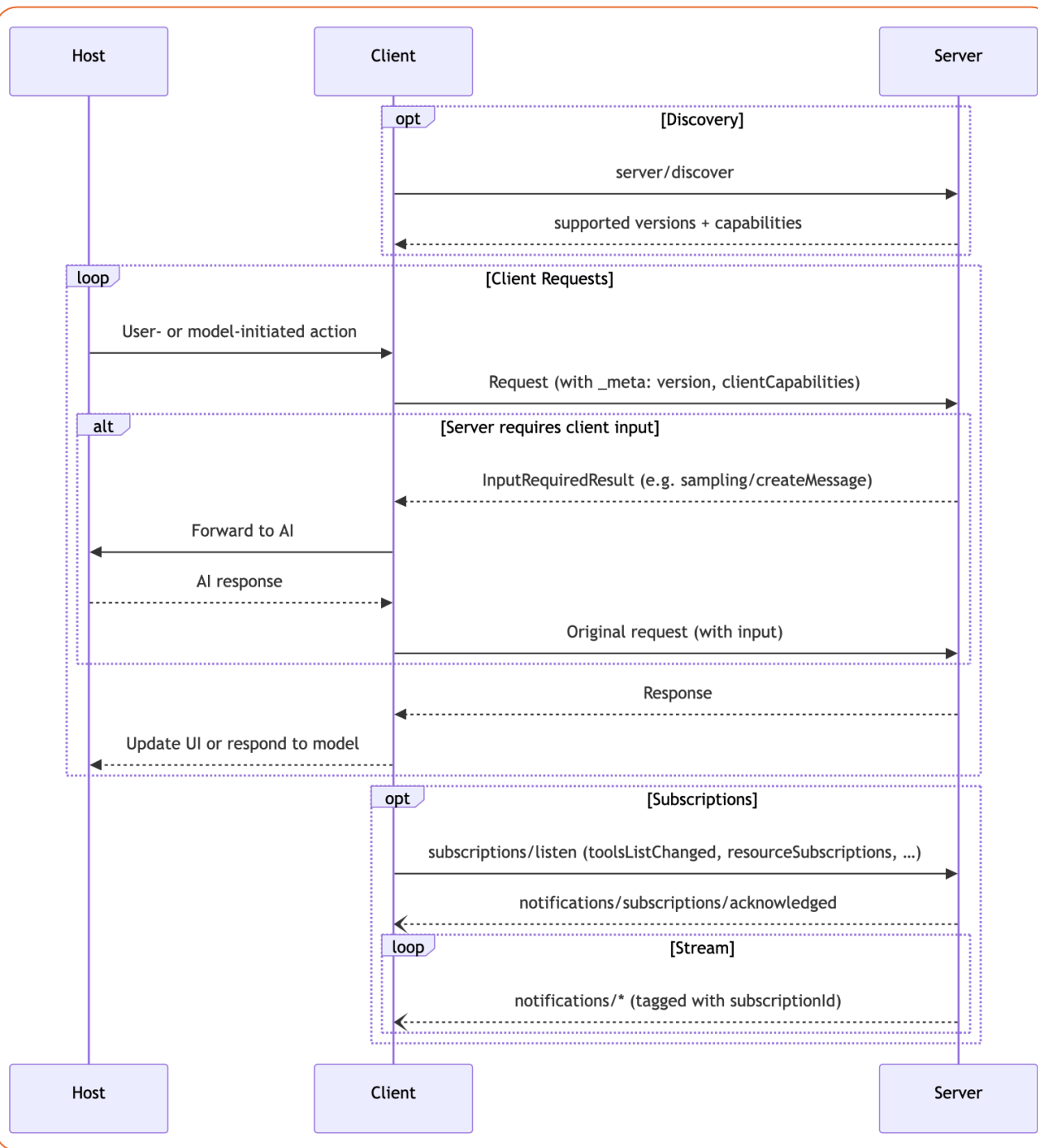
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Last commit date
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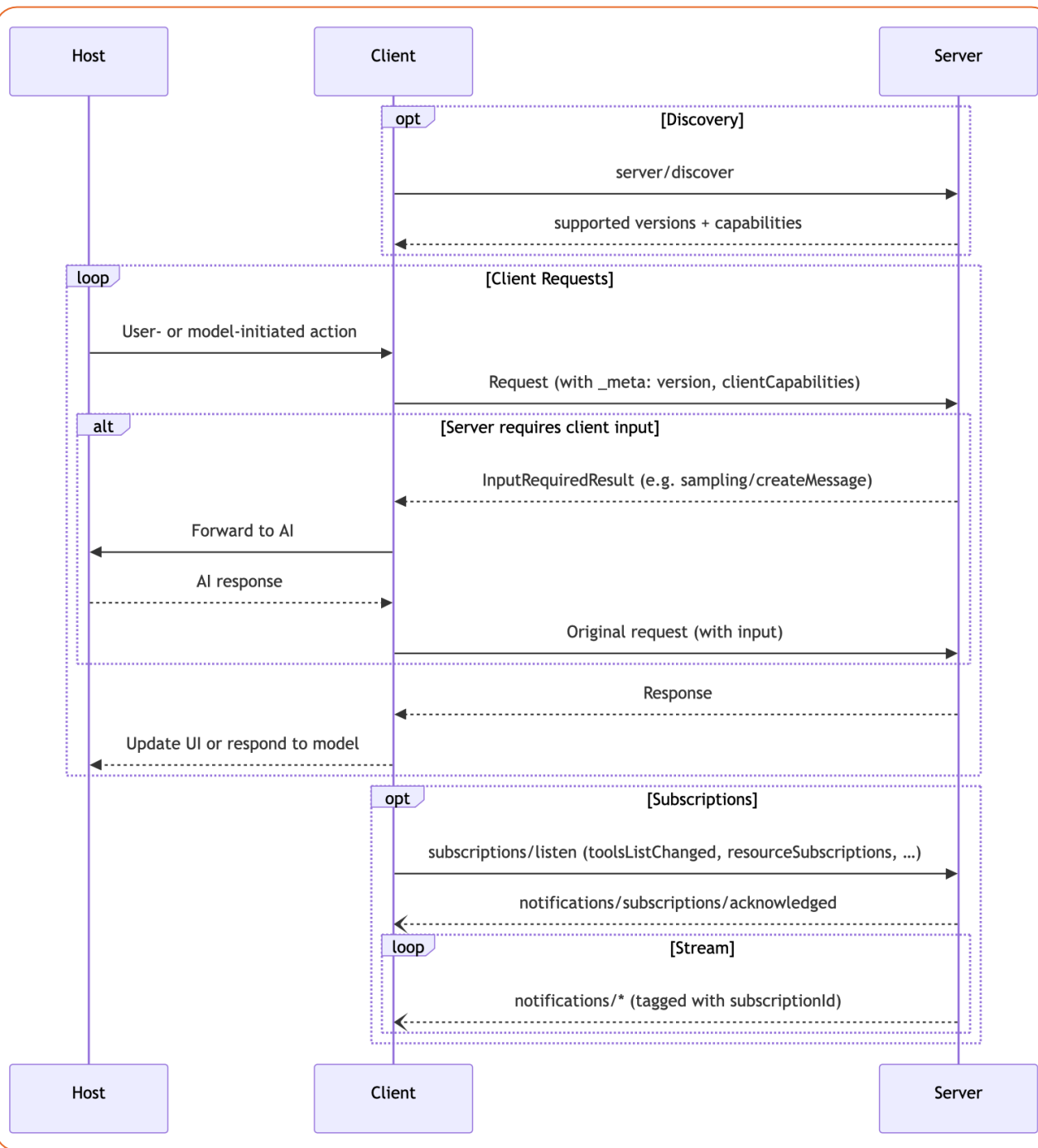




Host

The host process acts as the container and coordinator:

- Creates and manages multiple client instances
- Controls client connection permissions and lifecycle
- Enforces security policies and consent requirements
- Handles user authorization decisions
- Coordinates AI/LLM integration and sampling
- Manages context aggregation across clients



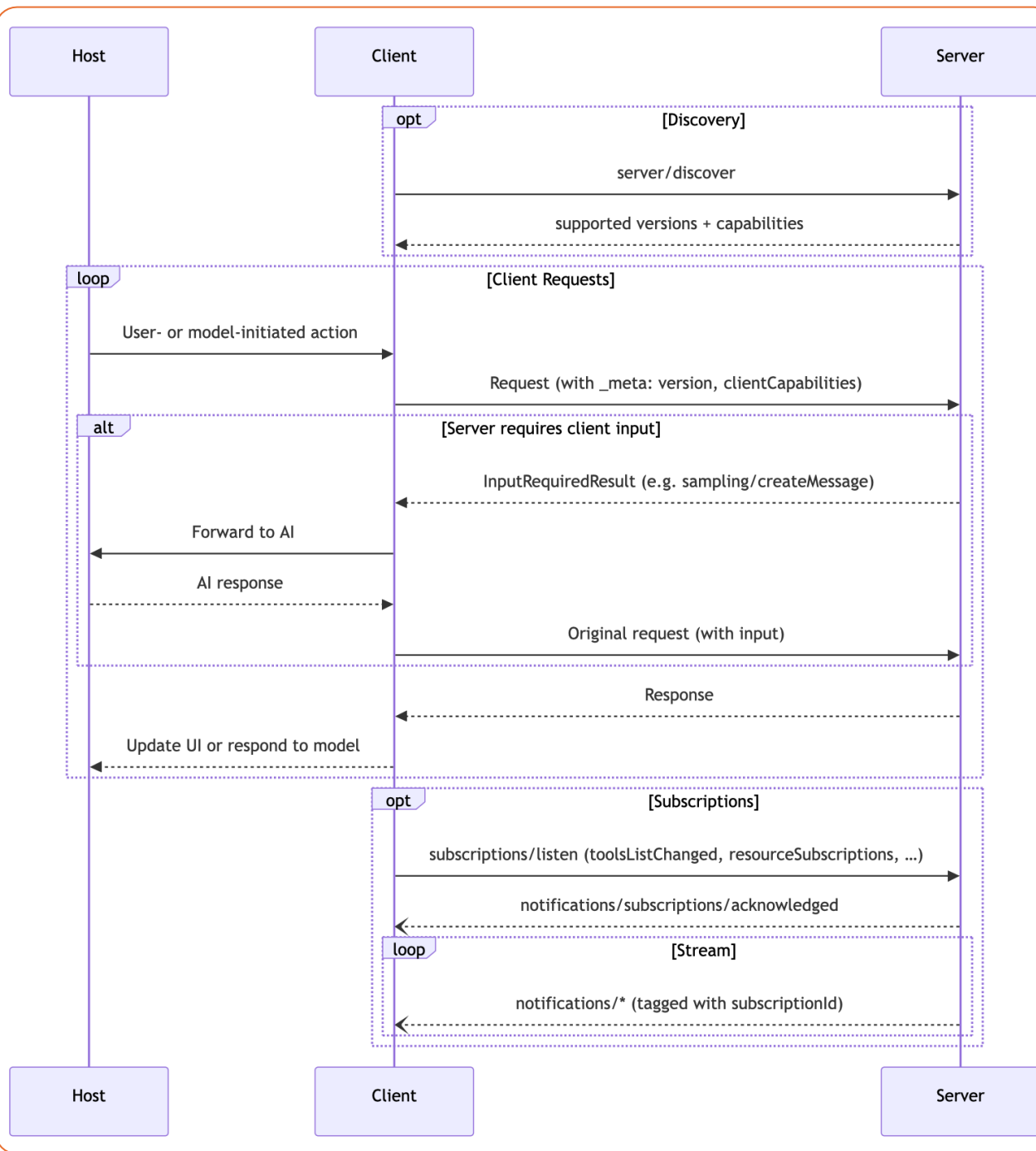
Host

The host process acts as the container and coordinator

Clients

Each client is created by the host and communicates with exactly one server:

- Communicates with exactly one server
- Attaches protocol version and capabilities to requests
- Routes protocol messages bidirectionally
- Manages subscriptions and notifications
- Maintains security boundaries between servers



Host

The host process acts as the container and coordinator

Clients

Each client is created by the host and communicates with exactly one server

Servers

Servers provide specialized context and capabilities:

- Expose resources, tools and prompts via MCP primitives
- Operate independently with focused responsibilities
- Request client input (sampling, elicitation, roots)

via **InputRequiredResult** within a reply

- Must respect security constraints
- Can be local processes or remote services

1. Deprecated

Feature	Deprecation SEP	Deprecated in	Migration path	Earliest removal
<u>Roots</u>	<u>SEP-2577</u>	2026-07-28	Pass directories or files via tool parameters, resource URIs, or server configuration	First revision released on or after 2027-07-28
<u>Sampling</u>	<u>SEP-2577</u>	2026-07-28	Integrate directly with LLM provider APIs	First revision released on or after 2027-07-28
<u>Logging</u>	<u>SEP-2577</u>	2026-07-28	Log to <code>stderr</code> for stdio transports; use <u>OpenTelemetry</u> for observability	First revision released on or after 2027-07-28
<u>Dynamic Client Registration</u>	<u>PR #2858</u>	2026-07-28	<u>Client ID Metadata Documents</u>	First revision released on or after 2027-07-28
<code>includeContext: "thisServer" / "allServers"</code> (<u>Sampling</u>)	<u>SEP-2596</u>	2025-11-25	Omit the field or use <code>"none"</code>	Follows Sampling (<u>SEP-2577</u>)
<u>HTTP+SSE transport</u>	<u>SEP-2596</u>	2025-03-26	<u>Streamable HTTP</u>	Three months after SEP-2596 reaches Final

The HTTP+SSE transport and the `includeContext` values were already described as deprecated before the lifecycle policy existed; SEP-2596 reclassifies them as Deprecated under its transition provisions.

2. Removed

No features have been removed under this policy yet. When a Deprecated feature is removed, its row moves to this section with a link to the changelog entry recording the removal.

Model Context Protocol servers

This repository is a collection of *reference implementations* for the [Model Context Protocol](#) (MCP), as well as references to community-built servers and additional resources.

Important

If you are looking for a list of MCP servers, you can browse published servers on [the MCP Registry](#). The repository served by this README is dedicated to housing just the small number of reference servers maintained by the MCP steering group.

Warning

The servers in this repository are intended as **reference implementations** to demonstrate MCP features and SDK usage. They are meant to serve as educational examples for developers building their own MCP servers, not as production-ready solutions. Developers should evaluate their own security requirements and implement appropriate safeguards based on their specific threat model and use case.

Model Context Protocol

This repository is a central hub for discovering and managing Model Context Protocol (MCP) servers. It provides references to community-maintained servers and documentation.

Important

If you are looking for a repository served by the MCP steering committee, please refer to the official MCP Registry.

Warning

The servers in this repository are not as production-ready as the official MCP Registry. Please use appropriate safeguards when interacting with these servers.

Official MCP Registry

Discover Model Context Protocol servers

[GitHub](#) [Docs](#) [API Reference](#)

Recently Updated

ai.naumu/mcp v0.18.1 Search, extend, and act on your team's Naumu knowledge graph: notes, threads, and nodes. Updated 9/24/2026	ai.openbash/openbash v1.0.0 Primary data as bash programs: trials, filings, customs, rentals, rank, OCR and SQL. Updated 9/24/2026	com.adbtd/mcp v1.0.0 Email infrastructure as a service: mailboxes, domains, dedicated IPs, sending, deliverability. Updated 9/24/2026
com.astranl/mcp v1.35.174 Verified local hands for AI agents: physical errands, funds held until agreed proof. Updated 9/24/2026	com.getabstract/mcp v1.0.0 Verified summaries of business books, articles and reports for workplace learning and AI upskilling Updated 9/24/2026	com.goaimoat/pricing-strategy v1.0.0 Pricing strategy, psychology tactics and competitor price positioning for cross-border sellers Updated 9/24/2026
com.gobananasai/go-bananas v1.0.0 AI image generation: characters, products, posters and edits, saved to your gallery. Updated 9/24/2026	com.gocepetrov/goce-petrov v1.2.0 Goce Petrov, senior brand designer for tech and AI startups: ask his agent or send him a brief. Updated 9/24/2026	com.goldandsilvercompany/porte v2.6.0 Official answers of Gold & Silver Company, precious metals refinery, Belgium (FR/EN/NL) Updated 9/24/2026
com.mailbox-mcp/mailbox-mcp v1.31.0 Read, send, file and search email in any Gmail, Microsoft 365 or IMAP mailbox, plus its calendar. Updated 9/24/2026	com.neuhausre/followupboss v1.5.3 Follow Up Boss CRM in plain English. 160 tools; bring your own API key. Updated 9/24/2026	com.scoreinnov/outils v1.3.0 Score innovation, benchmarks sectoriels et barometre cle-personne, en lecture seule. Updated 9/24/2026
com.tierux/mcp v0.1.0 Server-side purchase verification and entitlements for Google Play and Apple App Store over MCP. Updated 9/24/2026	de.meinbodenwert/bodenrichtwert v1.0.1 Amtliche Bodenrichtwerte (BORIS) und Flurstücke (ALKIS) der Bundesländer zu einer Adresse. Updated 9/24/2026	dev.recapp/recapp v1.0.0 Loom for agents: AI agents record narrated product demos, plus transcripts, summaries, and search. Updated 9/24/2026

Third-Party Servers

Official Integrations

Official integrations are maintained by companies building production ready MCP servers for their platforms.

- 🔗 **21st.dev Magic** - Create crafted UI components inspired by the best 21st.dev design engineers.
- 🔗 **ActionKit by Paragon** - Connect to 130+ SaaS integrations (e.g. Slack, Salesforce, Gmail) with Paragon's [ActionKit API](#).
- 🔗 **Adfin** - The only platform you need to get paid - all payments in one place, invoicing and accounting reconciliations with [Adfin](#).
- 🔗 **AgentOps** - Provide observability and tracing for debugging AI agents with [AgentOps API](#).
- 🔗 **AgentQL** - Enable AI agents to get structured data from unstructured web with [AgentQL](#).
- 🔗 **AgentRPC** - Connect to any function, any language, across network boundaries using [AgentRPC](#).
- 🔗 **Agentset** - RAG for your knowledge base connected to [Agentset](#).
- 🔗 **Aiven** - Navigate your [Aiven projects](#) and interact with the PostgreSQL®, Apache Kafka®, ClickHouse® and OpenSearch® services
- 🔗 **Alation** - Unlock the power of the enterprise Data Catalog by harnessing tools provided by the Alation MCP server.
- 🔗 **Alby Bitcoin Payments** - Connect any bitcoin lightning wallet to your agent to send and receive instant payments globally with your agent.
- 🔗 **Algolia** - Use AI agents to provision, configure, and query your [Algolia](#) search indices.
- 🔗 **Alibaba Cloud AnalyticDB for MySQL** - Connect to an [AnalyticDB for MySQL](#) cluster for getting database or table metadata, querying and analyzing data. It will be supported to add the OpenAPI for cluster operation in the future.
- 🔗 **Alibaba Cloud AnalyticDB for PostgreSQL** - An MCP server to connect to [AnalyticDB for PostgreSQL](#) instances, query and analyze data.
- 🔗 **Alibaba Cloud DataWorks** - A Model Context Protocol (MCP) server that provides tools for AI, allowing it to interact with the [DataWorks](#) Open API through a standardized interface. This implementation is based on the Alibaba Cloud Open API and enables AI agents to perform cloud resources operations seamlessly.
- 🔗 **Alibaba Cloud OpenSearch** - This MCP server equips AI Agents with tools to interact with [OpenSearch](#) through a standardized and extensible interface.
- 🔗 **Alibaba Cloud OPS** - Manage the lifecycle of your Alibaba Cloud resources with [CloudOps Orchestration Service](#) and Alibaba Cloud OpenAPI.
- 🔗 **Alibaba Cloud RDS** - An MCP server designed to interact with the Alibaba Cloud RDS OpenAPI, enabling programmatic management of RDS resources via an LLM.
- 🔗 **AlipayPlus** - Connect your AI Agents to AlipayPlus Checkout Payment.
- 🔗 **AlVoiceLab** - An AI voice toolkit with TTS, voice cloning, and video translation, now available as an MCP server for smarter agent integration.
- 🔗 **Alpaca** - Alpaca's MCP server lets you trade stocks and options, analyze market data, and build strategies through [Alpaca's Trading API](#)
- 🔗 **AlphaVantage** - Connect to 100+ APIs for financial market data, including stock prices, fundamentals, and more from [AlphaVantage](#)
- 🔗 **AltTester®** - Use AltTester® capabilities to connect and test your Unity or Unreal game. Write game test automation faster and smarter, using [AltTester](#) and the AltTester® MCP server.
- 🔗 **Antom** - Connect your AI Agents to Antom Checkout Payment.
- 🔗 **Anypoint** - An MCP server enabling AI assistants to interact with [Anypoint](#) - a local and collaborative wiki - to organize objects, lists, and more through natural language.
- 🔗 **Apache Doris** - MCP Server For [Apache Doris](#), an MPP-based real-time data warehouse.
- 🔗 **Apache IoTDB** - MCP Server for [Apache IoTDB](#) database and its tools
- 🔗 **Apache Pinot** - MCP server for running real - time analytics queries on Apache Pinot, an open-source OLAP database built for high-throughput, low-latency powering real-time applications.
- 🔗 **Apify** - Use 6,000+ pre-built cloud tools to extract data from websites, e-commerce, social media, search engines, maps, and more
- 🔗 **APIMatic MCP** - APIMatic MCP Server is used to validate OpenAPI specifications using [APIMatic](#). The server processes OpenAPI files and returns validation summaries by leveraging APIMatic's API.
- 🔗 **Apollo MCP Server** - Connect your GraphQL APIs to AI agents
- 🔗 **Aqara MCP Server** - Control [Aqara](#) smart home devices, query status, execute scenes, and much more using natural language.
- 🔗 **Archbee** - Write and publish documentation that becomes the trusted source for instant answers with AI. Stop cobbling tools and use [Archbee](#) — the first complete

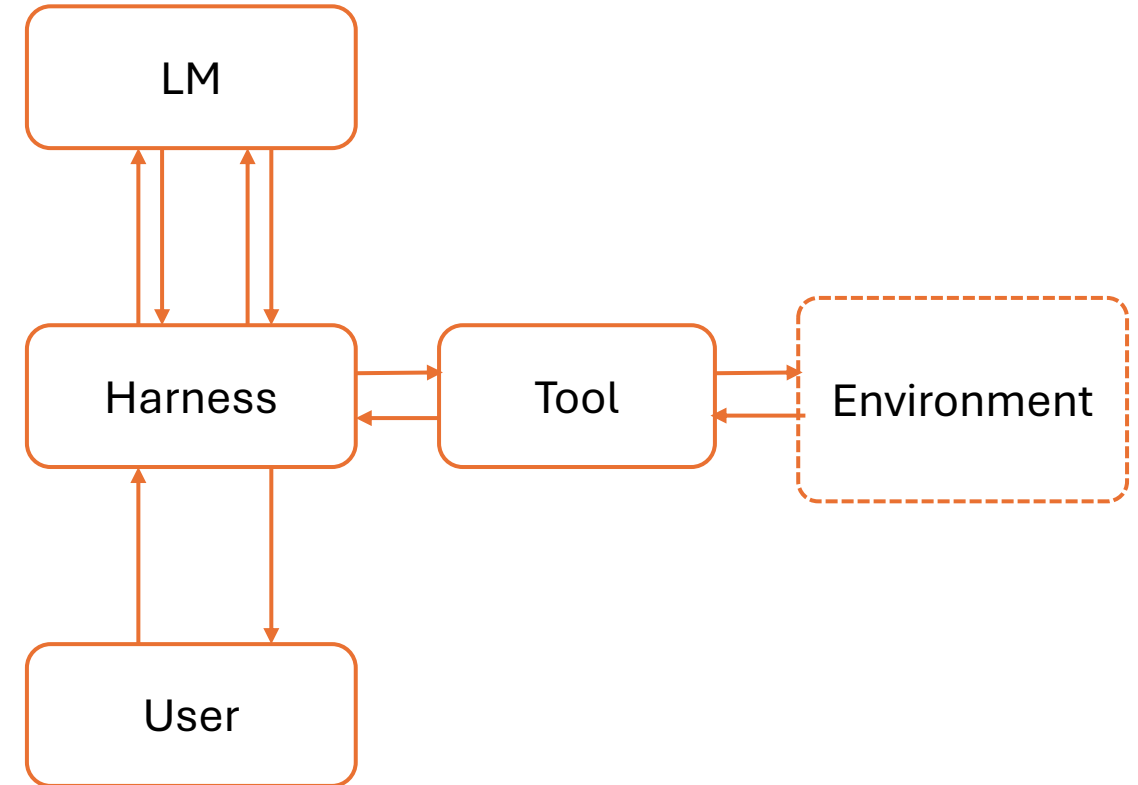
- 🔗 **Astra DB** - Comprehensive tools for managing collections and documents in a [AstraDB](#) NoSQL database with a full range of operations such as create, update, delete, find, and associated bulk actions.
- 🔗 **Atla** - Enable AI agents to interact with the [Atla API](#) for state-of-the-art LLMJ evaluation.
- 🔗 **Atlan** - The Atlan Model Context Protocol server allows you to interact with the [Atlan](#) servers through multiple tools.
- 🔗 **Atlassian** - Securely interact with Jira work items and Confluence pages, and search across both.
- 🔗 **AtomGit** - Official AtomGit server for integration with repository management, PRs, issues, branches, labels, and more.
- 🔗 **Audiense Insights** - Marketing insights and audience analysis from [Audiense](#) reports, covering demographic, cultural, influencer, and content engagement analysis.
- 🔗 **Auth0** - MCP server for interacting with your Auth0 tenant, supporting creating and modifying actions, applications, forms, logs, resource servers, and more.
- 🔗 **Authenticator App - 2FA** - A secure MCP (Model Context Protocol) server that enables AI agents to interact with the Authenticator App.
- 🔗 **AWS** - Specialized MCP servers that bring AWS best practices directly to your development workflow.
- 🔗 **Axiom** - Query and analyze your Axiom logs, traces, and all other event data in natural language
- 🔗 **Azure** - The Azure MCP Server gives MCP Clients access to key Azure services and tools like Azure Storage, Cosmos DB, the Azure CLI, and more.
- 🔗 **Azure DevOps** - Interact with Azure DevOps services like repositories, work items, builds, releases, test plans, and code search.
- 🔗 **BackdoCKET** - Search, Retrieve, and Update your [BackdoCKET](#) data. This currently includes Claims, Matters, Contacts, Tasks and Advanced Searches. To easily use the Remote Mcp Server utilize the following url: <https://ai.backdoCKET.com/mcp>
- 🔗 **Baidu Map - Baidu Map MCP Server** provides tools for AI agents to interact with Baidu Maps APIs, enabling location-based services and geospatial data analysis.
- 🔗 **Bankless Onchain** - Query Onchain data, like ERC20 tokens, transaction history, smart contract state.
- 🔗 **Baserow** - Query data from Baserow self-hosted or SaaS databases using MCP integration.
- 🔗 **BICScan** - Risk score / asset holdings of EVM blockchain address (EOA, CA, ENS) and even domain names.
- 🔗 **Bitrise** - Chat with your builds, CI, and [more](#).
- 🔗 **Boikot** - Learn about the ethical and unethical actions of major companies with [boikot.xyz](#).
- 🔗 **BoldSign** - Search, request, and manage e-signature contracts effortlessly with [BoldSign](#).
- 🔗 **Boost.space** - An MCP server integrating with [Boost.space](#) for centralized, automated business data from 2000+ sources.
- 🔗 **Box** - Interact with the Intelligent Content Management platform through Box AI.
- 🔗 **BrightData** - Discover, extract, and interact with the web - one interface powering automated access across the public internet.
- 🔗 **Browserbase** - Automate browser interactions in the cloud (e.g. web navigation, data extraction, form filling, and more)
- 🔗 **BrowserStack** - Access BrowserStack's [Test Platform](#) to debug, write and fix tests, do accessibility testing and more.
- 🔗 **Buildkite** - Exposing Buildkite data (pipelines, builds, jobs, tests) to AI tooling and editors.
- 🔗 **Buildable** (TypeScript) - Official MCP server for Buildable AI-powered development platform. Enables AI assistants to manage tasks, track progress, get project context, and collaborate with humans on software projects.
- 🔗 **BuiltWith** - Identify the technology stack behind any website.
- 🔗 **Burp Suite** - MCP Server extension allowing AI clients to connect to [Burp Suite](#)
- 🔗 **Cal.com** - Connect to the Cal.com API to schedule and manage bookings and appointments.
- 🔗 **Campertunity** - Search campgrounds around the world on campertunity, check availability, and provide booking links.
- 🔗 **Canva** - Provide AI - powered development assistance for [Canva](#) apps and integrations.
- 🔗 **Carbon Voice** - MCP Server that connects AI Agents to [Carbon Voice](#). Create, manage, and interact with voice messages, conversations, direct messages, folders, voice memos, AI actions and more in [Carbon Voice](#).
- 🔗 **Cartesia** - Connect to the [Cartesia](#) voice platform to perform text-to-speech, voice cloning etc.
- 🔗 **Cashfree** - [Cashfree Payments](#) official MCP server.
- 🔗 **CB Insights** - Use the [CB Insights](#) MCP Server to connect to [ChatCBI](#)

- 🔗 **Claudia** - Exposes [Claudia](#)'s media output, transformation, AI analysis, management, optimization and delivery as tools usable by AI agents
- 🔗 **Cloudsway SmartSearch** - Web search MCP server powered by Cloudsway, supporting keyword search, language, and safety options. Returns structured JSON results.
- 🔗 **Codacy** - Interact with [Codacy](#) API to query code quality issues, vulnerabilities, and coverage insights about your code.
- 🔗 **CodeLogic** - Interact with [CodeLogic](#), a Software Intelligence platform that graphs complex code and data architecture dependencies, to boost AI accuracy and insight.
- 🔗 **CoinGecko** - Official [CoinGecko API](#) MCP Server for Crypto Price & Market Data, across 200+ Blockchain Networks and 8M+ Tokens.
- 🔗 **Comet Opik** - Query and analyze your [Opik](#) logs, traces, prompts and all other telemetry data from your LLMs in natural language.
- 🔗 **Conductor** - Interact with Conductor (OSS and Orkes) REST APIs.
- 🔗 **Composio** - Use [Composio](#) to connect 100+ tools. Zero setup. Auth built-in. Made for agents, works for humans.
- 🔗 **Confluent** - Interact with Confluent Kafka and Confluent Cloud REST APIs.
- 🔗 **Contrast Security** - Brings Contrast's vulnerability and SCA data into your coding agent to quickly remediate vulnerabilities.
- 🔗 **Convex** - Introspect and query your apps deployed to Convex.
- 🔗 **Cortex** - Official MCP server for [Cortex](#).
- 🔗 **Couchbase** - Interact with the data stored in Couchbase clusters.
- 🔗 **CRIC Wuye AI** - Interact with capabilities of the CRIC Wuye AI platform, an intelligent assistant specifically for the property management industry.
- 🔗 **CrowdStrike Falcon** - Connects AI agents with the CrowdStrike Falcon platform for intelligent security analysis, providing programmatic access to detections, incidents, behaviors, threat intelligence, hosts, vulnerabilities, and identity protection capabilities.
- 🔗 **CTERA Edge Filer** - CTERA Edge Filer delivers intelligent edge caching and multiprotocol file access, enabling fast, secure access to files across core and remote sites.
- 🔗 **CTERA Portal** - CTERA Portal is a multi-tenant, multi-cloud platform that delivers a global namespace and unified management across petabytes of distributed content.
- 🔗 **Cycode** - Boost security in your dev lifecycle via SAST, SCA, Secrets & IaC scanning with [Cycode](#).
- 🔗 **Dart** - Interact with task, doc, and project data in [Dart](#), an AI-native project management tool
- 🔗 **Databricks** - Connect to data, AI tools & agents, and the rest of the Databricks platform using turnkey managed MCP servers. Or, host your own custom MCP servers within the Databricks security and data governance boundary.
- 🔗 **DataHub** - Search your data assets, traverse data lineage, write SQL queries, and more using [DataHub](#) metadata.
- 🔗 **Daytona** - Fast and secure execution of your AI generated code with [Daytona](#) sandboxes
- 🔗 **Debugg.AI** - Zero-Config, Fully AI-Managed End-to-End Testing for any code gen platform via [Debugg.AI](#) remote browsing test agents.
- 🔗 **DeepL** - Translate or rewrite text with [DeepL](#)'s very own AI models using [the DeepL API](#)
- 🔗 **Defang** - Deploy your project to the cloud seamlessly with the [Defang](#) platform without leaving your integrated development environment
- 🔗 **Detailer** - Instantly generate rich, AI-powered documentation for your GitHub repositories. Designed for AI agents to gain deep project context before taking action.
- 🔗 **DevCycle** - Create and monitor feature flags using natural language in your AI coding assistant.
- 🔗 **DevHub** - Manage and utilize website content within the [DevHub](#) CMS platform
- 🔗 **DevRev** - An MCP server to integrate with DevRev APIs to search through your DevRev Knowledge Graph where objects can be imported from diff. Sources listed [here](#).
- 🔗 **DexPaprika (CoinPaprika)** - Access real-time DEX data, liquidity pools, token information, and trading analytics across multiple blockchain networks with [DexPaprika](#) by CoinPaprika.
- 🔗 **Dolt** - The official MCP server for version-controlled [Dolt](#) databases.
- 🔗 **Dot (GetDot.ai)** - Fetch, analyze or visualize data from your favorite database or data warehouse (Snowflake, BigQuery, Redshift, Databricks, Clickhouse, ...) with [Dot](#), your AI Data Analyst. This remote MCP server is a one-click integration for user that have setup Dot.
- 🔗 **Drata** - Get hands-on with our experimental MCP server—bringing real-time compliance intelligence into your AI workflows.
- 🔗 **Dumpling AI** - Access data, web scraping, and document conversion APIs by [Dumpling AI](#)

- 🔗 **GitGuardian** - GitGuardian official MCP server - Scan projects using GitGuardian's industry-leading API, which features over 500 secret detectors to prevent credential leaks before they reach public repositories. Resolve security incidents directly with rich contextual data for rapid, automated remediation.
- 🔗 **GitHub** - GitHub's official MCP Server.
- 🔗 **GitKraken** - A CLI for interacting with GitKraken APIs. Includes an MCP server via `gk mcp` that not only wraps GitKraken APIs, but also Jira, GitHub, GitLab, and more.
- 🔗 **Glean** - Enterprise search and chat using Glean's API.
- 🔗 **Globalping** - Access a network of thousands of probes to run network commands like ping, traceroute, mtr, http and DNS resolve.
- 🔗 **gNucleus Text-To-CAD** - Generate CAD parts and assemblies from text using gNucleus AI models.
- 🔗 **Google Cloud Run** - Deploy code to Google Cloud Run
- 🔗 **GoLogin MCP server** - Manage your GoLogin browser profiles and automation directly through AI conversations!
- 🔗 **Google Maps Platform Code Assist** - Ground agents on fresh, official documentation and code samples for optimal geo-related guidance and code..
- 🔗 **gotoHuman** - Human-in-the-loop platform - Allow AI agents and automations to send requests for approval to your [gotoHuman](#) inbox.
- 🔗 **Grafana** - Search dashboards, investigate incidents and query datasources in your Grafana instance
- 🔗 **Grafbase** - Turn your GraphQL API into an efficient MCP server with schema intelligence in a single command.
- 🔗 **Grain** - Access your Grain meetings notes & transcripts directly in claude and generate reports with native Claude Prompts.
- 🔗 **Graphlit** - Ingest anything from Slack to Gmail to podcast feeds, in addition to web crawling, into a searchable [Graphlit](#) project.
- 🔗 **Gremlin** - The official [Gremlin](#) MCP server. Analyze your reliability posture, review recent tests and chaos engineering experiments, and create detailed reports.
- 🔗 **GreptimeDB** - Provides AI assistants with a secure and structured way to explore and analyze data in [GreptimeDB](#).
- 🔗 **GROWI** - Official MCP Server to integrate with GROWI APIs.
- 🔗 **Gyazo** - Search, fetch, upload, and interact with Gyazo images, including metadata and OCR data.
- 🔗 **Harper** - An MCP server providing an interface for MCP clients to access data within [Harper](#).
- 🔗 **Heroku** - Interact with the Heroku Platform through LLM-driven tools for managing apps, add-ons, dynos, databases, and more.
- 🔗 **HeyOnCall** - Page a human, sending critical or non-critical alerts to the free [HeyOnCall](#) iOS or Android apps.
- 🔗 **Hiveflow** - Create, manage, and execute agentic AI workflows directly from your assistant.
- 🔗 **Hive Intelligence** - Ultimate cryptocurrency MCP for AI assistants with unified access to crypto, DeFi, and Web3 analytics
- 🔗 **Hologres** - Connect to a [Hologres](#) instance, get table metadata, query and analyze data.
- 🔗 **Homebrew** Allows [Homebrew](#) users to run Homebrew commands locally.
- 🔗 **Honeycomb** Allows [Honeycomb](#) Enterprise customers to query and analyze their data, alerts, dashboards, and more; and cross-reference production behavior with the codebase.
- 🔗 **HubSpot** - Connect, manage, and interact with [HubSpot](#) CRM data
- 🔗 **Hugging Face** - Connect to the Hugging Face Hub APIs programmatically: semantic search for spaces and papers, exploration of datasets and models, and access to all compatible MCP Gradio tool spaces!
- 🔗 **Hunter** - Interact with the [Hunter API](#) to get B2B data using natural language.
- 🔗 **Hyperbolic** - Interact with Hyperbolic's GPU cloud, enabling agents and LLMs to view and rent available GPUs, SSH into them, and run GPU-powered workflows for you.
- 🔗 **Hyperbrowser - Hyperbrowser** is the next-generation platform empowering AI agents and enabling effortless, scalable browser automation.
- 🔗 **IBM wxflows** - Tool platform by IBM to build, test and deploy tools for any data source
- 🔗 **Inbox Zero** - AI personal assistant for email [Inbox Zero](#)
- 🔗 **Inflectra Spira** - Connect to your instance of the SpiraTest, SpiraTeam or SpiraPlan application lifecycle management platform by [Inflectra](#)
- 🔗 **Inkeep** - RAG Search over your content powered by [Inkeep](#)
- 🔗 **Integration App** - Interact with any other SaaS applications on behalf of your customers.
- 🔗 **IP2Location.io** - Interact with IP2Location.io API to retrieve the geolocation information for an IP address.
- 🔗 **IPLocate** - Look up IP address geolocation, network information, detect proxies and VPNs, and find abuse contact details using [IPLocate.io](#)

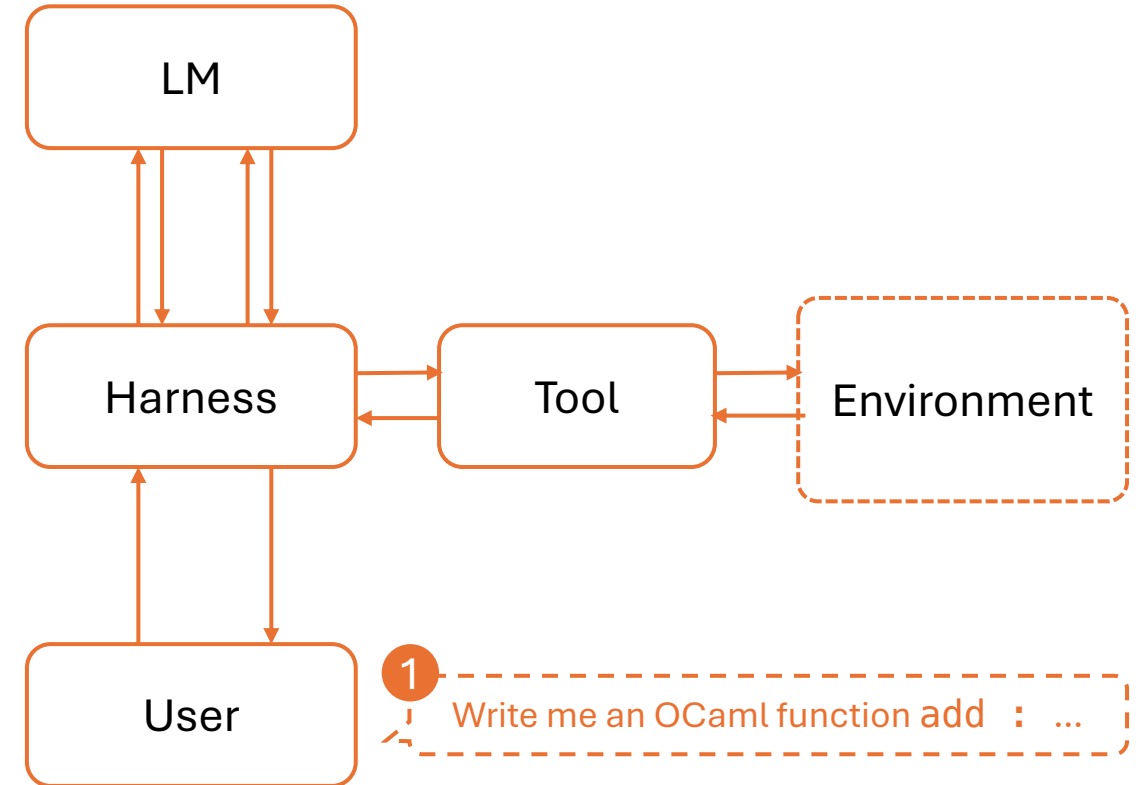
Sequential Thinking Tool

- **thought** (string)
 - the current step you want to record
- **nextThoughtNeeded** (boolean)
 - whether you intend to add another step
- **thoughtNumber** (int ≥ 1)
 - index of this step
- **totalThoughts** (int ≥ 1)
 - your *current* plan for how many steps you'll need



Sequential Thinking Tool

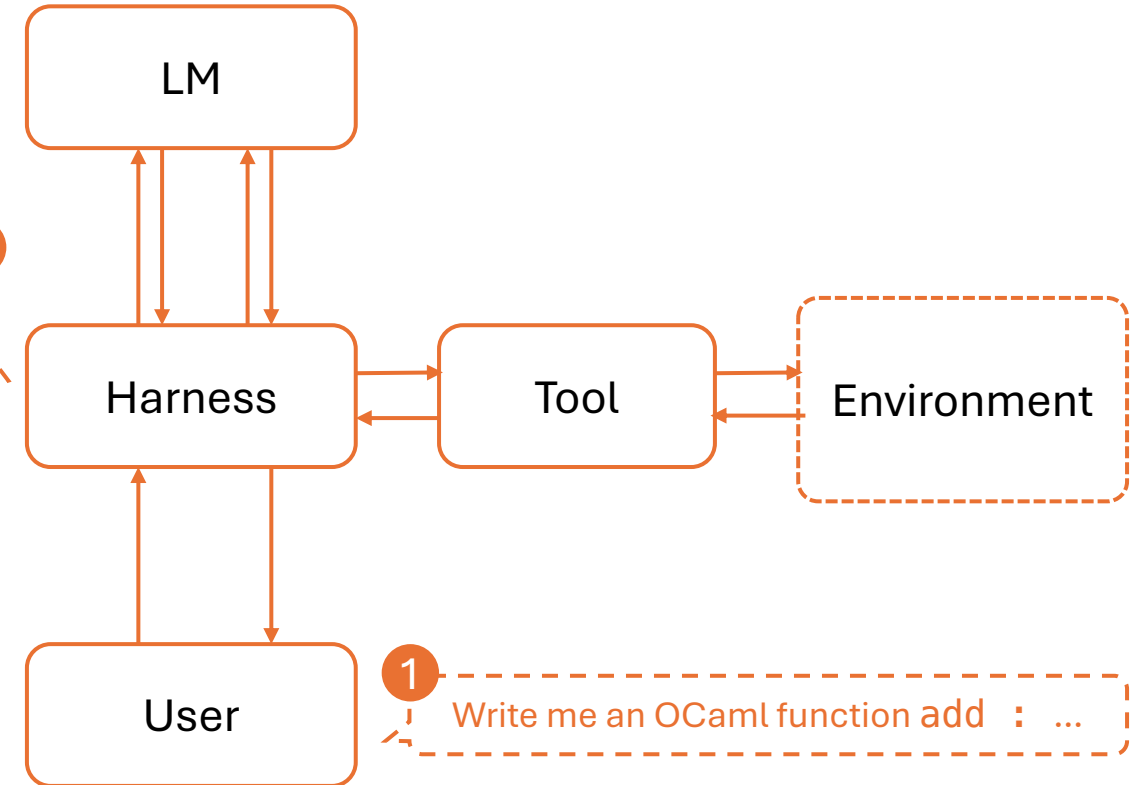
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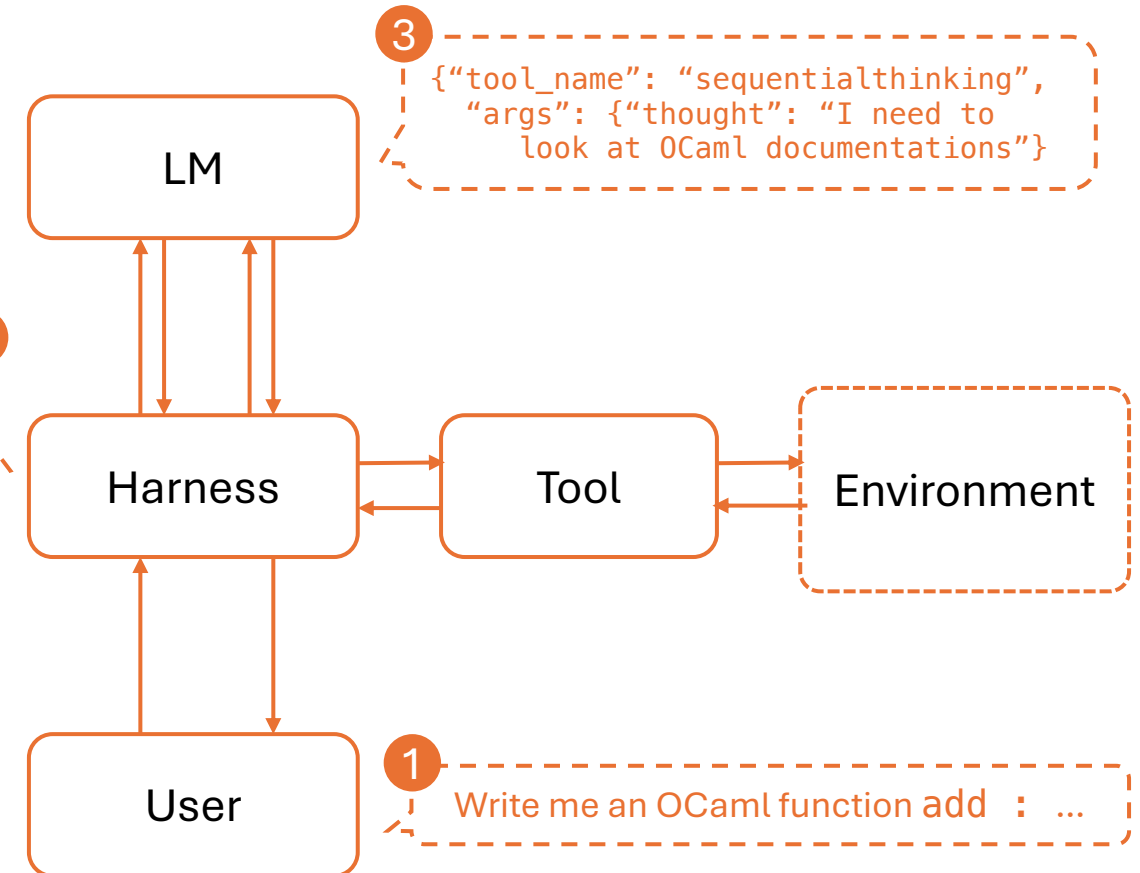
The user wants to write a function in OCaml. Here are the available tools: [..., {"name": "sequentialthinking"}]



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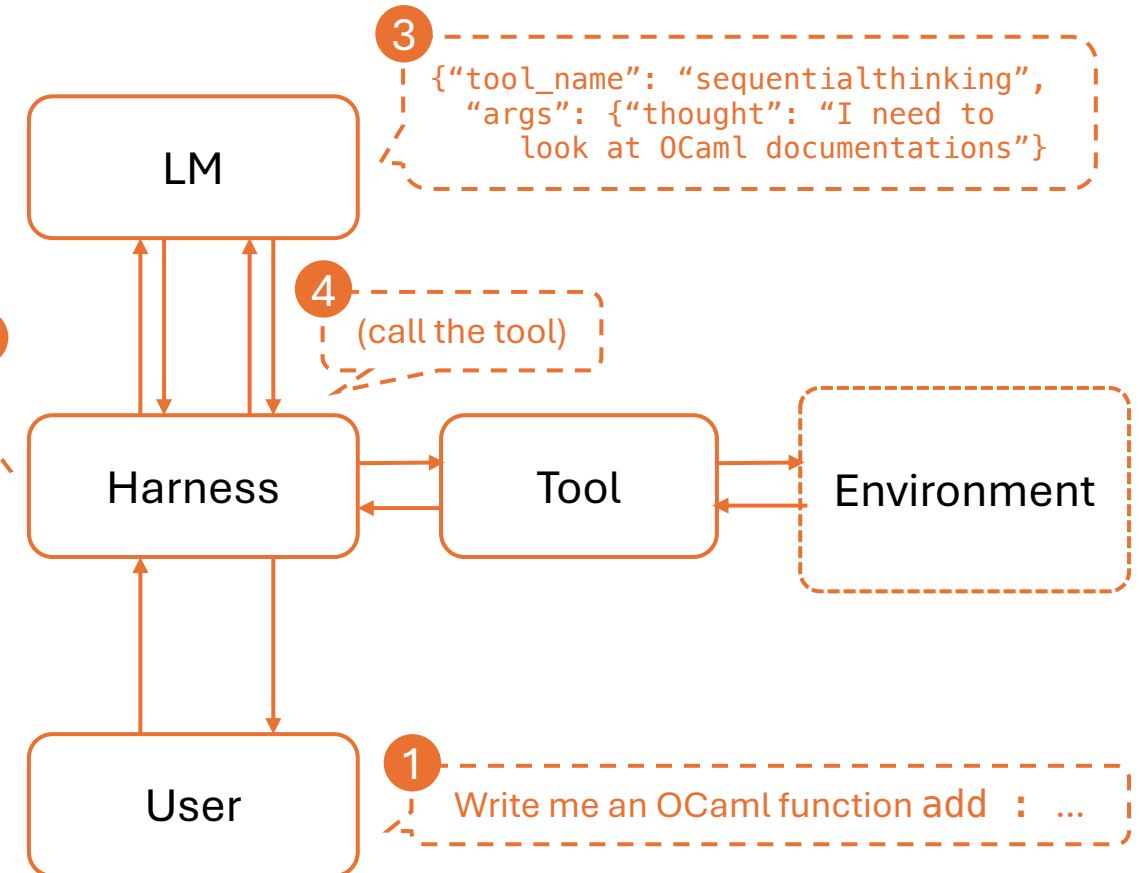
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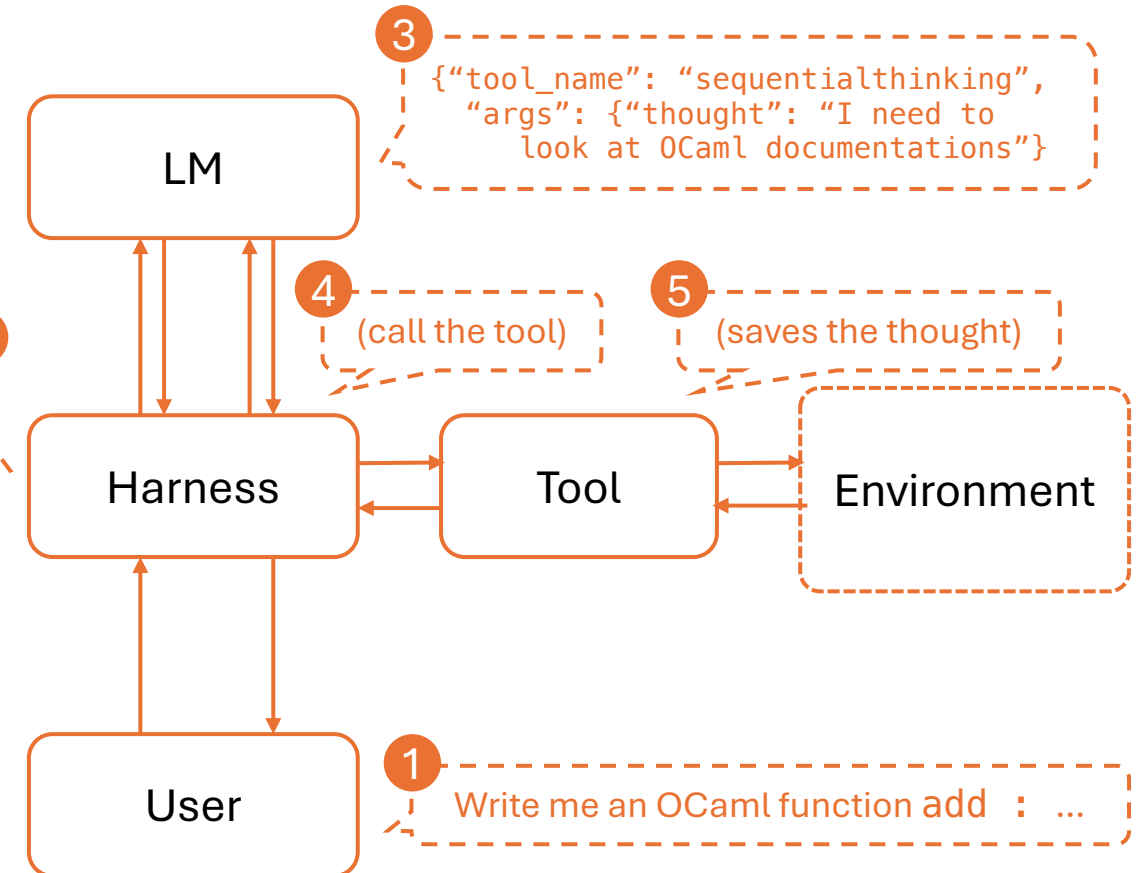
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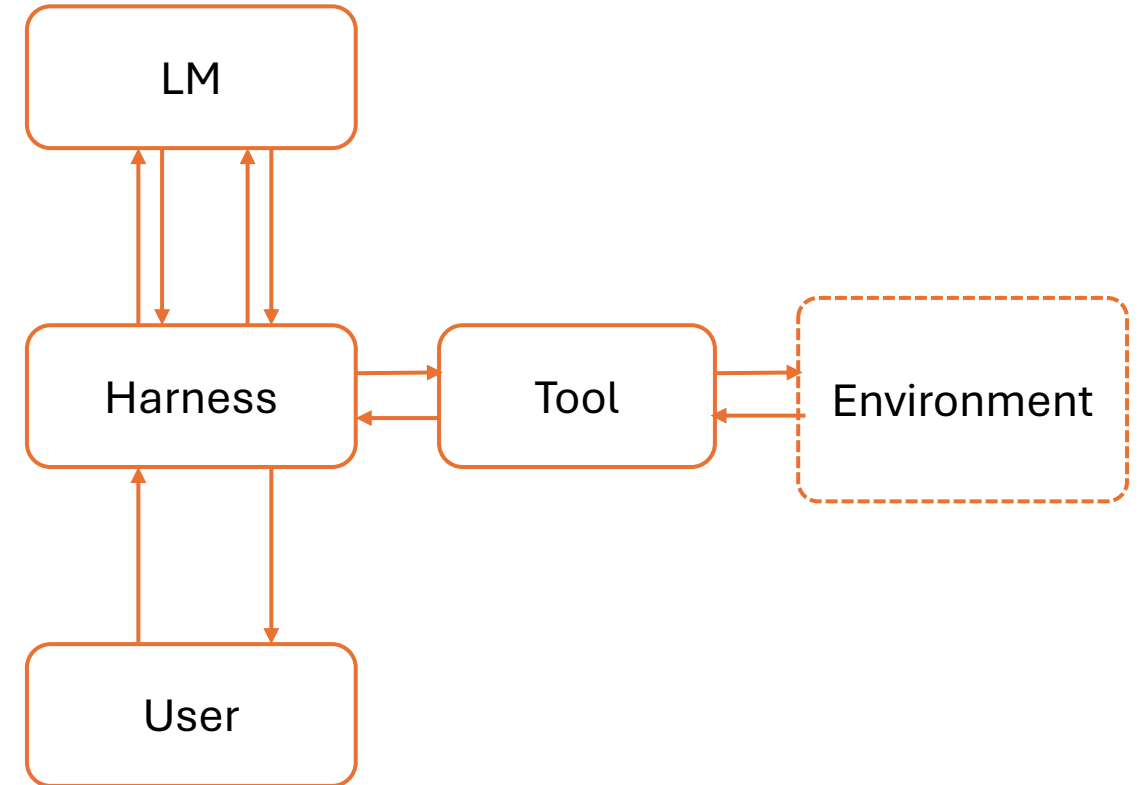
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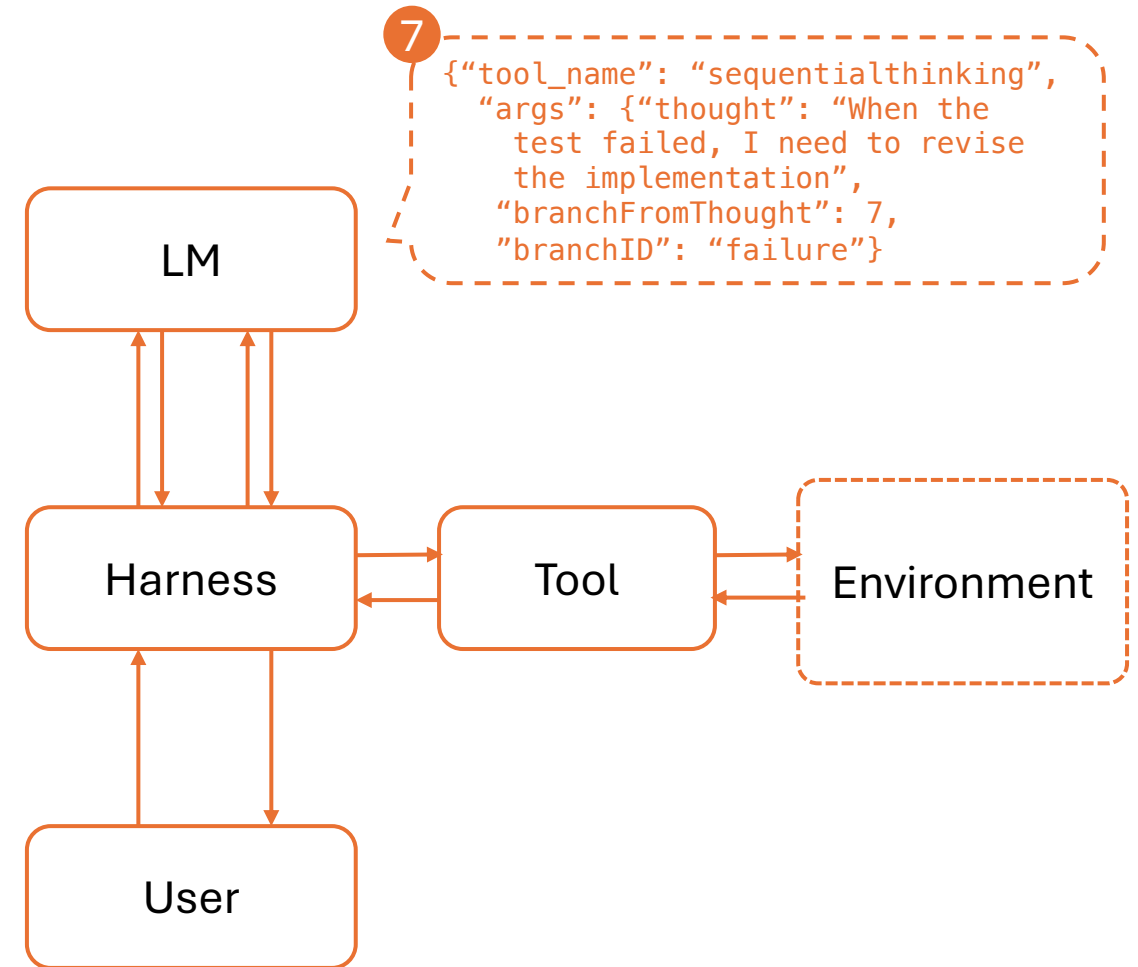
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- **branchID** (string)



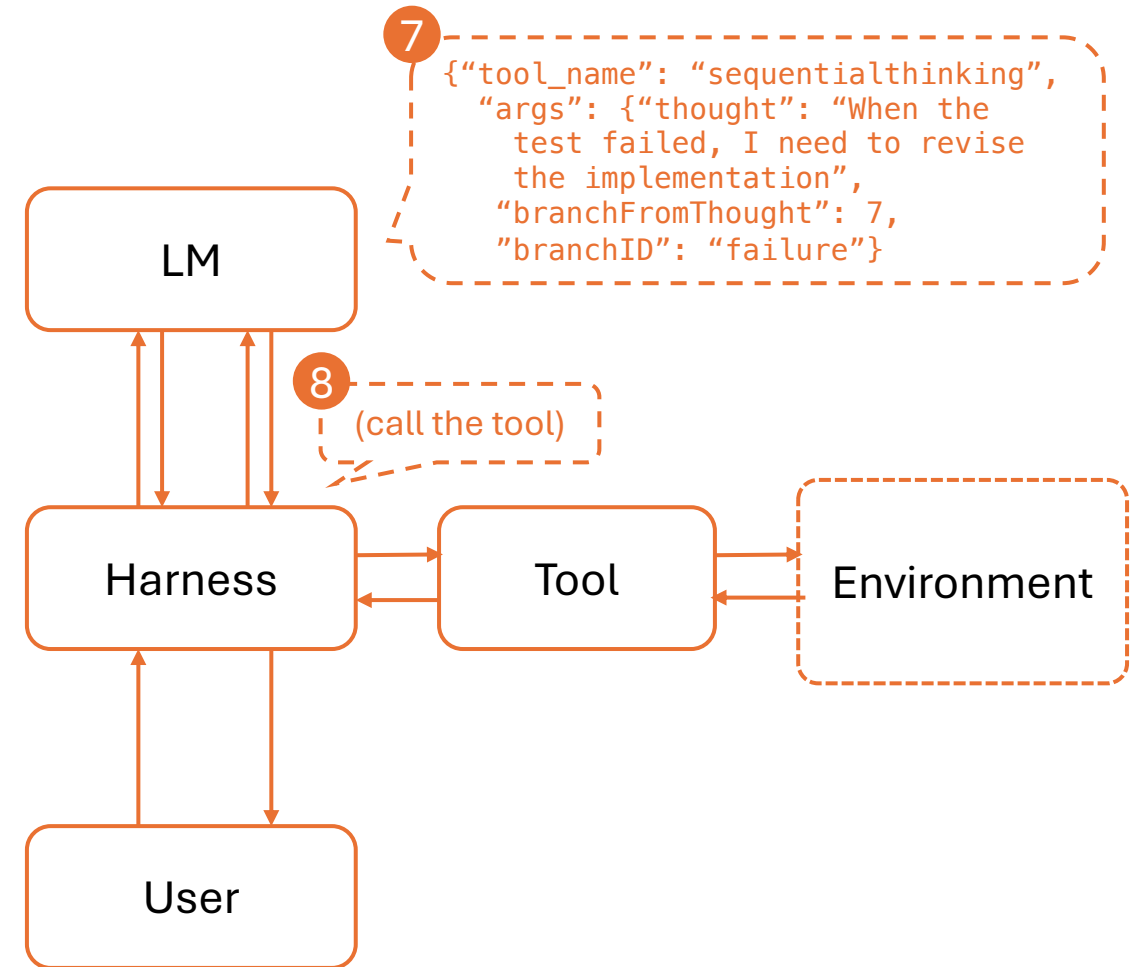
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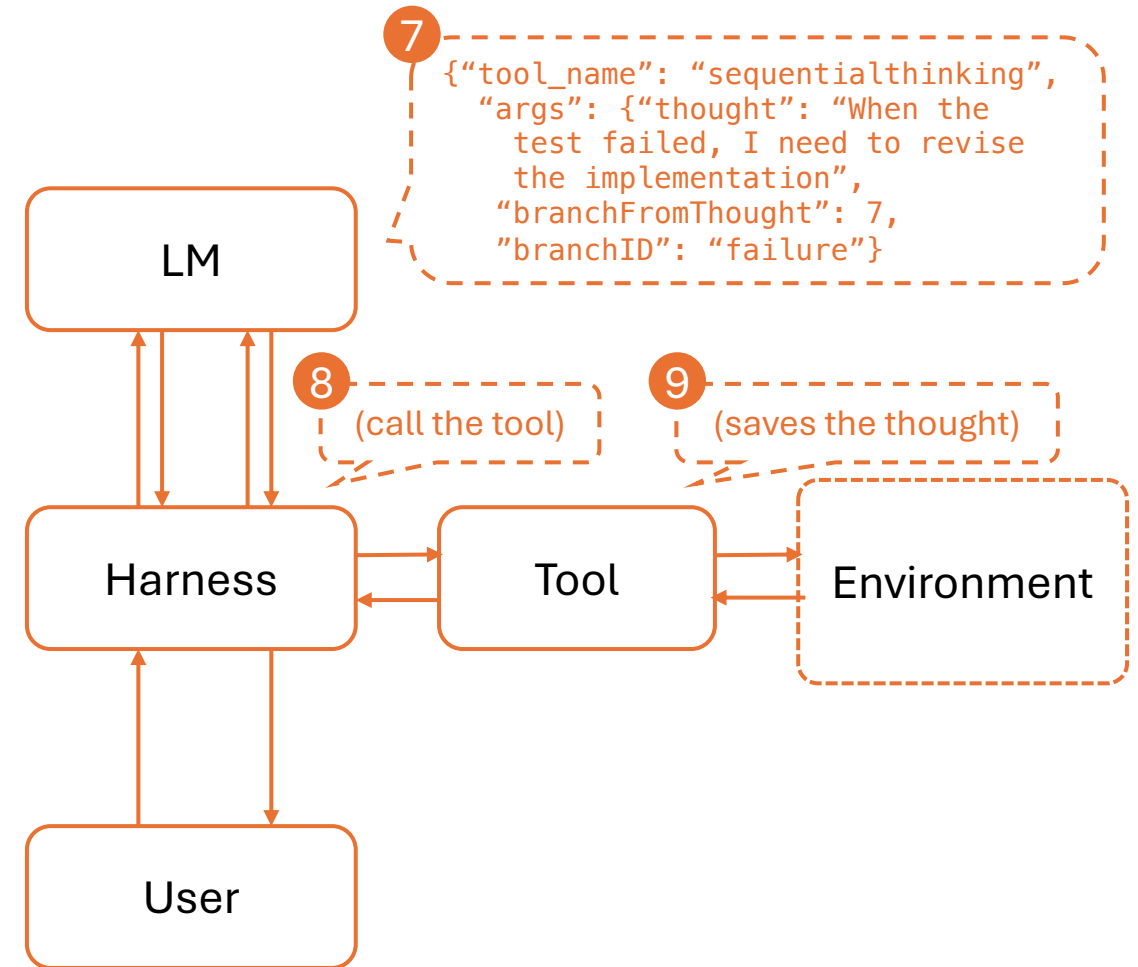
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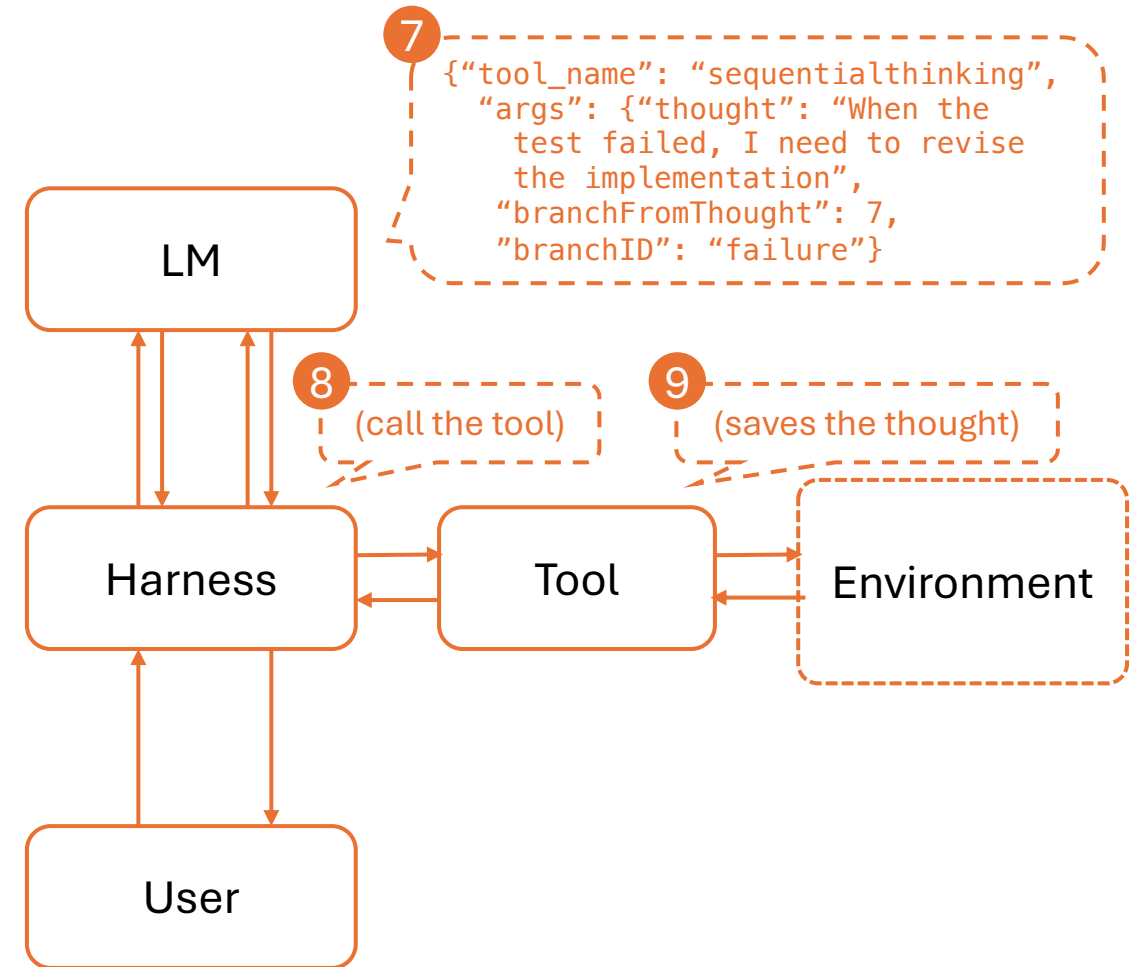
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☰ To-dos 7



- ➔ Create main package structure and exports in `__init__.py`
- ☐ Implement base Tool class and tool registry system
- ☐ Create LLM interface and JSON parsing system
- ☐ Implement core Agent class with plan/act loop
- ☐ Create CLI interface for waa command
- ☐ Set up logging system
- ☐ Create tool implementations (todo, fs, server, tests)

Now let me start implementing the high-level structure. First, I'll create the main package structure:

🔗 `__init__.py` +17 -1 •



☰ 1 of 7 Done View all



- ✓ ~~Create main package structure and exports in `__init__.py`~~
- ☐ Implement base Tool class and tool registry system

REACT: SYNERGIZING REASONING AND ACTING IN LANGUAGE MODELS

Shunyu Yao^{*,1}, Jeffrey Zhao², Dian Yu², Nan Du², Izhak Shafran², Karthik Narasimhan¹, Yuan Cao²

¹Department of Computer Science, Princeton University

²Google Research, Brain team

¹{shunyuy, karthikn}@princeton.edu

²{jeffreyzhao, dianyu, dunan, izhak, yuancao}@google.com

REACT: SYNERGIZING REASONING AND ACTING IN LANGUAGE MODELS

Shunyu Yao^{*1}, Jeffrey Zhao², Dian Yu², Nan Du², Izhak Shafran², Karthik Narasimhan¹, Yuan Cao²

²{je

(1) Hotspot QA

Question: Aside from the Apple Remote, what other device can control the program Apple Remote was originally designed to interact with?

(1a) Standard

Answer: iPod



(1b) CoT (Reason Only)

Thought: Let's think step by step. Apple Remote was originally designed to interact with Apple TV. Apple TV can be controlled by iPhone, iPad, and iPod Touch. So the answer is iPhone, iPad, and iPod Touch.

Answer: iPhone, iPad, iPod Touch



(1c) Act-Only

Act 1: Search[Apple Remote]
Obs 1: The Apple Remote is a remote control ...
Act 2: Search[Front Row]
Obs 2: Could not find [Front Row]. Similar: ...
Act 3: Search[Front Row (software)]
Obs 3: Front Row is a discontinued media center software ...
Act 4: Finish[yes]



(1d) ReAct (Reason + Act)

Thought 1: I need to search Apple Remote and find the program it was originally designed to interact with.
Act 1: Search[Apple Remote]
Obs 1: The Apple Remote is a remote control introduced in October 2005 by Apple ... originally designed to control the Front Row media center program ...
Thought 2: Apple Remote was originally designed to control the Front Row media center program. I need to search Front Row next and find what other device can control it.
Act 2: Search[Front Row]
Obs 2: Could not find [Front Row]. Similar: ['Front Row Seat to Earth', 'Front Row Motorsports', 'Front Row (software)', ...]
Thought 3: Front Row is not found. I need to search Front Row (software) .
Act 3: Search[Front Row (software)]
Obs 3: Front Row is a discontinued media center software ...
Thought 4: Front Row (software) is controlled by an Apple Remote or the keyboard function keys. So the answer is keyboard function keys.
Act 4: Finish[keyboard function keys]



REACT: SYNERGIZING REASONING AND ACTING IN LANGUAGE MODELS

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Act 1: Search[Apple Remote]
Obs 1: The Apple Remote is a remote control introduced in October 2005 by Apple ... originally designed to control the

Prompt Method ^a	HotpotQA (EM)	Fever (Acc)
Standard	28.7	57.1
CoT (Wei et al., 2022)	29.4	56.3
CoT-SC (Wang et al., 2022a)	33.4	60.4
Act	25.7	58.9
ReAct	27.4	60.9
CoT-SC → ReAct	34.2	64.6
ReAct → CoT-SC	35.1	62.0
Supervised SoTA ^b	67.5	89.5

Table 1: PaLM-540B prompting results on HotpotQA and Fever.

^aHotpotQA EM is 27.1, 28.9, 33.8 for Standard, CoT, CoT-SC in Wang et al. (2022b).
^b(Zhu et al., 2021; Lewis et al., 2020)

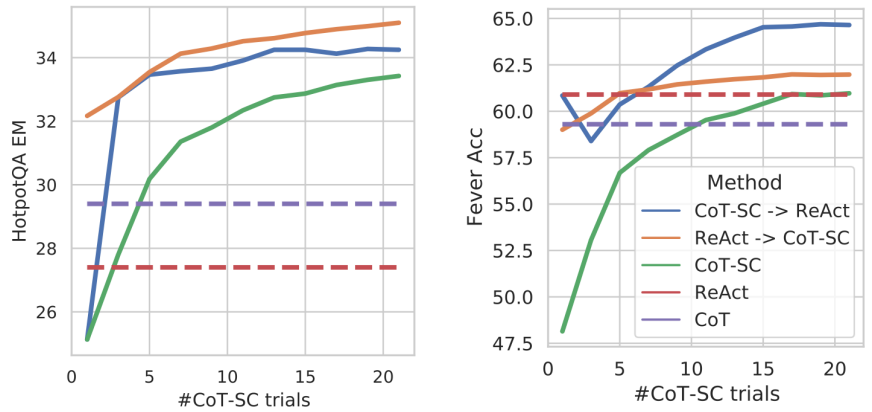


Figure 2: PaLM-540B prompting results with respect to number of CoT-SC samples used.

Tools Related to Software Development

- ~~Sequential thinking~~
 - ~~Planning, deliberate thinking~~
- File systems (FS)
 - Creating, removing, editing programs
 - Drafting test cases and setting up testing environment
- Retrieval augmented generation (RAG) databases
 - Context management
 - Documentations, examples, code snippets

File System (FS) for Code Implementation

- Create, read, write, or edit files
 - Managing project folder structures
 - Put implementation in relevant files, write test cases, etc.
- Search for files
 - Good when we do not know what exact files to work on
 - Often substituted by command line tools such as “grep”

Can you look at core/scallop-fir/docs/ and help me implement a type_checker

I'll help you implement a type checker for the Scallop language. Let me first explore the documentation and understand the current structure of the project.

Listed docs

Read grammar.md

Searched What is the current

Let me explore the existing

Read test_type_inference.rs

Read test_type_bounds.rs

Searched Where are type def

Let me explore the source c

Listed src

Read lib.rs

Grepped type.*check|TypeChecker|type_infer

I can see that the type checking functionalit
s. Let me explore the AST types to understa

Let me explore more of the type system structure:

Read mod.rs

Read type_decl.rs

Read named_type.rs

Now let me create a comprehensive plan for implementing the type checker:

1 of 9 Done

✓ Explore AST structure and understand type system components

➔ Create the analysis module with type checking infrastructure

○ Implement

Now let me create the analysis module with the type checking infrastructure:

mod.rs (new) +3 •

type_inference.rs (new) +510 •

Now let me update the lib.rs to enable t

lib.rs +1 -1 •

Let me check for any compilation errors

No linter errors found in analysis

I need to fix the generic type parameter structure. Let me check the actual structure:

type_inference.rs +1 -1 •

Now let me check the ADT declaration structure:

Read adt_decl.rs

I need to fix the ADT checking code:

type_inference.rs +7 -2 •

Let me check the alias type declaration structure:

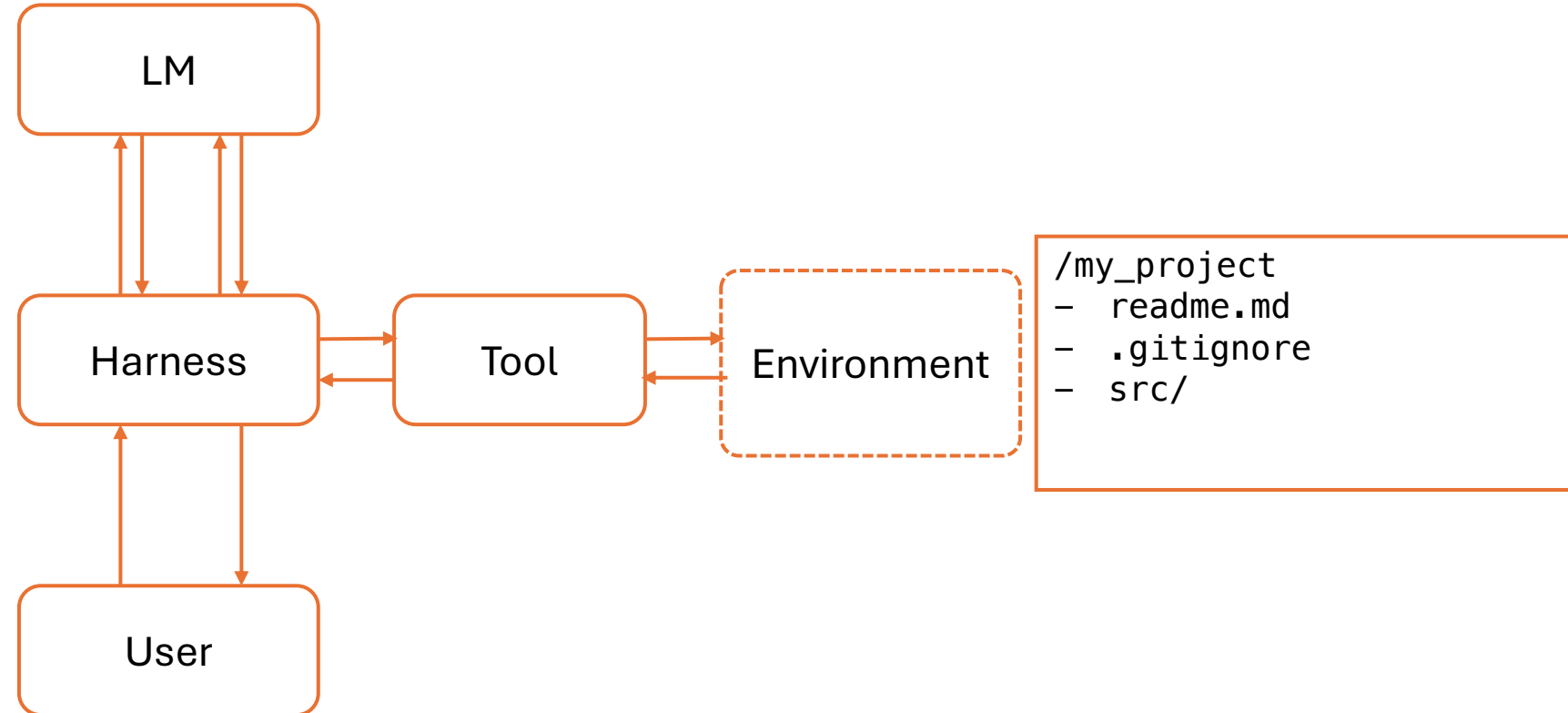
Read alias_type_decl.rs

I need to fix the alias type declaration checking:

type_inference.rs +7 -1 •

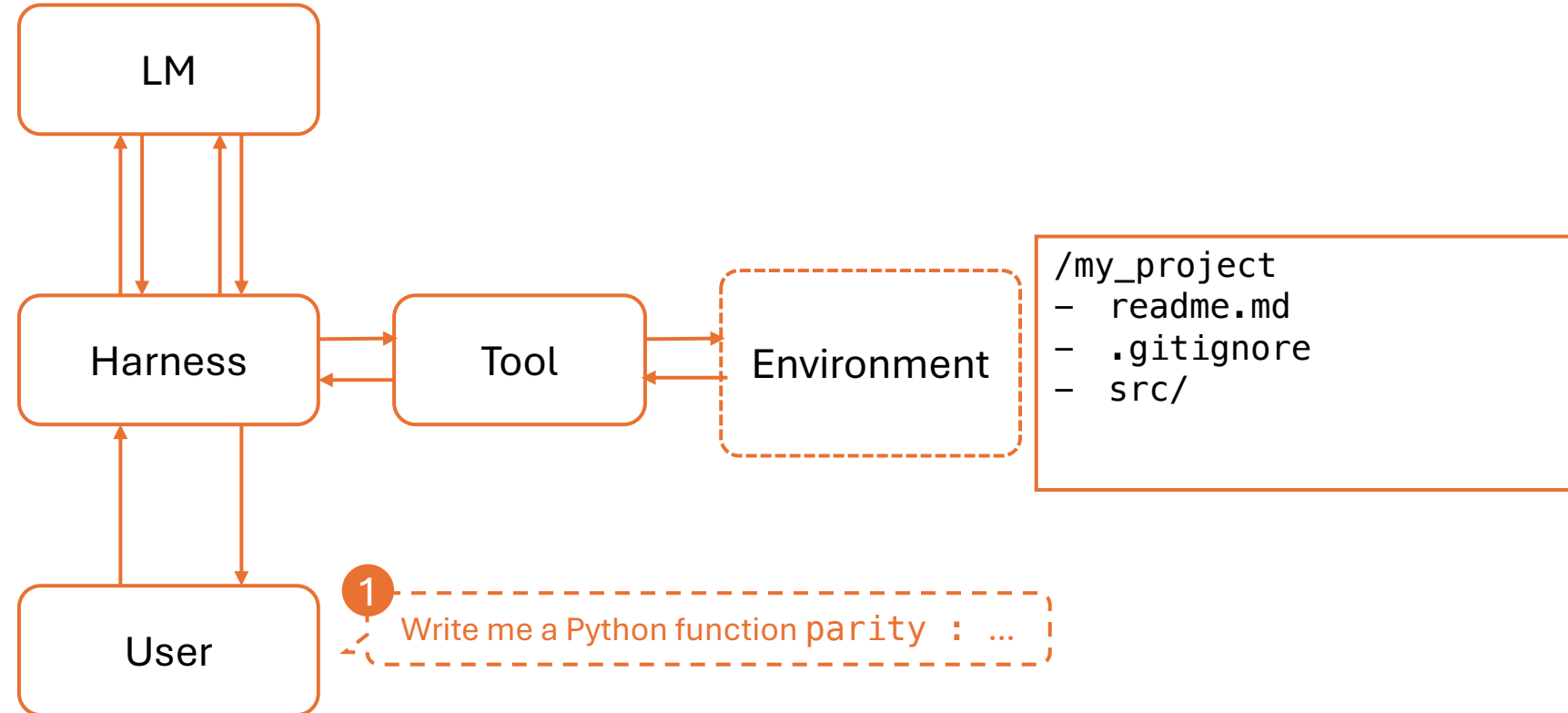
File System (FS) for Code Implementation

- Create, read, write, or edit files



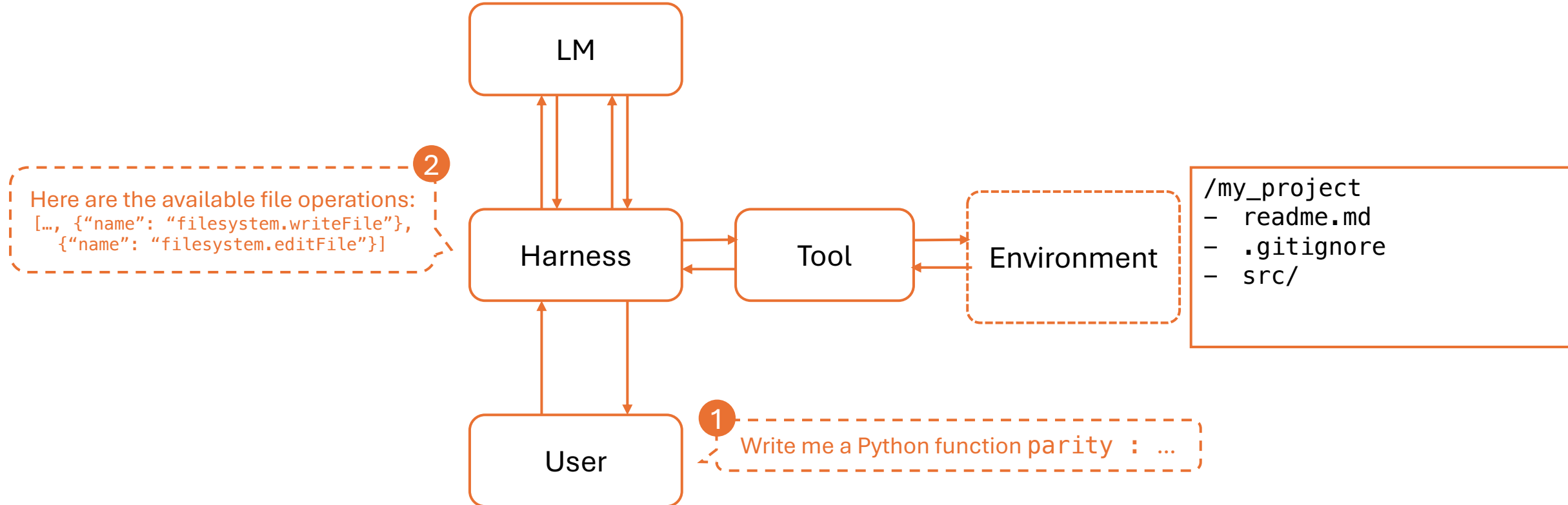
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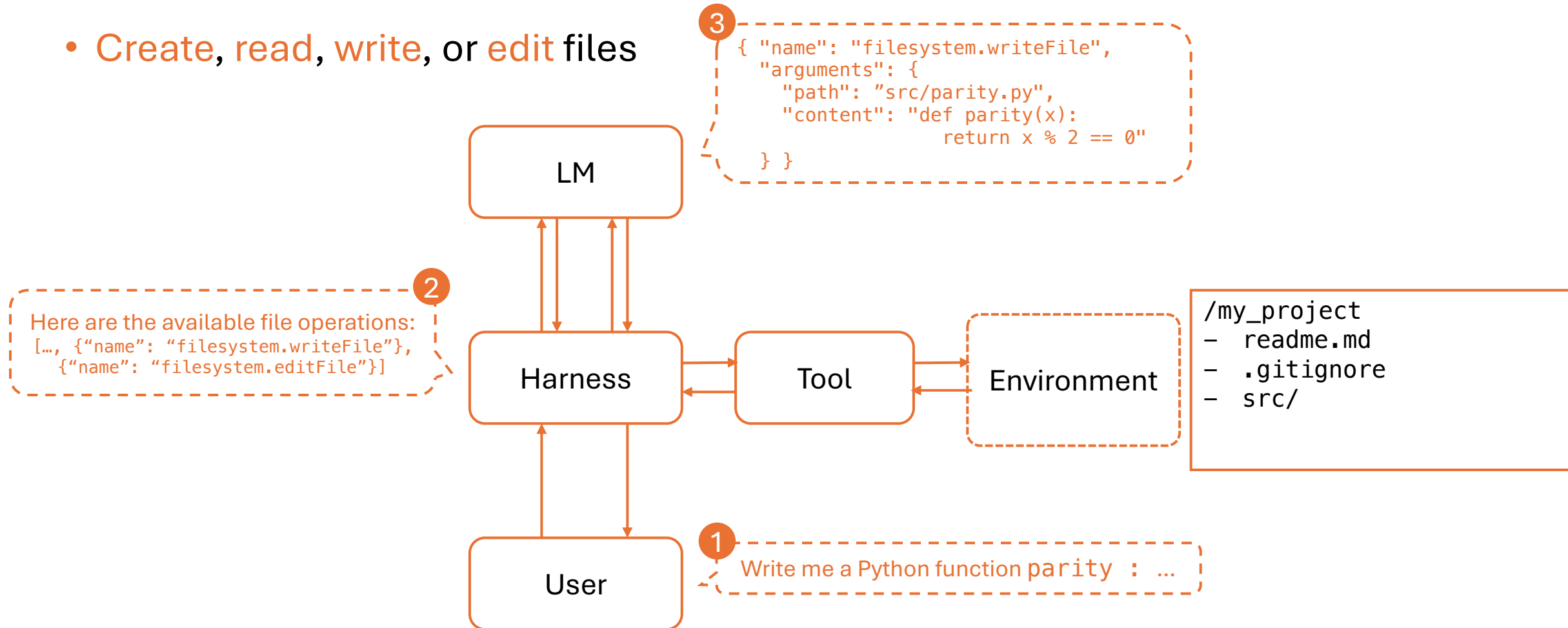
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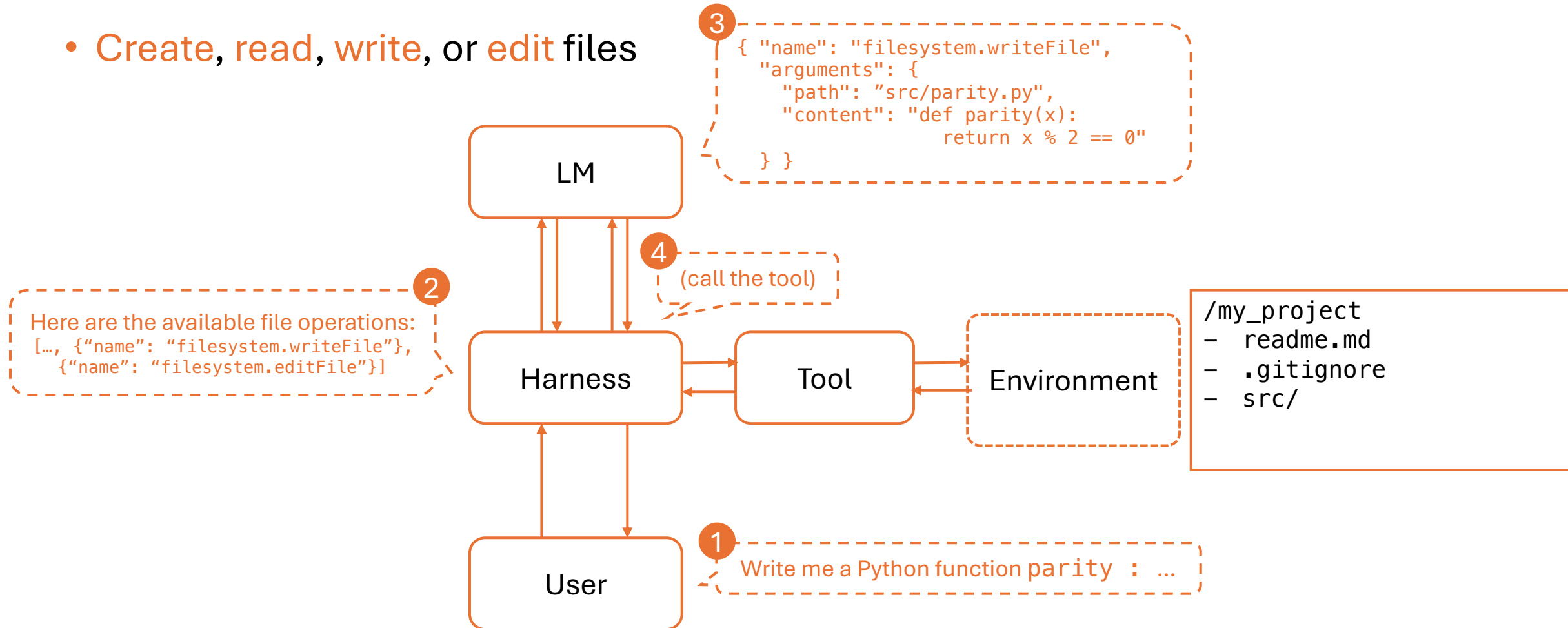
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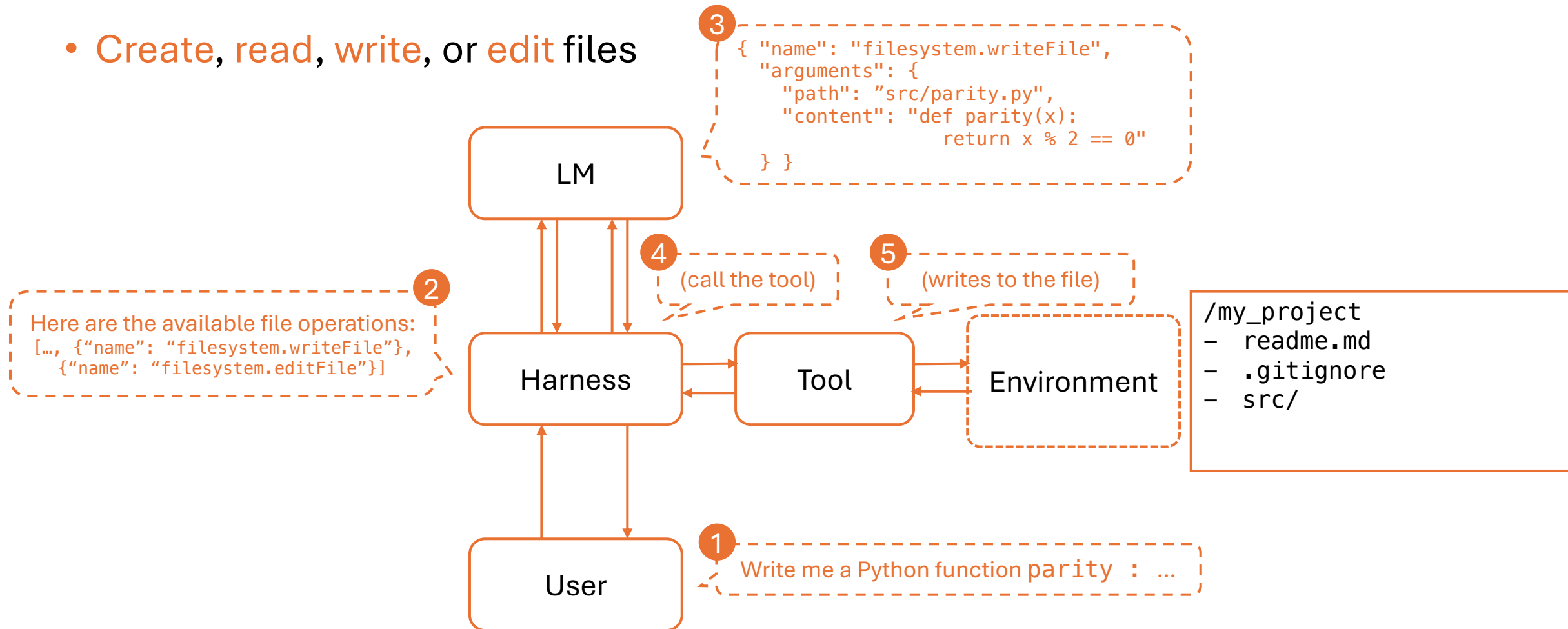
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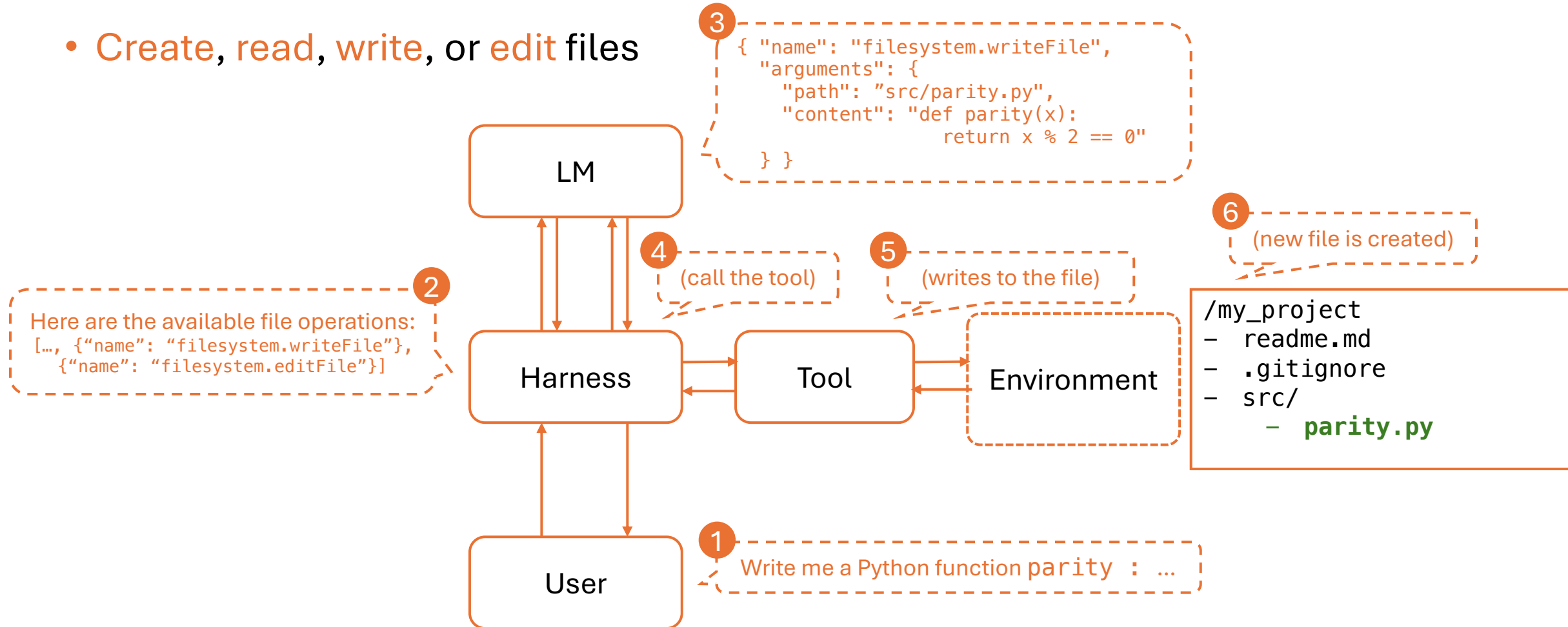
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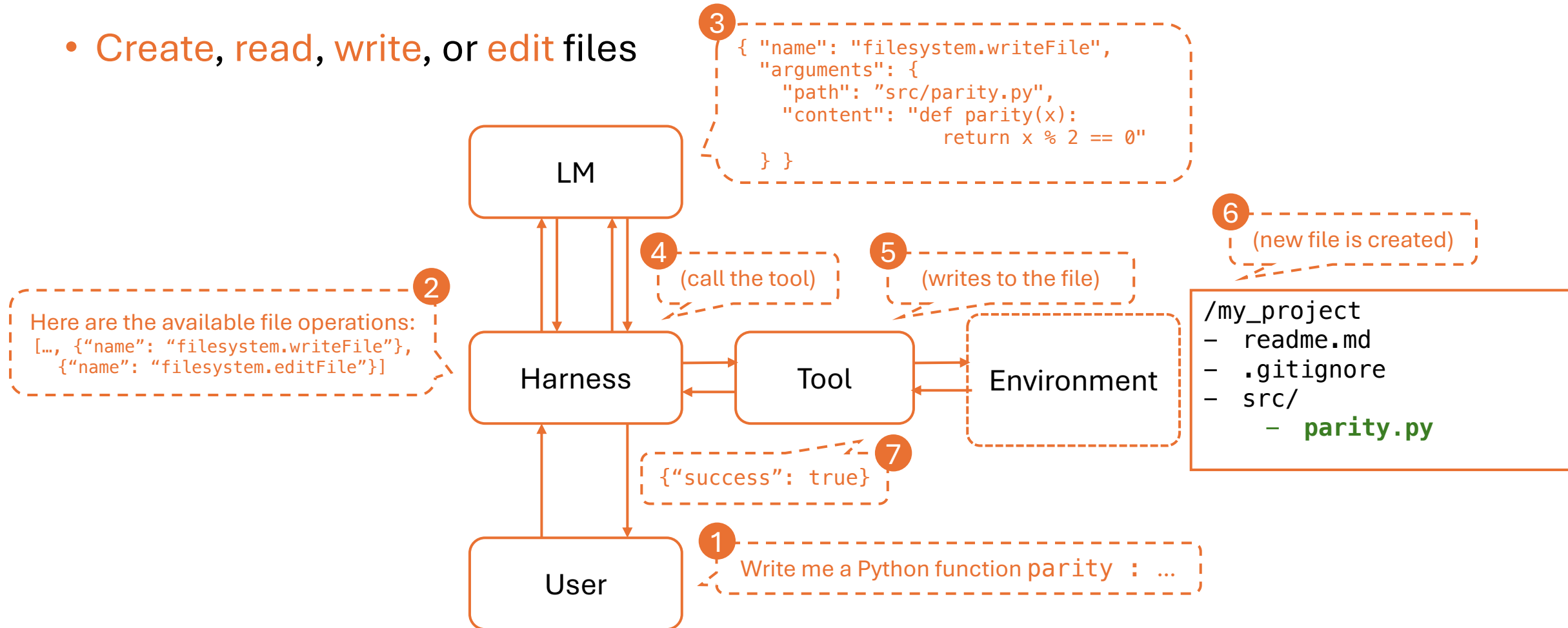
File System (FS) for Code Implementation

- Create, read, write, or edit files



File System (FS) for Code Implementation

- Create, read, write, or edit files



```
{
  name: "read_text_file",
  description:
    "Read the complete contents of a file from the file system as text. " +
    "Handles various text encodings and provides detailed error messages " +
    "if the file cannot be read. Use this tool when you need to examine " +
    "the contents of a single file. Use the 'head' parameter to read only " +
    "the first N lines of a file, or the 'tail' parameter to read only " +
    "the last N lines of a file. Operates on the file as text regardless of extension. " +
    "Only works within allowed directories.",
  inputSchema: zodToJsonSchema(ReadTextFileArgsSchema) as ToolInput,
},
```

```
{
```

```
{
  name: "create_directory",
  description:
    "Create a new directory or ensure a directory exists. Can create multiple " +
    "nested directories in one operation. If the directory already exists, " +
    "this operation will succeed silently. Perfect for setting up directory " +
    "structures for projects or ensuring required paths exist. Only works within allowed directories.",
  inputSchema: zodToJsonSchema(CreateDirectoryArgsSchema) as ToolInput,
},
```

```
ing file with new content. " +
es without warning. " +
rks within allowed directories.",
ToolInput,
```

```
{
  name: "directory_tree",
  description:
    "Get a recursive tree view of files and directories as a JSON structure. " +
    "Each entry includes 'name', 'type' (file/directory), and 'children' for directories. " +
    "Files have no children array, while directories always have a children array (which may be empty). " +
    "The output is formatted with 2-space indentation for readability. Only works within allowed directories.",
  inputSchema: zodToJsonSchema(DirectoryTreeArgsSchema) as ToolInput,
},
```

File System (FS) for Code Implementation

- **Edit file**: how do we specify the “edit” action?
 - Choice 1: Rewrite the entire file
 - Will waste a lot of tokens during iterative refinement
 - **Edit**: new file content **S**
 - Choice 2: Rewrite given lines by line numbers
 - Need to also display “line numbers” when reading the file
 - **Edit**: change lines **X** to **Y** by the string **S**
 - Choice 3: Replace a string with another string
 - Is a bit fragile; the pattern string must be chosen carefully:
 - If it’s too generic, the tool may unintentionally modify multiple occurrences.
 - If there is no exact match, nothing can be edited.
 - **Edit**: replace string **X** with string **Y**

File System (FS) for Code Implementation

- **Edit file**: how do we specify the “edit” action?
 - Choice 1: Rewrite the entire file
 - Will waste a
 - **Edit**: new file content **S**
 - Choice 2: Rewrite given lines by line numbers
 - Need to also display “line numbers” when reading the file
 - **Edit**: change lines **X** to **Y** by the string **S**
 - Choice 3: Replace a string with another string
 - Is a bit fragile; the pattern string must be chosen carefully:
 - If it’s too generic, the tool may unintentionally modify multiple occurrences.
 - If there is no exact match, nothing can be edited.
 - **Edit**: replace string **X** with string **Y**

SWE-agent: Agent-Computer Interfaces Enable Automated Software Engineering

John Yang* Carlos E. Jimenez* Alexander Wettig Kilian Lieret
Shunyu Yao Karthik Narasimhan Ofir Press

Princeton Language and Intelligence, Princeton University

- Choice 2: Rewrite given lines by line numbers
 - Need to also display “line numbers” when reading the file
 - **Edit**: change lines **X** to **Y** by the string **S**
- Choice 3: Replace a string with another string
 - Is a bit fragile; the pattern string must be chosen carefully:
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 - **Edit**: replace string **X** with string **Y**

- Edit file: how

- Choice 1: Replace

- Will waste

- Edit: new f

- Choice 2: Rewrite given lines by line numbers

- Need to also display “line numbers” when reading the file

- Edit: change lines X to Y by the string S

- Choice 3: Replace a string with another string

- Is a bit fragile; the pattern string must be chosen carefully:

- If it’s too generic, the tool may unintentionally modify multiple occurrences.

- If there is no exact match, nothing can be edited.

- Edit: replace string X with string Y

Observation (File Viewer)

```
[File: solvers/diophantine.py (3093 lines total)]  
(400 lines above)  
401: elif total_degree == 3:  
402:     diop_type = "cubic_thue"  
403:  
404: elif (total_degree > 3):  
405:     if all(coeff[k] == 1):  
406:         diop_type = 'general_sum_even'  
407:  
408: if diop_type is not None:  
409:     return var, coeff, diop_type  
(2684 lines below)
```

(a) Observation from the file viewer.

Response (Thought + Edit Action)

We need to add a check to ensure that all powers are the same. Let's edit the code to include this check.

```
edit 404:407  
    elif (total_degree > 3 and  
          len(set(k.exp[k.is_Pow])) == 1):  
        if all(coeff[k] == 1):  
            diop_type = 'general_sum_even'  
end_of_edit
```

(b) Action using the edit interface.

Figure 3: The file viewer and edit command are integrated. (a) The file viewer shows the agent the open file’s content with line numbers. (b) The agent invokes the edit function to replace lines 404-407 in the open file. After the edit, the file viewer shows the agent the now updated version of the file.

File System (FS) for Code Implementation

- **Edit file**: how do we specify the “edit” action?
 - Choice 1: Rewrite the entire file
 - Will waste a
 - **Edit**: new file content **S**
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 - **Edit**: replace string **X** with string **Y**

File System (FS) for Code Implementation

- Edit file: how do we specify the “edit” action?

- Choice

- Will

- Edit:

- Choice

- Need

- Edit:

```
102     const EditOperation = z.object({
103       oldText: z.string().describe('Text to search for – must match exactly'),
104       newText: z.string().describe('Text to replace with')
105     });
106
107     const EditFileArgsSchema = z.object({
108       path: z.string(),
109       edits: z.array(EditOperation),
110       dryRun: z.boolean().default(false).describe('Preview changes using git-style diff format')
111     });
```

- Choice 3: Replace a string with another string

- Is a bit fragile; the pattern string must be chosen carefully:
 - If it's too generic, the tool may unintentionally modify multiple occurrences.
 - If there is no exact match, nothing can be edited.
 - Edit: replace string X with string Y

SWE-agent: Agent-Computer Interfaces Enable Automated Software Engineering

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SWE-agent: Agent-Computer Interfaces Enable Automated Software Engineering

John Yang* Ca
Shuny
Princeton

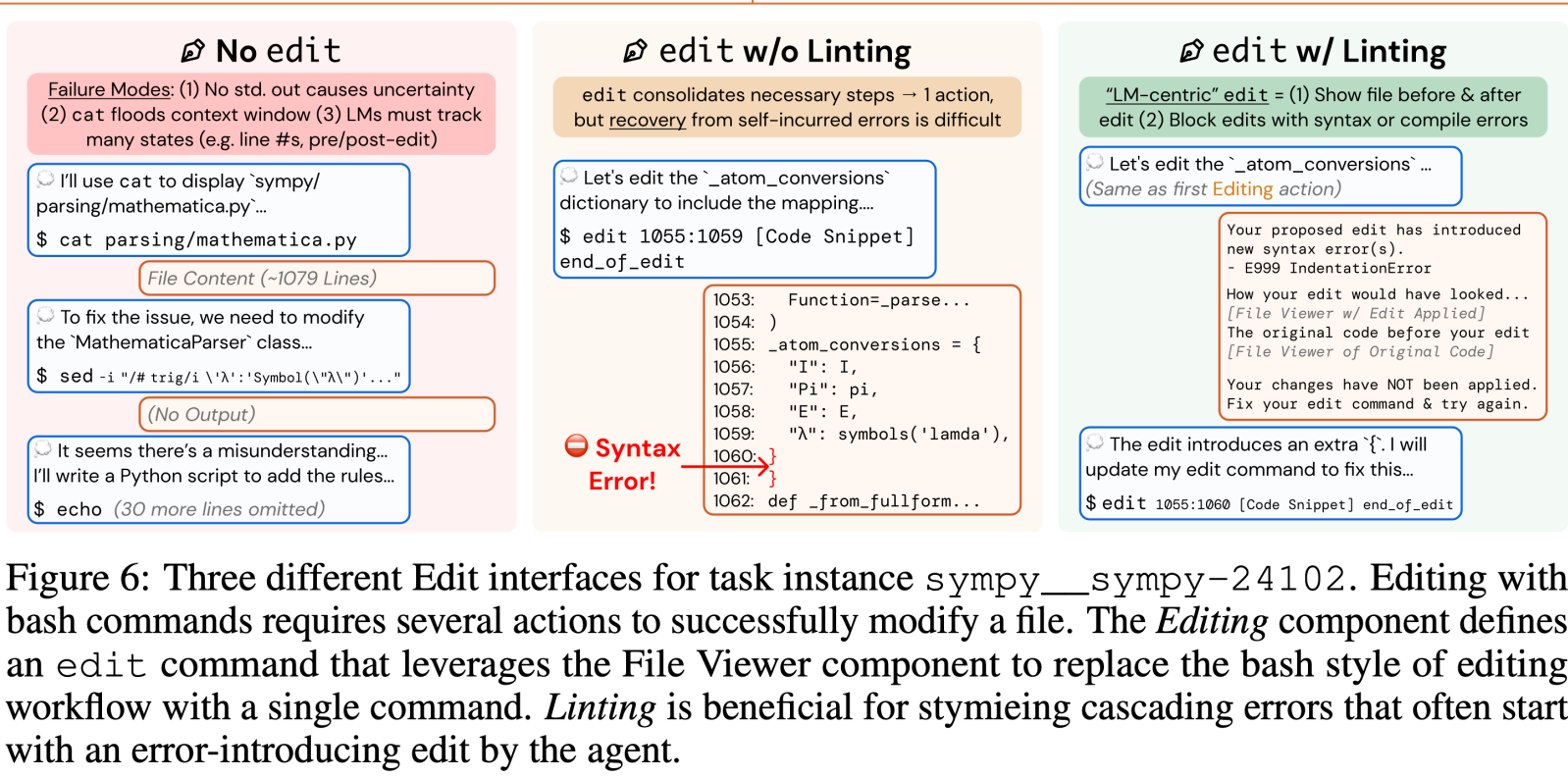
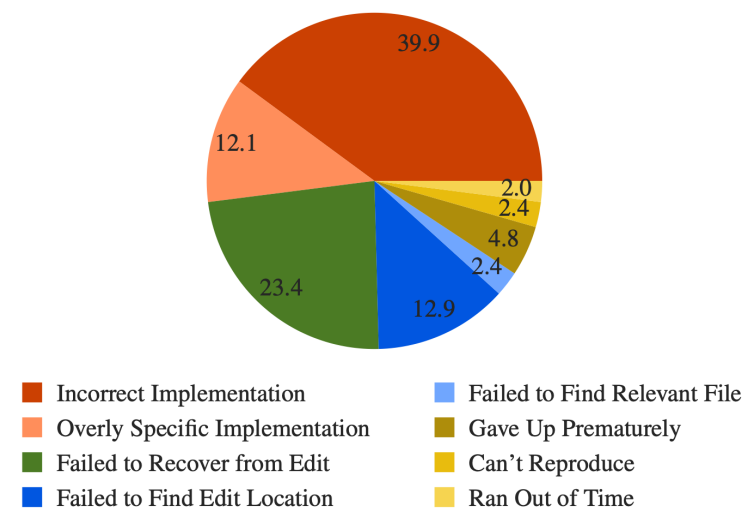
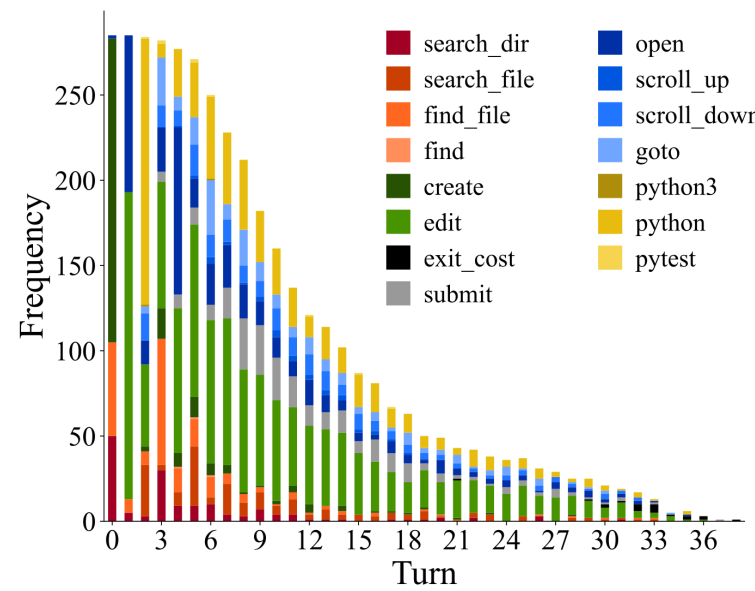
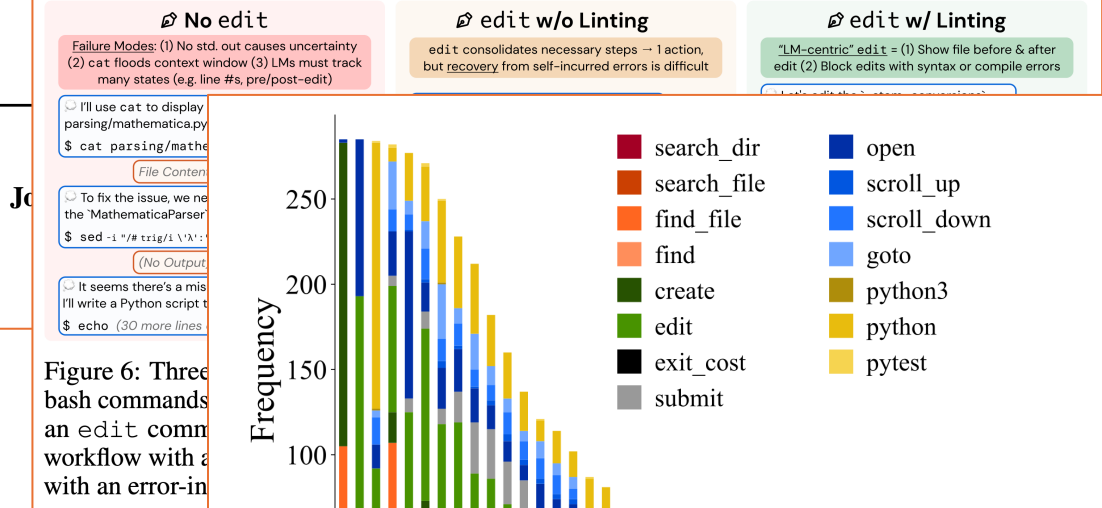


Figure 6: Three different Edit interfaces for task instance `sympy__sympy-24102`. Editing with bash commands requires several actions to successfully modify a file. The *Editing* component defines an `edit` command that leverages the File Viewer component to replace the bash style of editing workflow with a single command. *Linting* is beneficial for stymieing cascading errors that often start with an error-introducing edit by the agent.

SWE-agent: Agent-Computer Interfaces Enable



Can you look at core/scallop-fir/docs/ and help me implement a type_checker

I'll help you implement a type checker for the Scallop language. Let me first explore the documentation and understand the current structure of the project.

Listed docs

Read grammar.md

Searched What is the current

Let me explore the existing

Read test_type_inference.rs

Read test_type_bounds.rs

Searched Where are type def

Let me explore the source c

Listed src

Read lib.rs

Grepped type.*check|TypeChecker|type_infer

I can see that the type checking functionalit
s. Let me explore the AST types to understa

Let me explore more of the type system structure:

Read mod.rs

Read type_decl.rs

Read named_type.rs

Now let me create a comprehensive plan for implementing the type checker:

1 of 9 Done

✓ Explore AST structure and understand type system components

➔ Create the analysis module with type checking infrastructure

○ Implement

Now let me create the analysis module with the type checking infrastructure:

mod.rs (new) +3 •

type_inference.rs (new) +510 •

Now let me update the lib.rs to enable t

lib.rs +1 -1 •

Let me check for any compilation errors

No linter errors found in analysis

I need to fix the generic type parameter structure. Let me check the actual structure:

type_inference.rs +1 -1 •

Now let me check the ADT declaration structure:

Read adt_decl.rs

I need to fix the ADT checking code:

type_inference.rs +7 -2 •

Let me check the alias type declaration structure:

Read alias_type_decl.rs

I need to fix the alias type declaration checking:

type_inference.rs +7 -1 •

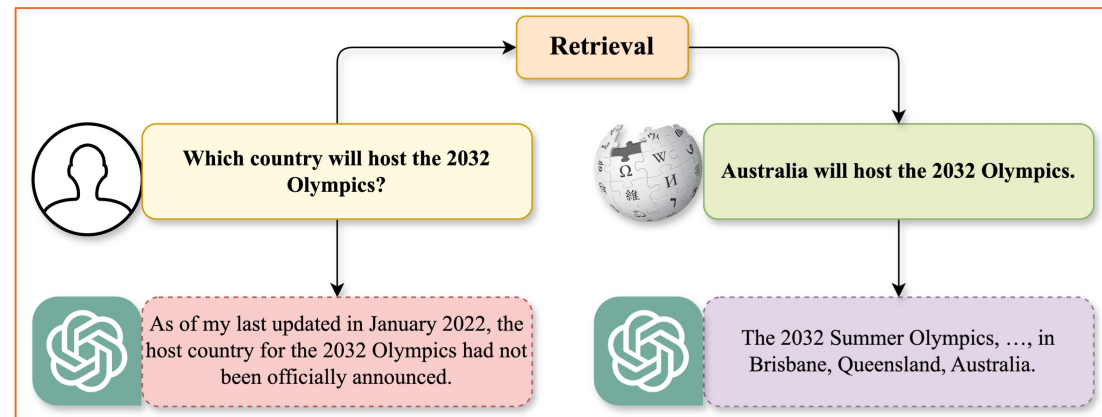
Tools Related to Software Development

- ~~Sequential thinking~~
 - ~~Planning, deliberate thinking~~
- ~~File systems (FS)~~
 - ~~Creating, removing, editing programs~~
 - ~~Drafting test cases and setting up testing environment~~
- Retrieval augmented generation (RAG) databases
 - Context management
 - Documentations, examples, code snippets

Retrieval Augmented Generation (RAG)

- Intuition:

- We do not solely rely on Language Model's internal knowledge
- We assume that there is an **external database** of knowledge
 - Each piece of knowledge is called a “**document**”
- Given a task, we **retrieve** some documents from this database
- Based on the documents, we ask LM to continue **generating** the answers



Retrieval Augmented Generation (RAG)

Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks

Patrick Lewis^{†‡}, Ethan Perez^{*},

Aleksandra Piktus[†], Fabio Petroni[†], Vladimir Karpukhin[†], Naman Goyal[†], Heinrich Küttler[†],

Mike Lewis[†], Wen-tau Yih[†], Tim Rocktäschel^{†‡}, Sebastian Riedel^{†‡}, Douwe Kiela[†]

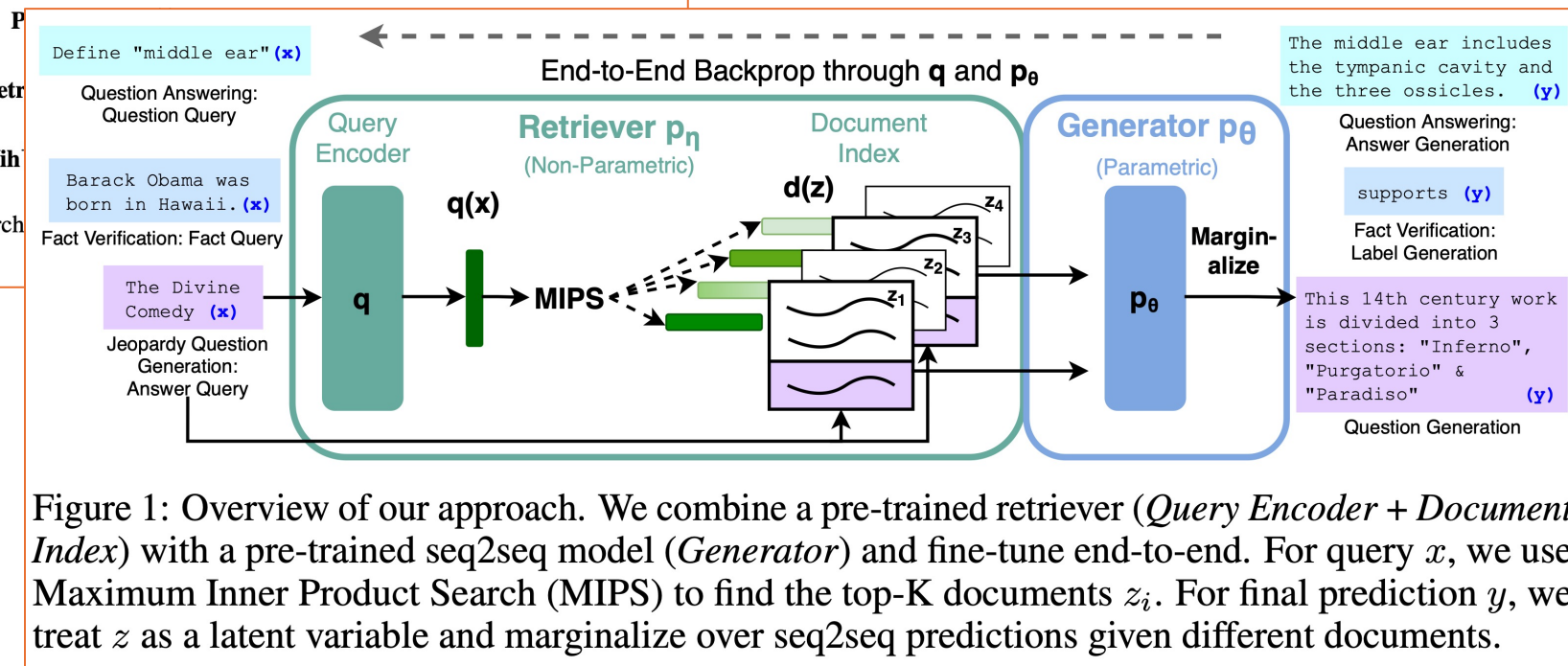
[†]Facebook AI Research; [‡]University College London; ^{*}New York University;
plewis@fb.com

Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks

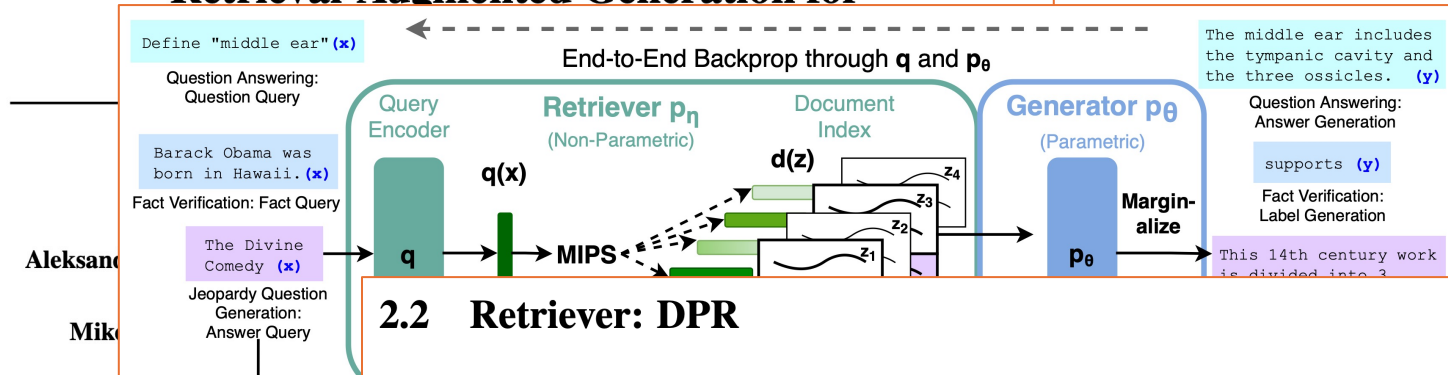
Aleksandra Piktus[†], Fabio Petr

Mike Lewis[†], Wen-tau Yih

[†]Facebook AI Research



Retrieval-Augmented Generation for



2.2 Retriever: DPR

The retrieval component $p_\eta(z|x)$ is based on DPR [26]. DPR follows a bi-encoder architecture:

$$p_\eta(z|x) \propto \exp(\mathbf{d}(z)^\top \mathbf{q}(x)) \quad \mathbf{d}(z) = \text{BERT}_d(z), \quad \mathbf{q}(x) = \text{BERT}_q(x)$$

where $\mathbf{d}(z)$ is a dense representation of a document produced by a $\text{BERT}_{\text{BASE}}$ *document encoder* [8], and $\mathbf{q}(x)$ a query representation produced by a *query encoder*, also based on $\text{BERT}_{\text{BASE}}$. Calculating $\text{top-k}(p_\eta(\cdot|x))$, the list of k documents z with highest prior probability $p_\eta(z|x)$, is a Maximum Inner Product Search (MIPS) problem, which can be approximately solved in sub-linear time [23]. We use a pre-trained bi-encoder from DPR to initialize our retriever and to build the document index. This retriever was trained to retrieve documents which contain answers to TriviaQA [24] questions and Natural Questions [29]. We refer to the document index as the *non-parametric memory*.

The Survey of Retrieval-Augmented Text Generation in Large Language Models

YIZHENG HUANG and JIMMY X. HUANG, York University, Canada

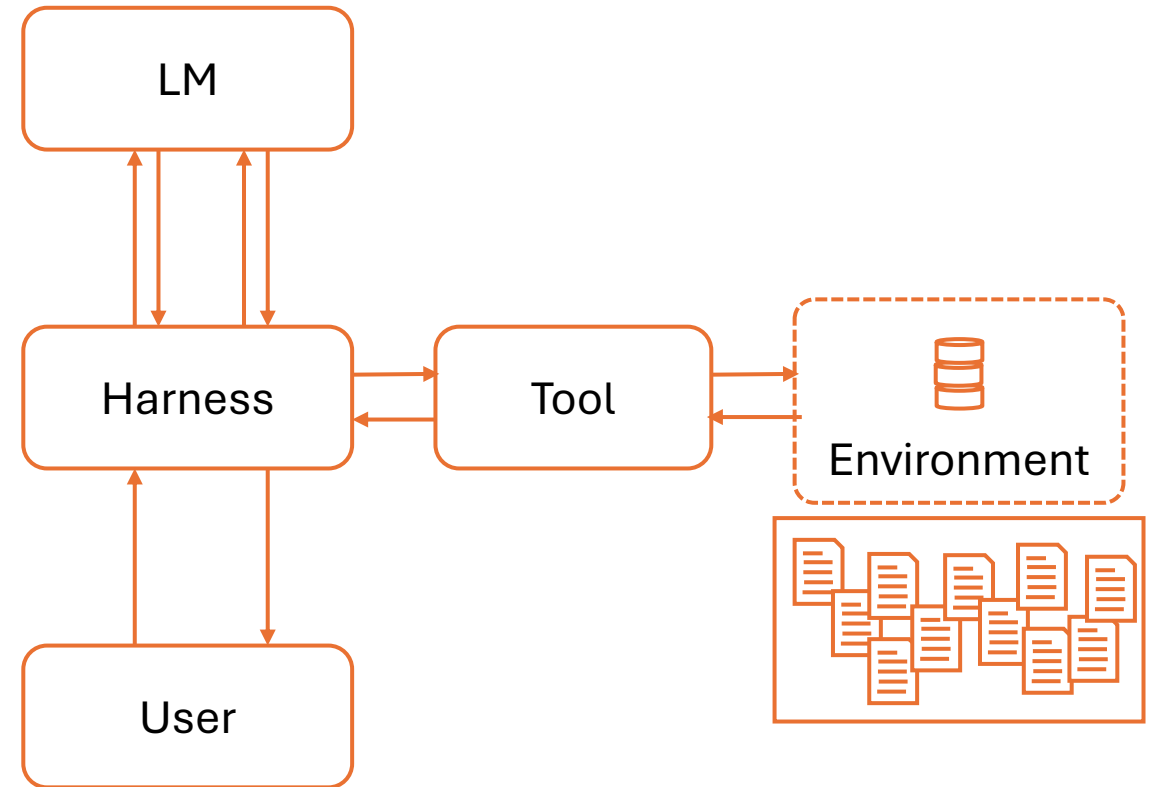
Retrieval-Augmented Generation (RAG) merges retrieval methods with deep learning advancements to address the static limitations of large language models (LLMs) by enabling the dynamic integration of up-to-date external information. This methodology, focusing primarily on the text domain, provides a cost-effective solution to the generation of plausible but possibly incorrect responses by LLMs, thereby enhancing the accuracy and reliability of their outputs through the use of real-world data. As RAG grows in complexity and incorporates multiple concepts that can influence its performance, this paper organizes the RAG paradigm into four categories: pre-retrieval, retrieval, post-retrieval, and generation, offering a detailed perspective from the retrieval viewpoint. It outlines RAG's mechanics and discusses the field's progression through the analysis of significant studies. Additionally, the paper introduces evaluation methods for RAG, addressing the challenges faced and proposing future research directions. By offering an organized framework and categorization, the study aims to consolidate existing research on RAG, clarify its technological underpinnings, and highlight its potential to broaden the adaptability and applications of LLMs.

Retrieval Augmented Generation for Code

- Vector database as environment:
 - Each document z in the database is associated with an embedding $d(z)$
 - Querying: embedding-based similarity search
 - Assume query x , query embedding is $q(x)$
 - Similarity score is cosine similarity between $q(x)$ and $d(z)$:

$$\text{sim}(x, z) = \frac{q(x) \cdot d(z)}{\|q(x)\| \cdot \|d(z)\|}$$

- We can add, update, or remove documents using document metadata



Retrieval Augmented Generation for Code

- Vector database as environment:

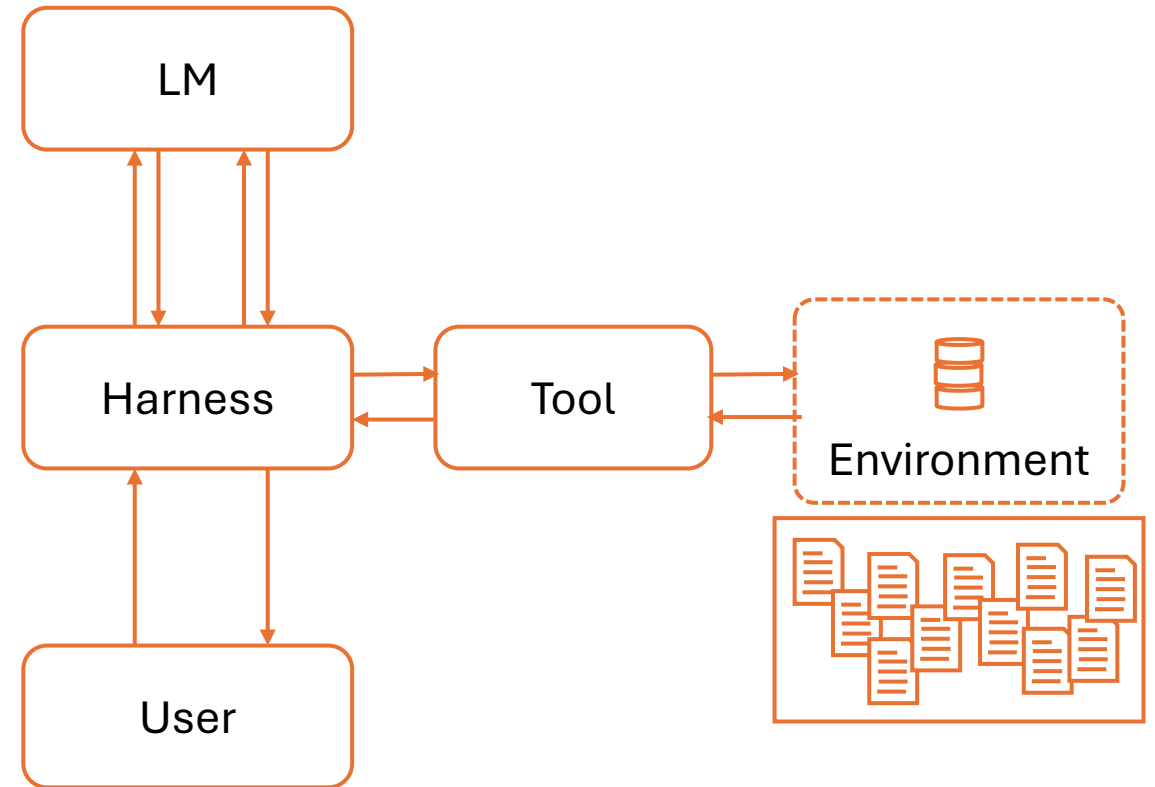
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1

Write me a **Dafny** function Max : ...



Retrieval Augmented Generation for Code

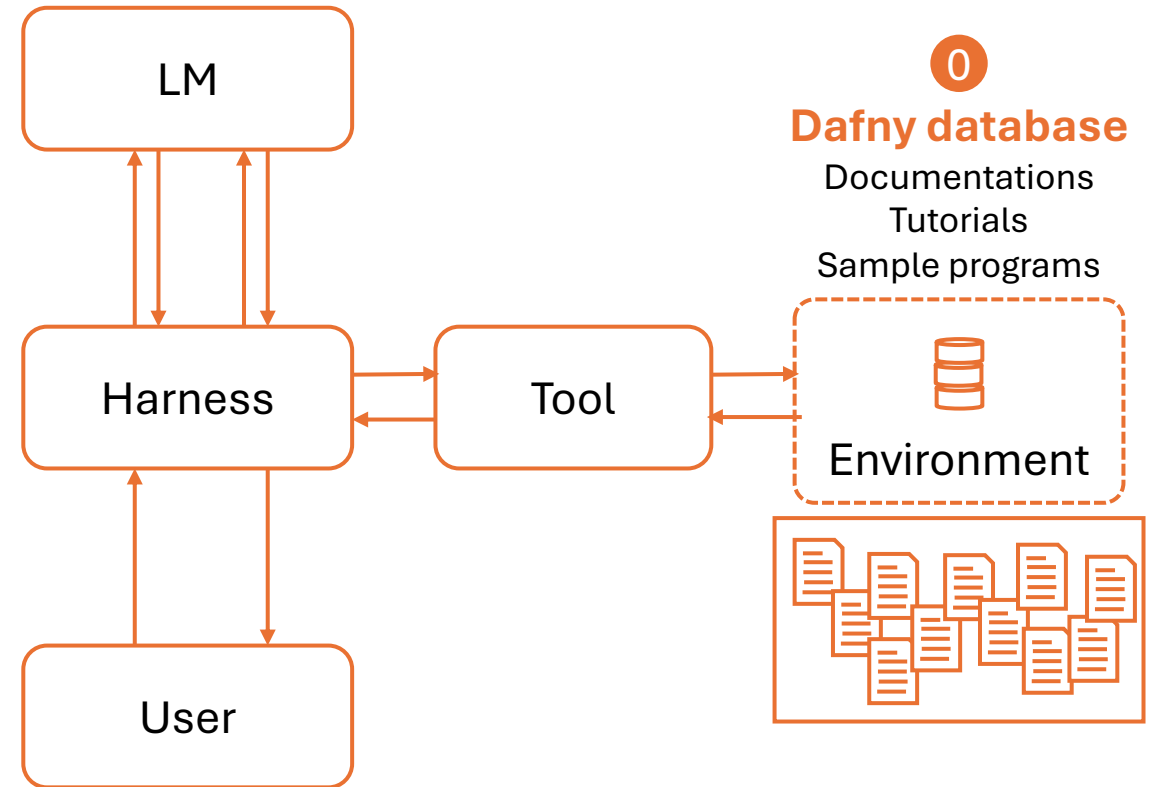
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Retrieval Augmented Generation for Code

- Vector database as environment:

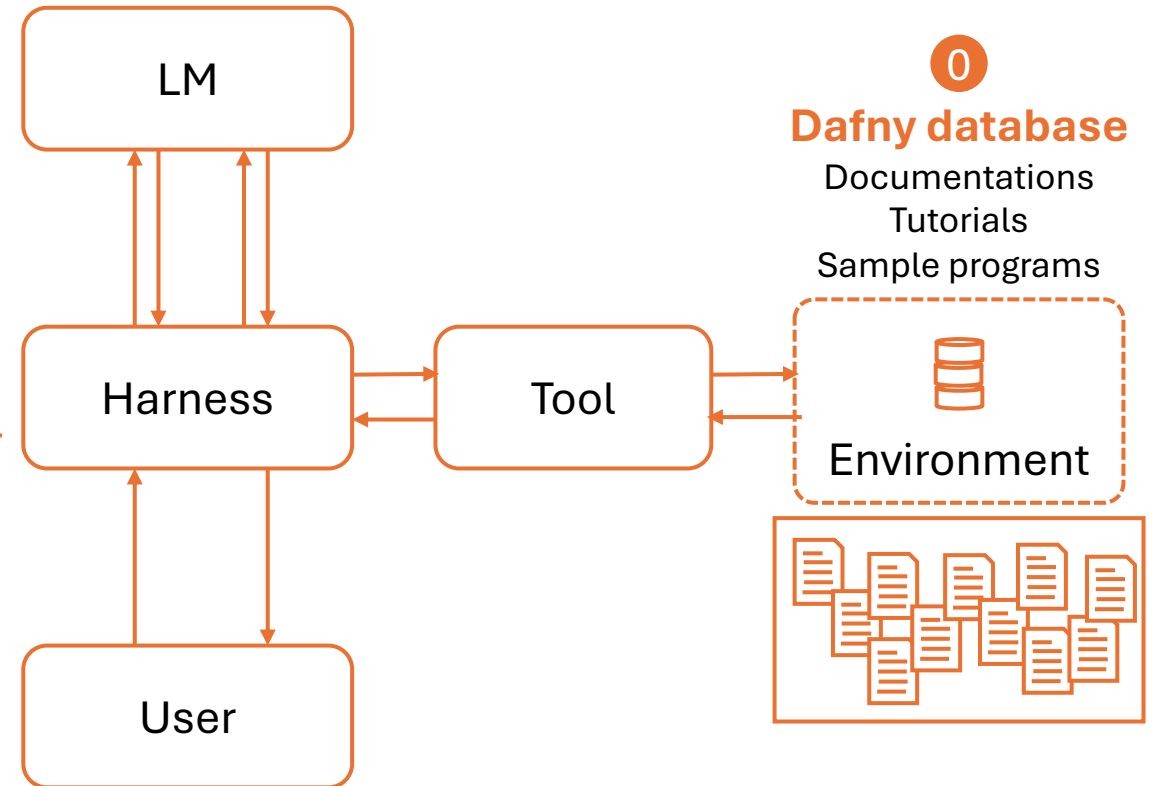
- Each document z in the database is associated with an embedding $d(z)$
- Querying: embedding-based similarity search
 - Assume query x , query embedding is $q(x)$
 - Similarity score is cosine similarity between

2 Here are the available tools:
[..., {"name": "chroma.query"}, ...]

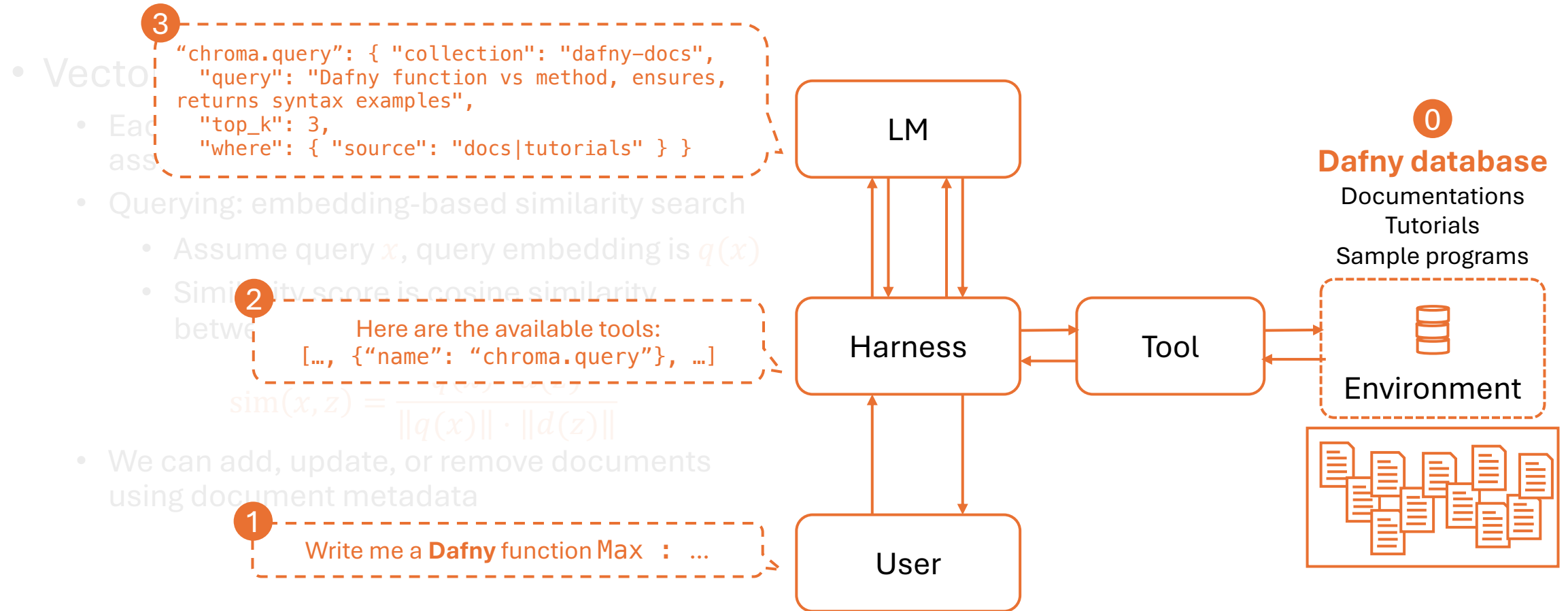
$$\text{sim}(x, z) = \frac{q(x) \cdot d(z)}{\|q(x)\| \cdot \|d(z)\|}$$

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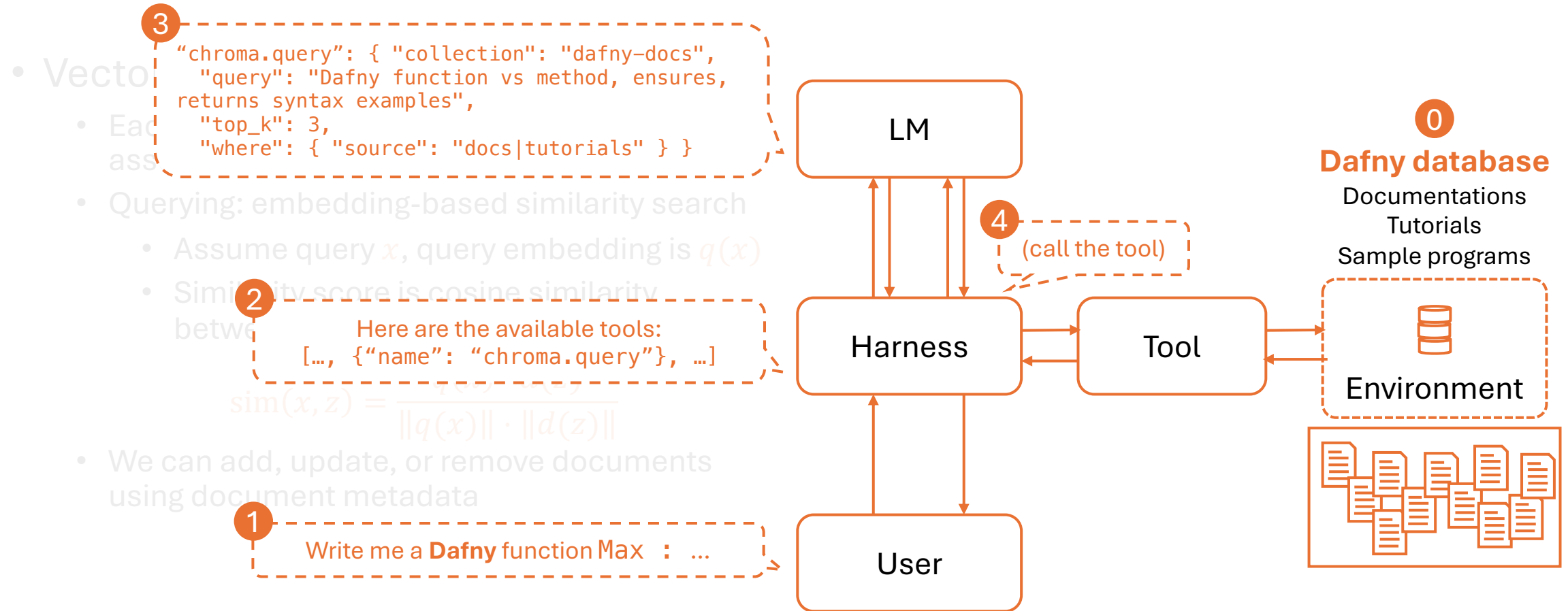
1 Write me a **Dafny** function Max : ...



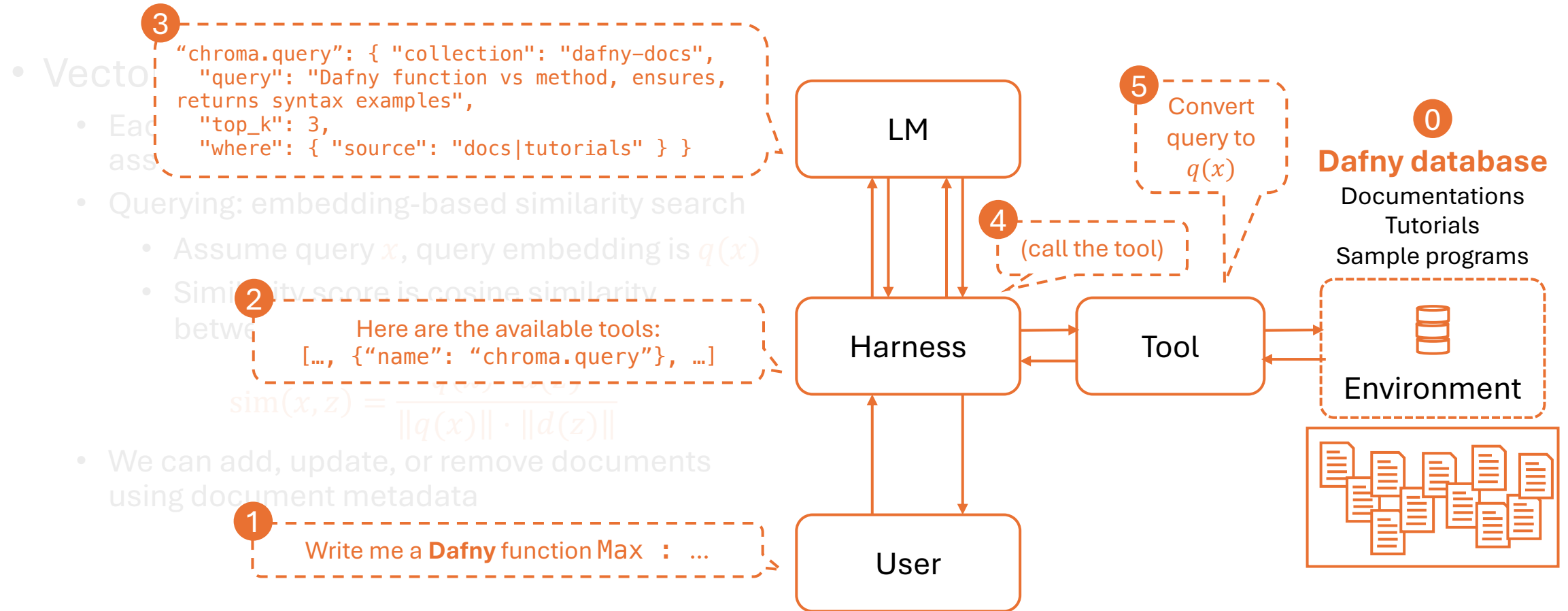
Retrieval Augmented Generation for Code



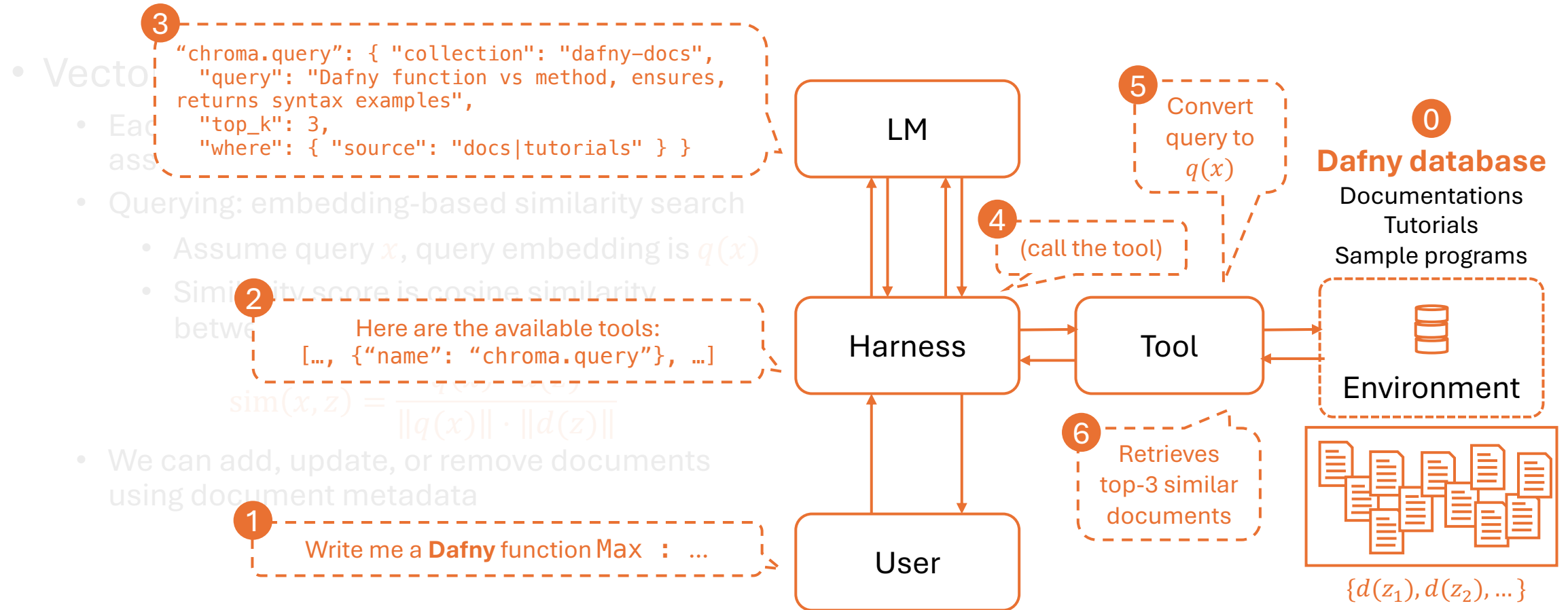
Retrieval Augmented Generation for Code



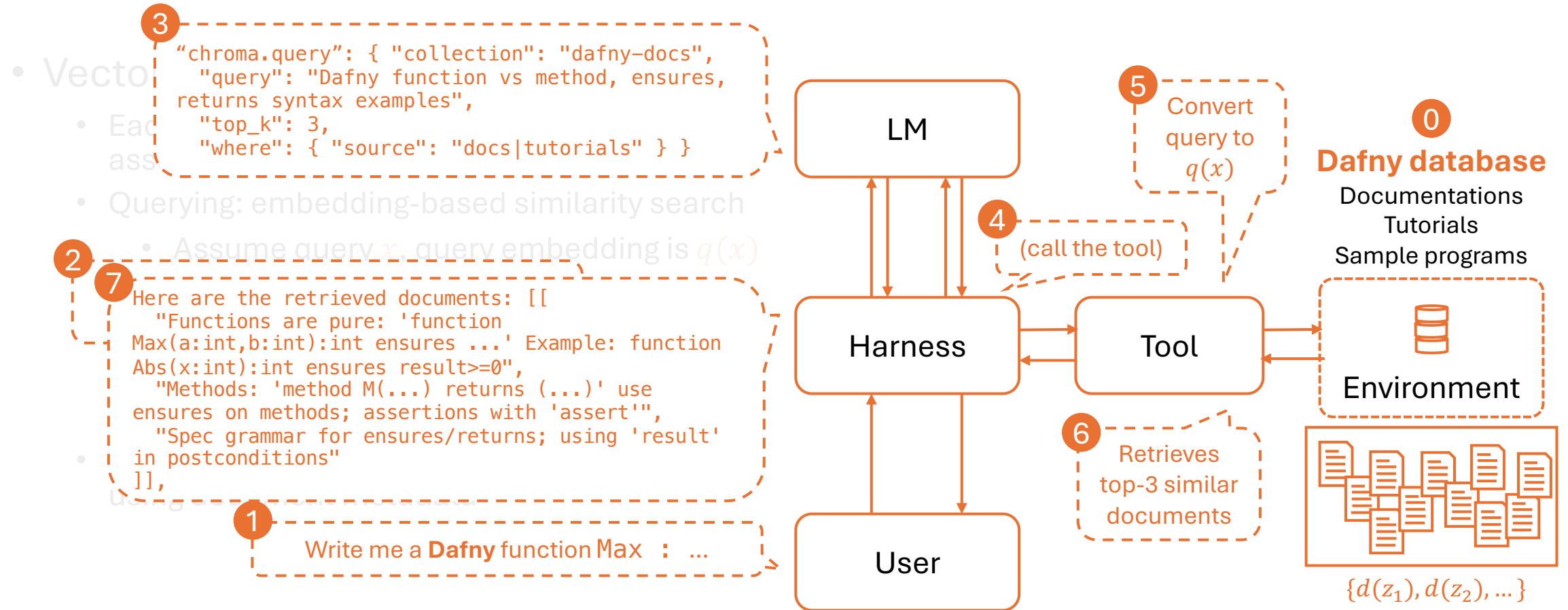
Retrieval Augmented Generation for Code



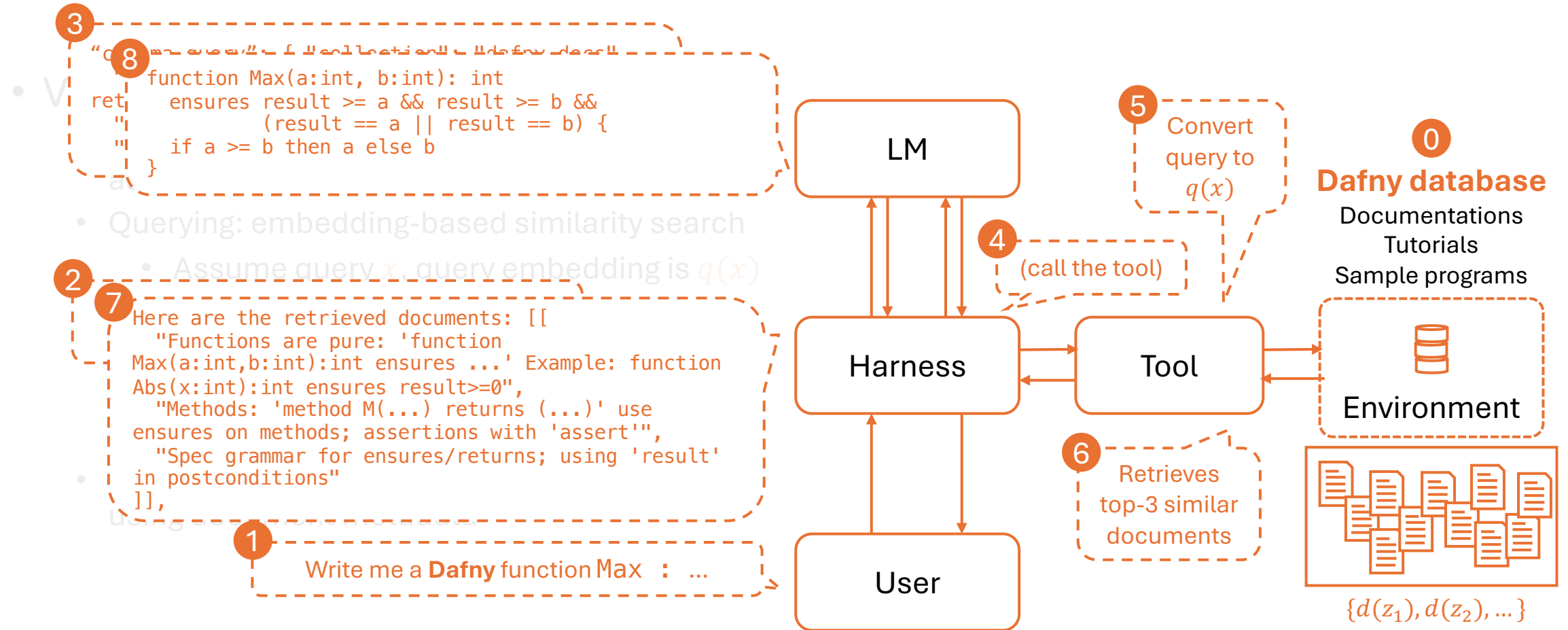
Retrieval Augmented Generation for Code



Retrieval Augmented Generation for Code



Retrieval Augmented Generation for Code

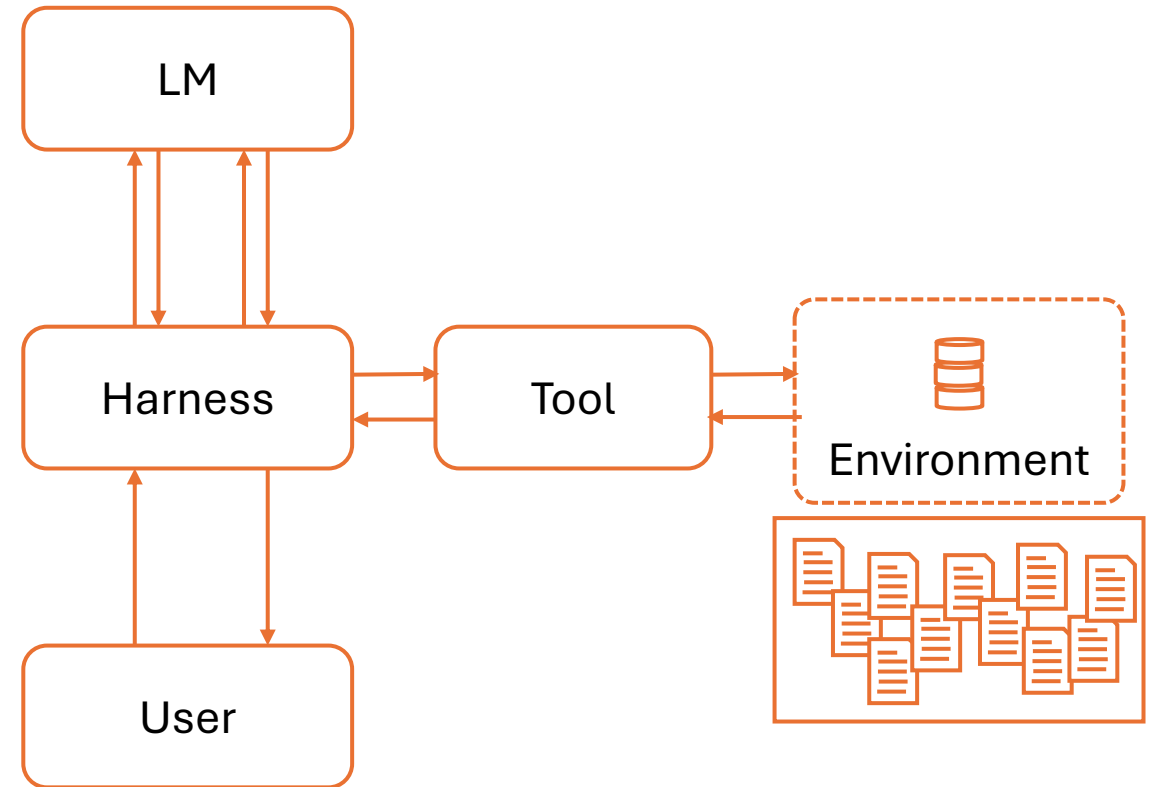


Retrieval Augmented Generation for Code

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$$\text{sim}(x, z) = \frac{q(x) \cdot d(z)}{\|q(x)\| \cdot \|d(z)\|}$$

- We can add, update, or remove documents using document metadata



CODERAG-BENCH: Can Retrieval Augment Code Generation?

Zora Zhiruo Wang^{♠*} **Akari Asai**^{◇*}

Xinyan Velocity Yu[♡] **Frank F. Xu**[♠] **Yiqing Xie**[♠]

Graham Neubig[♠] **Daniel Fried**[♠]

[♠]Carnegie Mellon University [◇]University of Washington

[♡]University of Southern California

<https://code-rag-bench.github.io/>

CODERAG-BENCH: Can Retrieval Augment Code Generation?

Zora Zhiruo Wang^{✉*} Akari Asai^{◇*}

Xinyan Velocity

Graham

[✉] Carnegie Mellon U

Unive

<https://carnegie Mellon.edu/>

8 Coding Tasks for Code RAG

Basic Programming x 3

```
def has_close_elements(
    numbers: List[float], threshold: float)
    -> bool:
    """ Check if in given list of numbers,
    are any two numbers closer to each
    other than given threshold.
    """
```

Open-Domain x 2

```
df.columns = np.concatenate(
    [df.iloc[0, :2], df.columns[2:]]
)
df =
df.iloc[1:].reset_index(drop=True)
return df
```

```
import
numpy as
np
import
pandas
```

Repository-Level x 2

```
def main():
    utils.copy_img(data, img_dir)
    utils.downscale(img_dir, config)
    # call COLMAP executable
    >> _____
```

```
run_colmap(
    image_path, ...)
def downscale(
    img_dir, config):
    ...
    return img
```

Code Retrieval x 1

```
Q: Print a log message to
standard error.
def print_log(text, *colors):
    sys.stderr.write(
        sprint(text, *colors))
```

5 Document Sources for Retrieval

Programming Solutions

```
def truncate_number(number) -> float:
    """Return the decimal part. """
    return number % 1.0
```

Library Documentation

```
numpy.dot(a, b, out=None)
Dot product of two arrays.
• If both a and b are 1-D arrays ...
```

Online Tutorials

Example #1: Using shutil.move() to move file from source to destination

```
... import shutil
dst = shutil.move(src, dst_path) ...
```

oneuser/codemodels

StackOverflow Posts

Pandas map multiple columns based on specific conditions

```
# fill NaN values with old values
out["nid"] = out["nid"].fillna(out["id"])
```

GitHub Repositories

Reproducible retrieval and end-to-end evaluation

Models

VOYAGE AI
OpenAI
elasticsearch

starcoder
deepseek coder

Datastore

Retriever

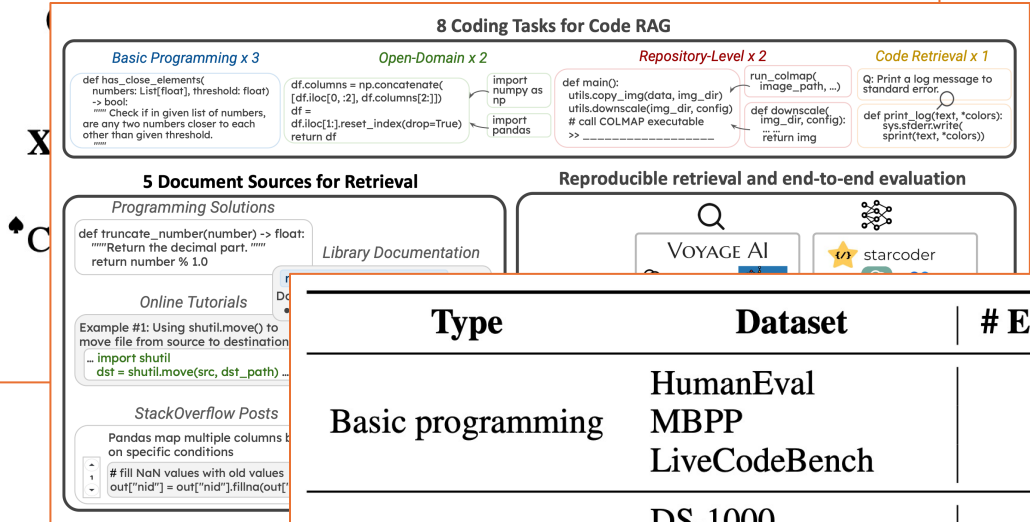
LM

Evaluation

✓ Evaluation against canonical docs

✓ Execution-based end-to-end evaluation

CodeRAG-BENCH:



Type	Dataset	# Examples	# Corpus	Ground-Truth Docs	Evaluation
Basic programming	HumanEval	164	164	program solutions	execution
	MBPP	500	500	program solutions	execution
	LiveCodeBench	400	-	-	execution
Open-domain	DS-1000	1000	34,003	docs	execution
	ODEX	945	34,003	docs, stackoverflow	execution
Repository-level	RepoEval (function)	373	237	github repository	execution
	SWE-bench-Lite	300	40,868	github repository	execution
Code retrieval	CodeSearchNet-Py	22,177	22177	CSN functions	ndcg@10

Method	Basic Programming			Open-Domain			Repo-Level	
	HumanEval	MBPP	LCB	DS-1000	ODEX	ODEX-hard	RepoEval	SWE-bench
w/ StarCoder2-7B								
None	31.7	2.4	1.5	29.2	14.6	10.3	26.5	0.0
BM25	43.9	51.8	1.0	36.7	14.1	13.8	36.7	0.0
GIST-large	38.7	50.4	0.5	35.9	17.3	13.8	40.8	0.3
Voyage, code	39.0	52.6	0.3	36.0	15.3	10.3	45.8	0.3
OpenAI, small	39.0	52.6	1.5	35.5	15.9	17.2	51.2	0.0
OpenAI, rerank	34.8	53.4	0.5	33.4	14.1	17.2	53.9	0.3
Gold	94.5	34.8	-	30.0	17.5	17.2	42.0	0.7
w/ DeepseekCoder-7B-instruct								
None	70.1	60.8	30.5	41.4	39.2	17.2	28.2	0.7
BM25	68.9	60.0	31.8	36.6	37.8	20.7	37.3	0.0
GIST-large	66.3	56.6	33.8	35.9	34.9	20.7	44.5	0.3
Voyage, code	66.5	56.4	31.8	35.9	39.4	17.2	46.6	0.3
OpenAI, small	68.9	58.6	32.0	35.5	37.1	20.7	55.2	0.3
OpenAI, rerank	53.0	60.6	31.5	36.5	37.1	24.1	55.5	0.3
Gold	87.8	63.6	-	43.2	41.7	24.1	48.1	0.0
w/ GPT-3.5-turbo								
None	72.6	70.8	35.3	43.7	41.7	17.2	23.9	2.3
BM25	73.2	72.4	35.5	36.9	41.0	24.1	30.8	6.7
GIST-large	73.2	68.2	34.8	36.7	36.2	13.8	38.3	19.3
Voyage, code	75.0	66.8	34.5	37.4	41.0	20.7	43.2	15.7
OpenAI, small	73.8	68.4	35.8	36.9	40.3	17.2	48.0	21.0
OpenAI, rerank	64.0	72.6	33.5	37.4	40.5	17.2	49.6	21.7
Gold	91.5	72.6	-	42.9	40.3	24.1	39.1	30.7

Table 6: Performance of retrieval-augmented code generation, with top retrieval and generation models. We bold-type the best RACG results. We test gpt-4o on SWE-bench to show non-trivial results than gpt-3.5-turbo. Note that we exclude code canonical answer from the retrieval corpora for basic programming tasks.

CODERAG-BENCH:

8 Coding Tasks for Code RAG

Basic Programming x 3

```
def has_close_elements(
    numbers: List[float], threshold: float)
    -> bool:
    """ Check if in
    are any two num
    other than given
    """
```

Open-Domain x 2

```
df.columns = np.concatenate(
    [df.columns, df.columns, ...])
```

Repository-Level x 2

```
import numpy as
def main():
    util.copy_image(data_image_dir)
```

Code Retrieval x 1

```
Q: Print a log message to
standard error.
```

Type

Basic programming

Open-domain

Repository

Code retrieval

StackOverflow posts

Method

None

BM25

GIST-large

Voyage, code

OpenAI, small

OpenAI, rerank

Gold

Table 6: Performance of different methods. Bold-type the best RAG method that we exclude code completion.

● Tutorials ● Library Documentation
● StackOverflow posts ● Github files

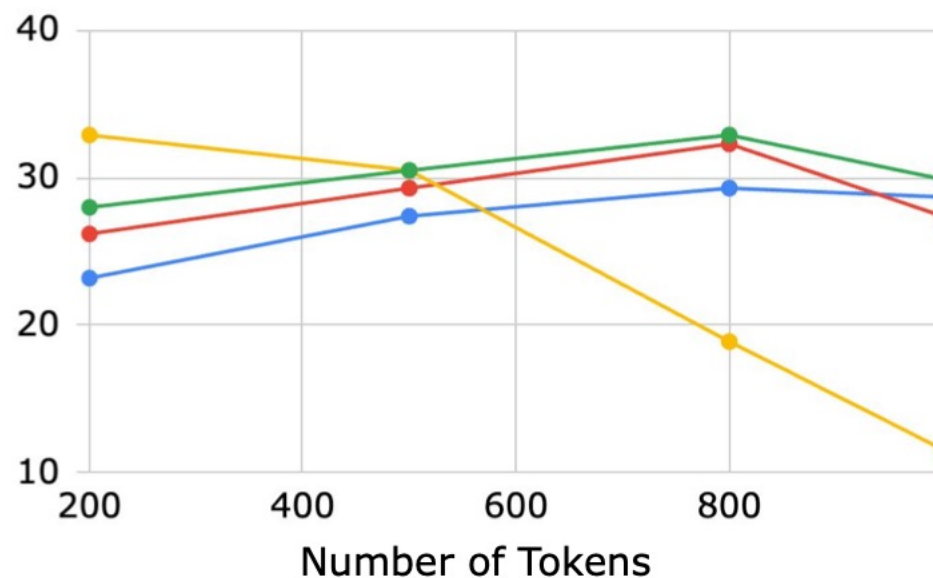


Figure 2: Performance with different chunking sizes.

Note: chunking size is the maximum number of tokens per document; A larger document (e.g., StackOverflow post) will be chunked into multiple smaller documents of chunking size

High Level Picture

